User Interface Design	23CSE209
LAB MANUAL	

Submitted to	
Name	Dr. Rajkumar Batchu
Department	CSE
Designation	Assistant Professor
Submitted by	
Name	PRANAY KINTALI
Roll No	AV.SC.U4.CSE25209
Year/Sem/Section	1st year | SEM-2 | CSE-C

5.	Week5A. Creating a card layout using <div> tag consider three cards and a button using html and CSS B. Navigation using iframe (mini website) upon clicking each buttons the details about that button should be display on the same page C. Design amrita login page by showing different campus on home page upon clicking on each campus the details about each campus should be display on the same page use html and css to design a stylish webpage.	**40-60**	
6.	Week6.A. Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. B. Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.	**61-77**	

7.	Week7. Create a Github Repository and upload all related files of lab manual.	**78-83**	
8.	Week8.A. Create a webpage that displays a square that continuously rotates 360 degree using CSS Animation. B. Create a ball that moves up and down continuously using CSS Animation. C. Create a circle that moves horizontally and rotates at the same time using CSS Animation.	**84-89**	
9.	**Week-9:Create an webpage using html, CSS, java script based on the given conditions:** **a) Create a button for each program, based on the button clicked the corresponding program has to be executed and should display the output.** **b) The logics for each button include finding an even numbers, odd number, prime or not, factorial of a number, Armstrong or not.** **c)Use appropriate designs to display the output.**	**90-93**	

10.	**Week-10:10(A): write JavaScript conditional statement to find the sign of product of three numbers using functions. Display analertboxwithspecified sign 10(B) write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result 10(C) write a java script conditional statement to find largest of find largest of five numbers using function and onclick event . display an alert box to show the result 10(D) write a java script for loop that will iterate from 0 to 15. For each iteration, it will check if the current number odd or even, and display a message on screen 10(E): write a java script program to read 6 subject's marks from the student then computes the average marks of students then, this average is used to determine the corresponding grade.**	**94-102**	
11.	**Week-11:** **A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters. B. A registration form should validate email format before submission. If invalid, the form should not be submitted. C. A system should check password strength while typing and display whether it is weak or strong. D. A form should validate a mobile number when the user leaves the input**	**103-110**	

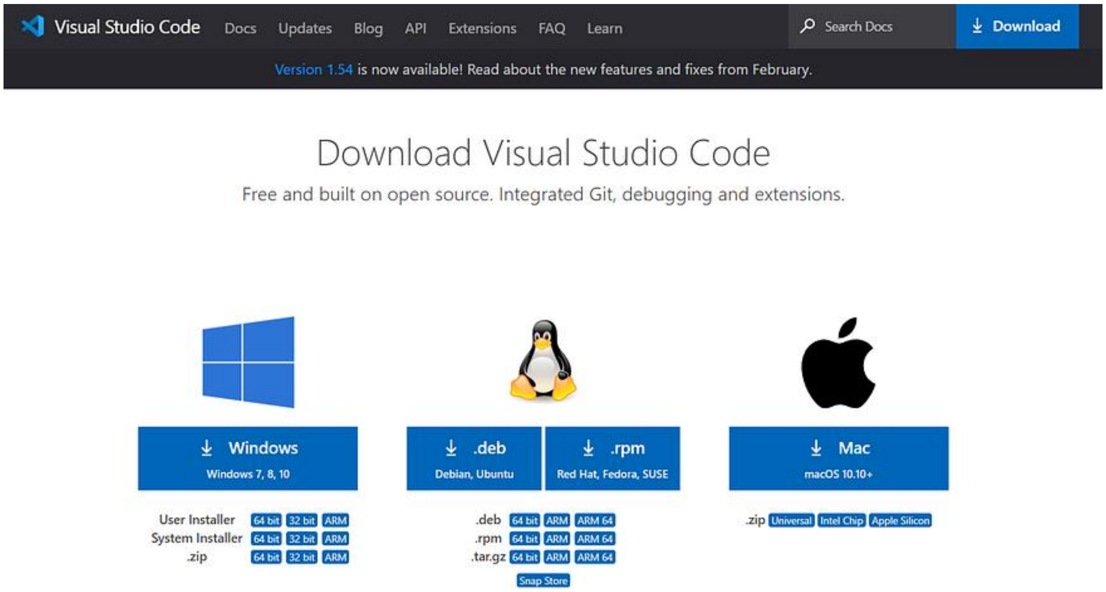
	field. The number must be exactly 10 digits. **E. Design an HTML page, such that the registration form must validate: Name (not empty) Email (valid format) Password (minimum 6 characters)**		
12.	**WEEK-12:** **Steps to install and set up electron**	**111-126**	

WEEK	***1***

1.Aim: To Download and Install VS code

Procedure:

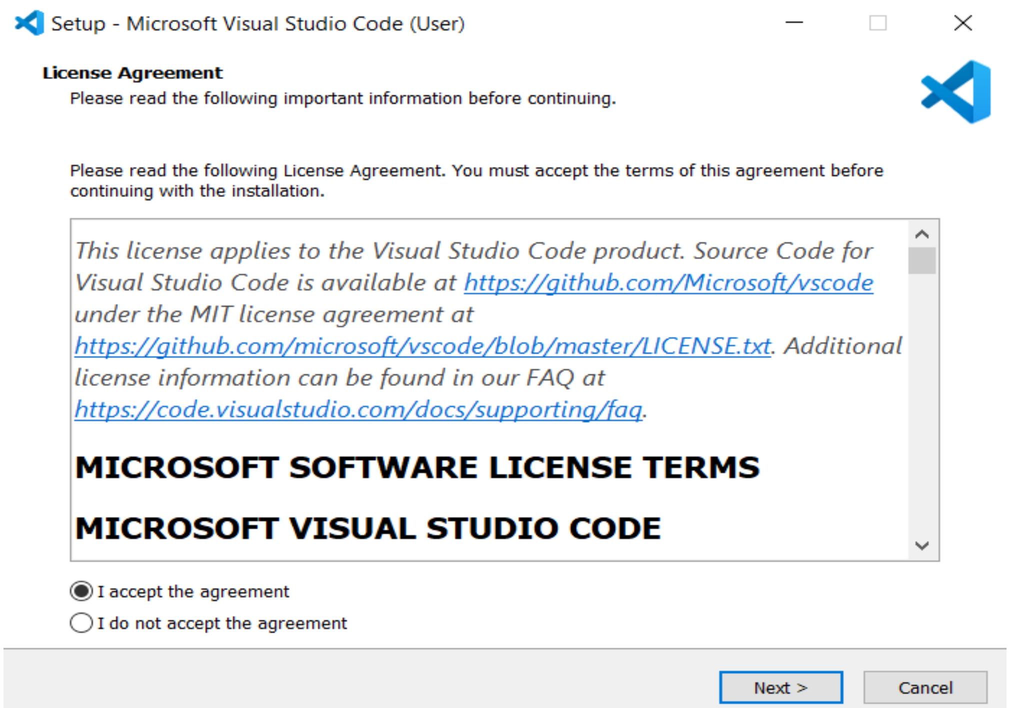
Step-1: Visit the official VS code website to get the downloadable file.



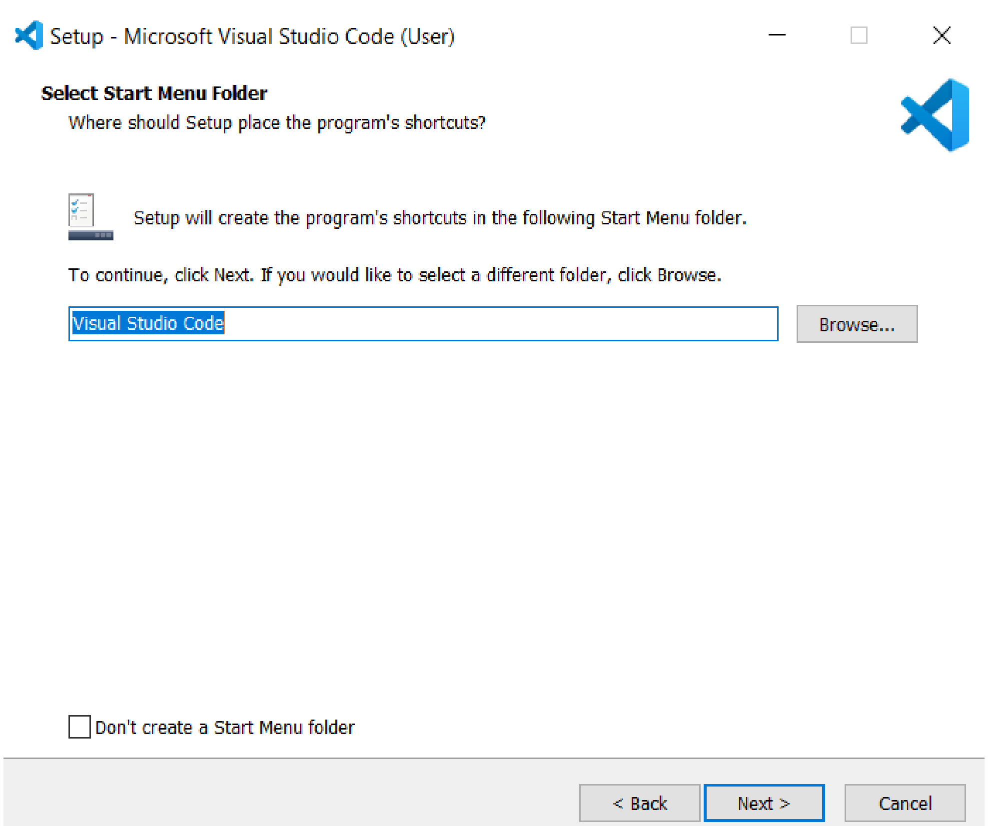
<u>Step-2:</u> Click on the download Option based on the Operating system.

<u>Step-3:</u> Click on the downloaded file to Install.

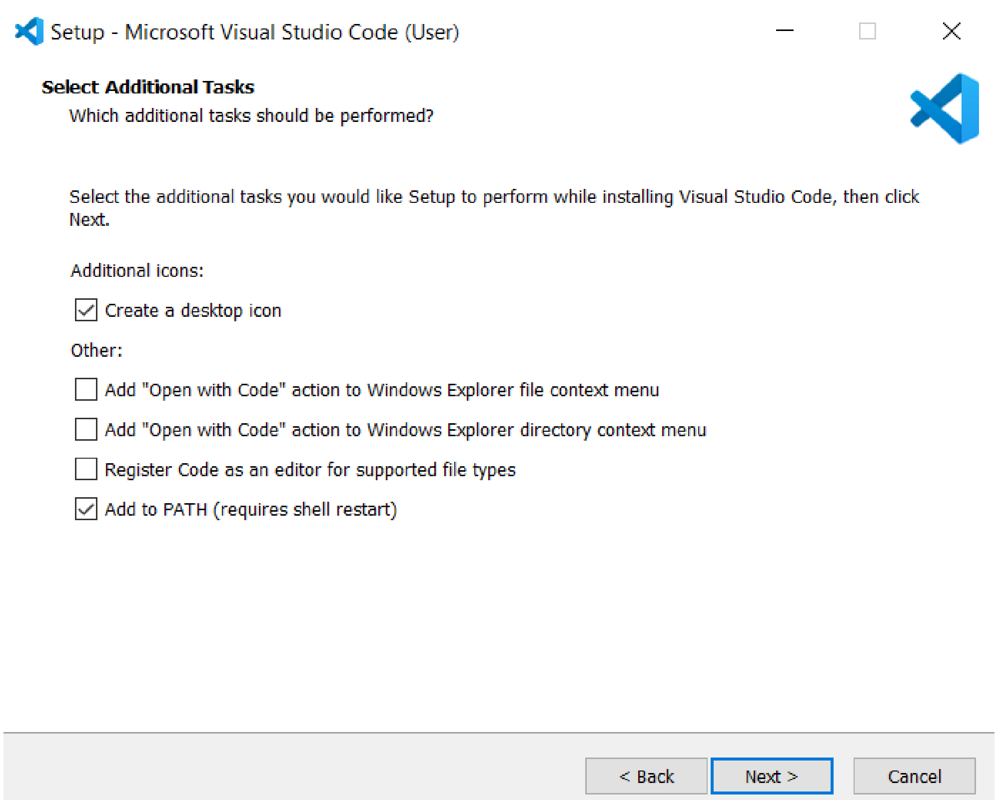
<u>Step-4:</u> Accept the agreement and click on “Next”.



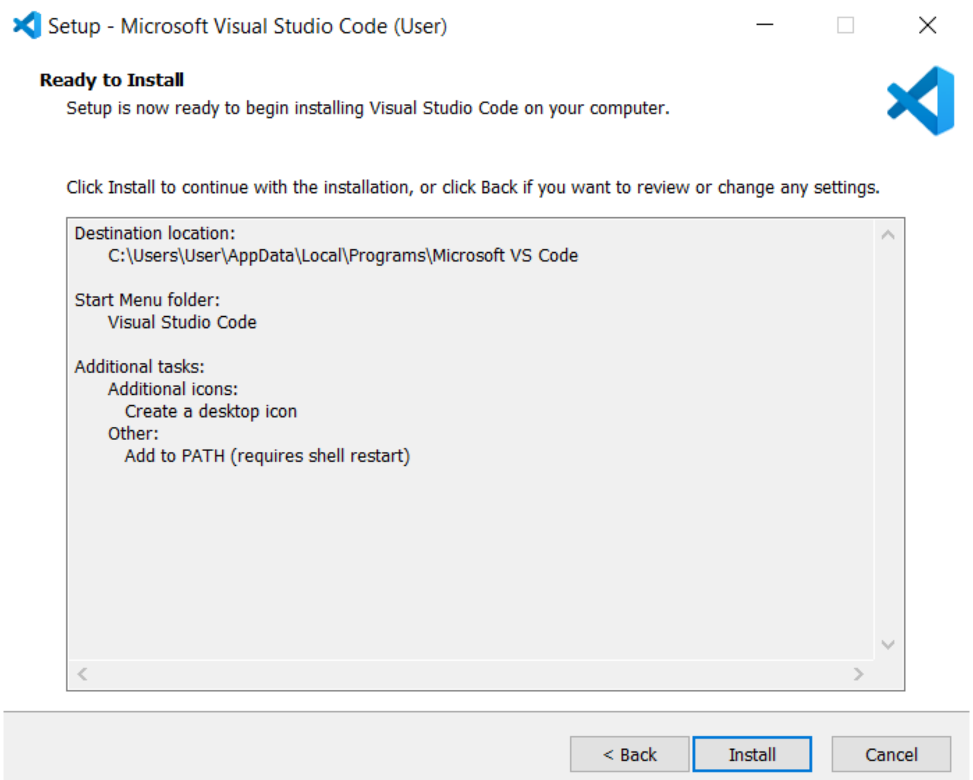
<u>Step-5</u>: Select a folder. Click "Next".



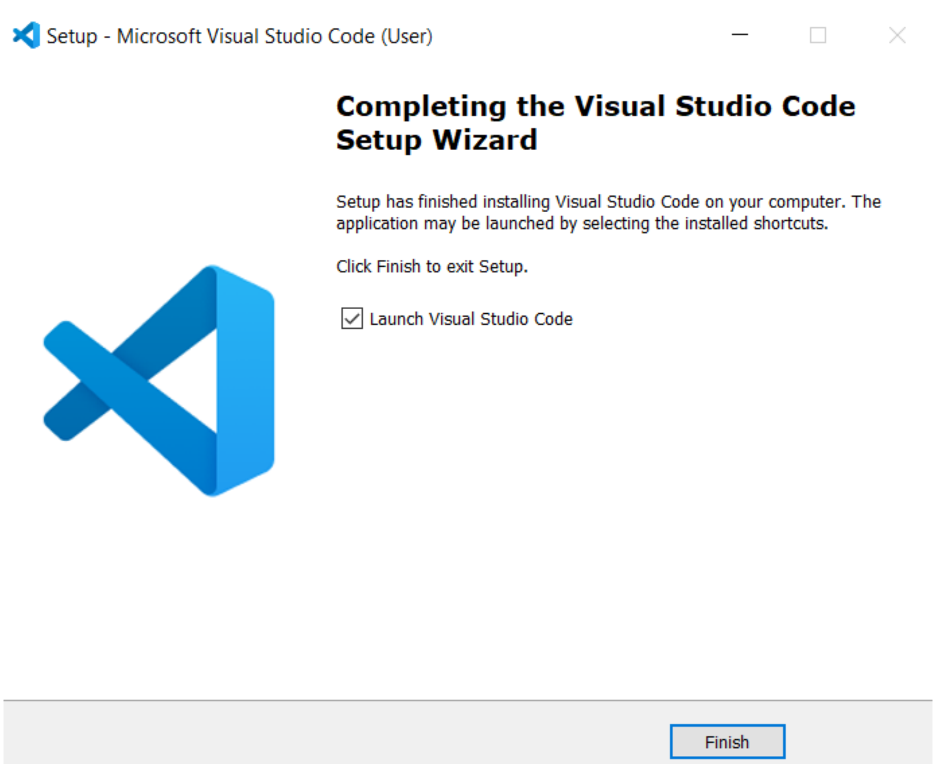
<u>Step-6</u>: Select the required options and click “Next”.



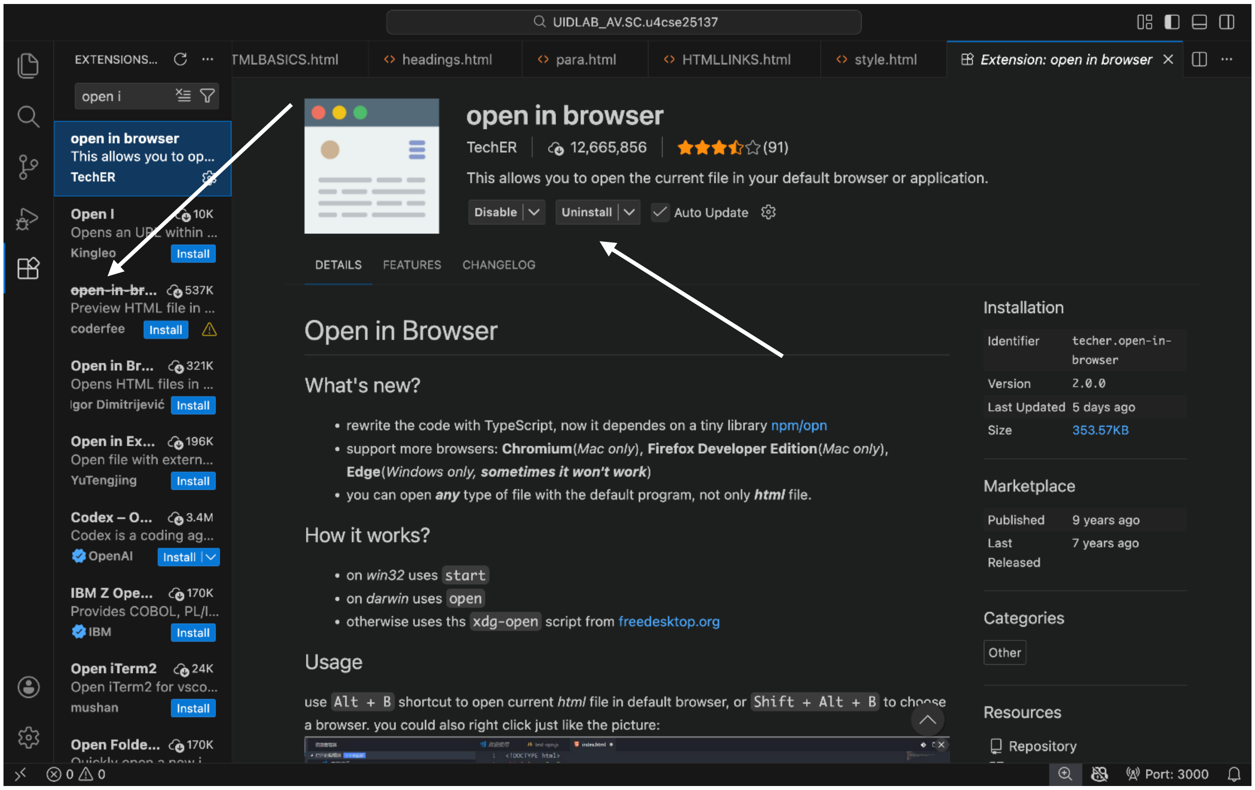
<u>Step-7</u>: Select “Install”.



<u>Step-8</u>: Click "Finish" after installing successfully.



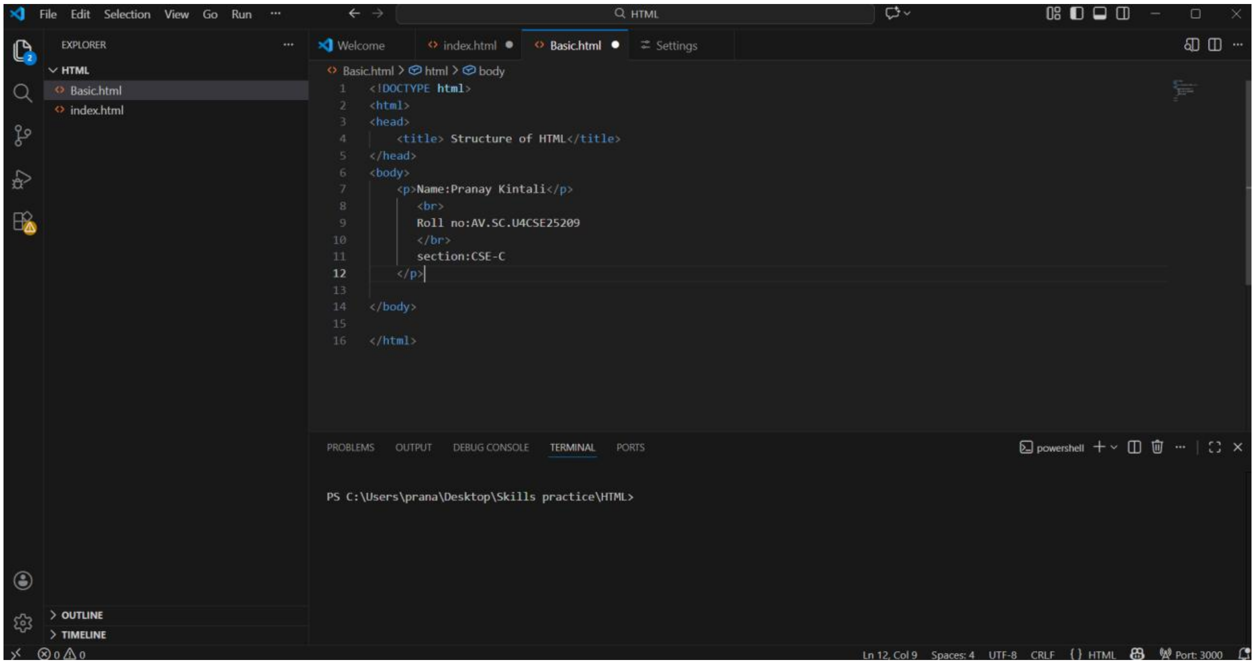
<u>Step-9:</u> Open VS code and go to Extensions tab on the left side panel, search for "Open in browser" extension and download that extension.



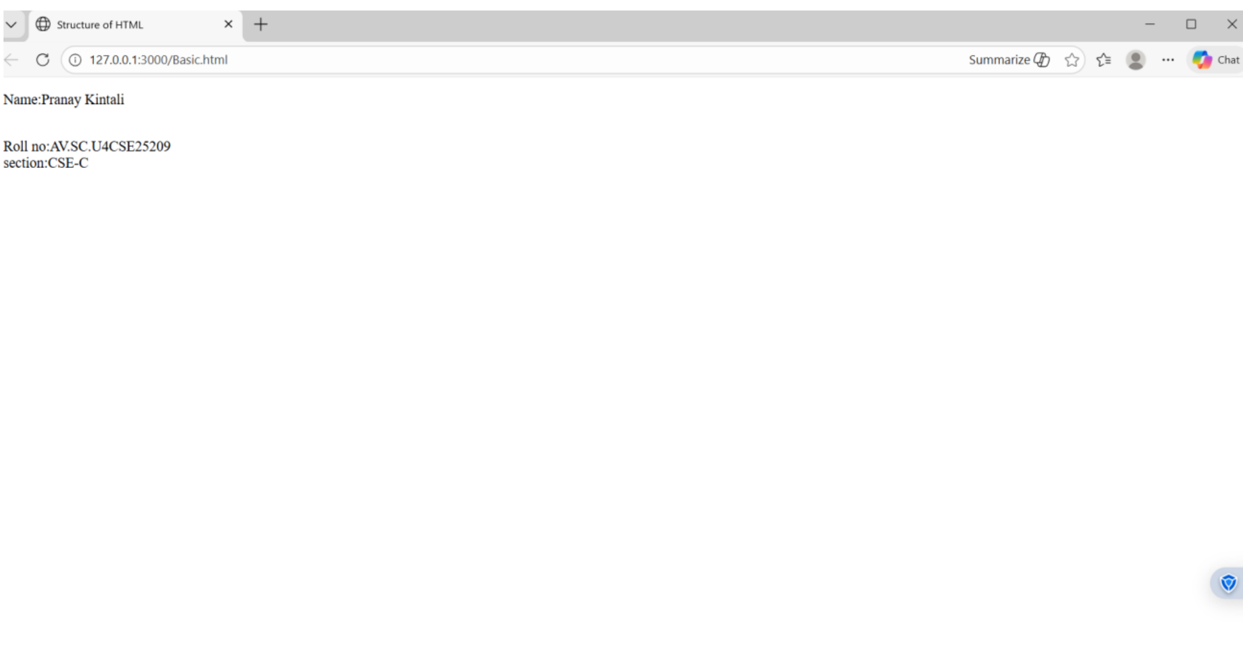
<u>2.</u>Aim: To understand the basic structure of HTML

Program:

Input:



Output:



Explanation:

<!DOCTYPE html>

This tells the browser that the document is written in HTML5.

<html></html>

This is the main element of the HTML document.

Everything on the webpage goes inside this tag.

<head></head>

The head section contains meta information about the page.

Things here are not shown directly on the webpage.

This contains information that is applied to whole webpage like images or colour to the web page.

<title></title>

Sets the title of the webpage.

This text appears on the Browser tab.

<body></body>

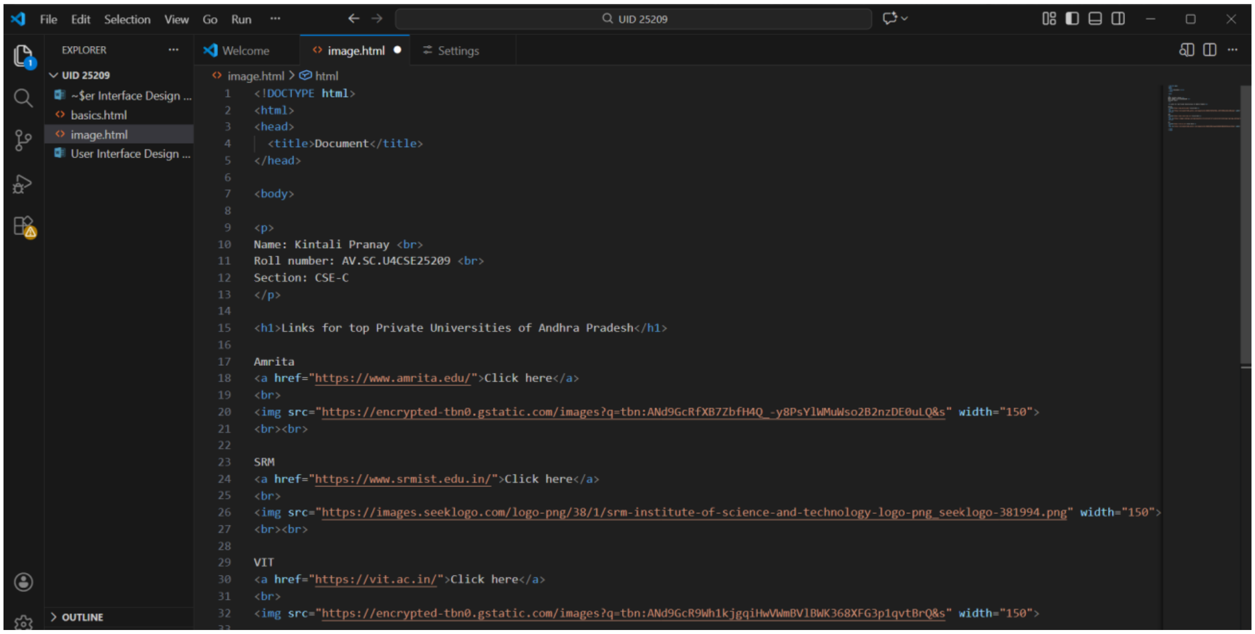
This is the visible part of the webpage.

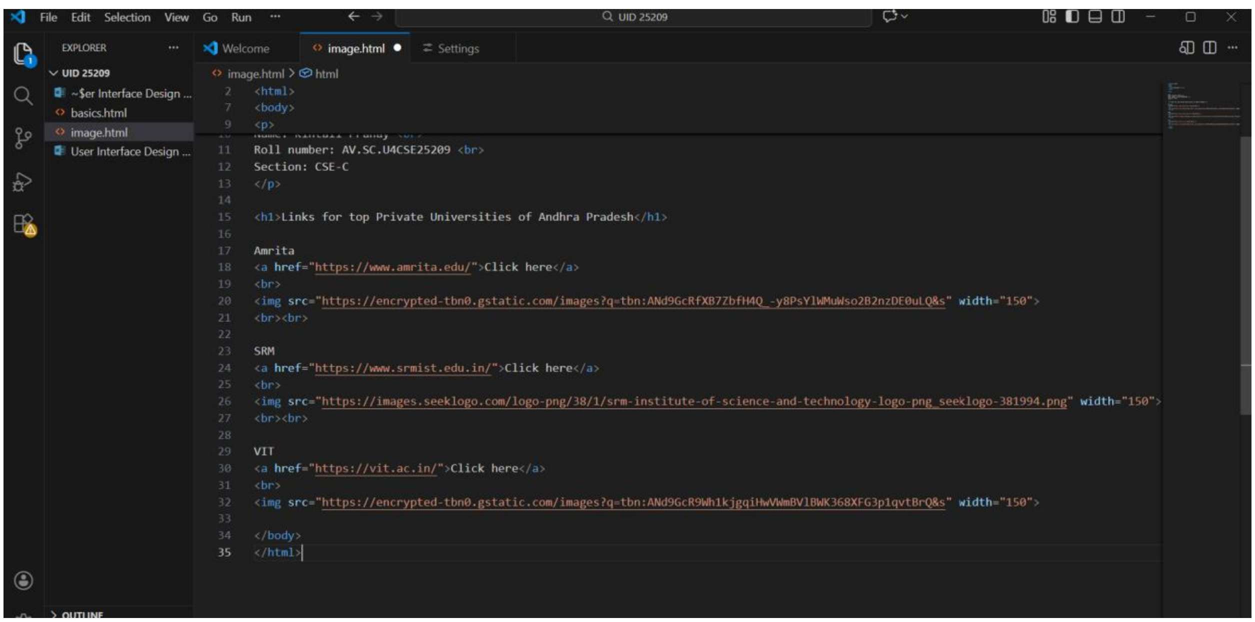
Everything the user sees (text, images, buttons, forms, etc.) goes here.

3.**Aim:** To understand the basic html tags

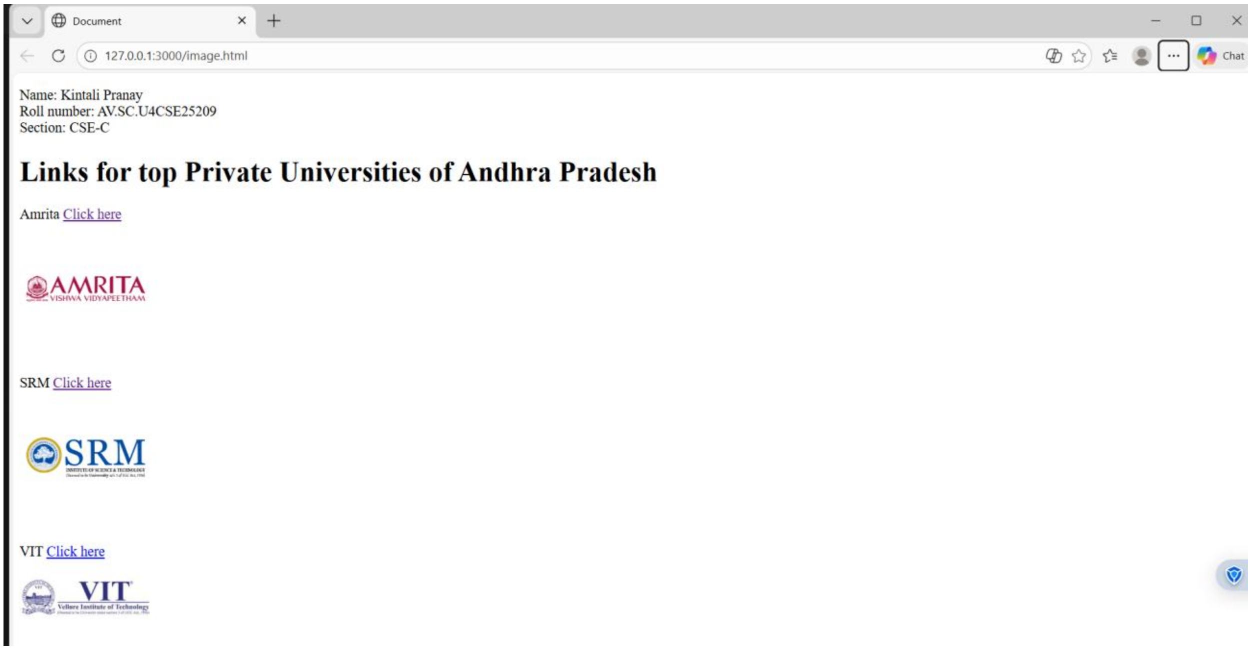
Program:

Input:





Output:



Explanation:

. <head> : the <head> element will contain meta data about the HTML web page
. <title> : the <title> element specifies a title for the HTML Page
. <body> : the <body> element contains all visible content all visible content like text , image , links , tables , etc
. <h1> : the <h1> element defines a large heading
. <p> : the <p> element defines a paragraph
.
 : the
 element defines a line break. It is an empty tag with no end tag
. style : the HTML style attribute is used a add style to an element such as colour , font , size and more
. <img> : the <img> tag is a fundamental HTML element used to embed an image into web page
. <a> :the HTML <a> element stands for “anchor”, is used to define hyper link
. href : the HTML href attribute is used to add link to the html web page

Week	2

2.A)Aim : desing a basic web page with headings ,paragraph,table, by adding styles

Program:

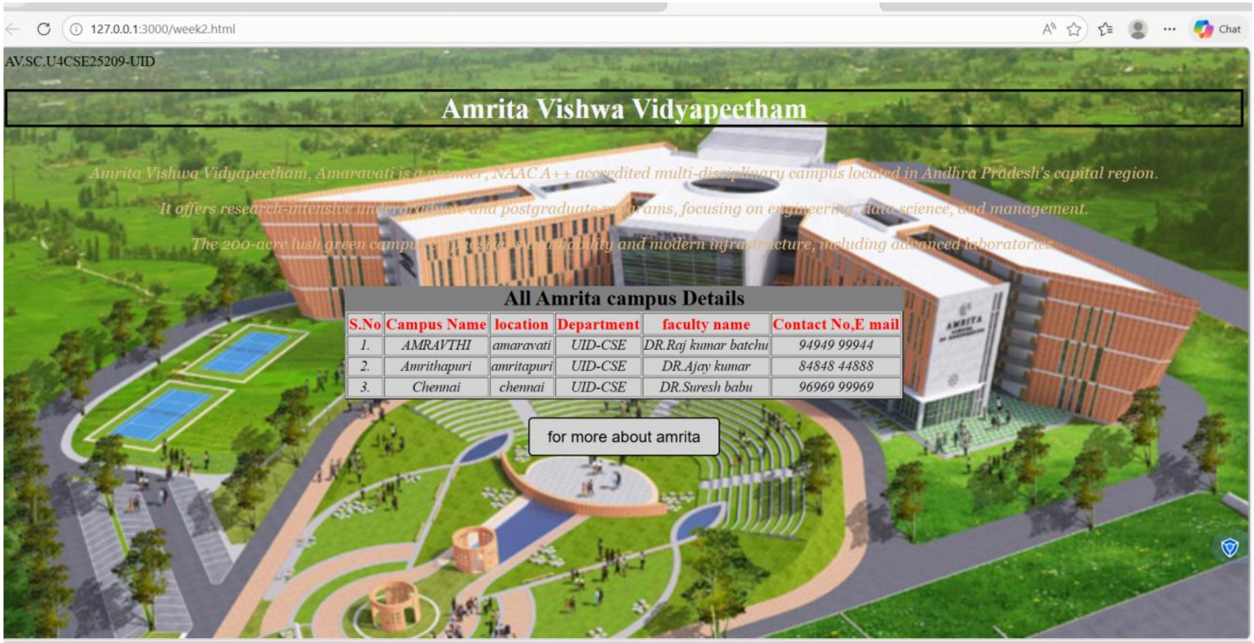
```
AV.SC.U4CSE25209-UID
<!DOCTYPE html>
<html>

<head>
    <title>PRANAY web page</title>
</head>

<body style="background-color: black; background-image: url(https://drupal.mbauniverse.com/sites/default/files/coll

    <h1 style="text-align: center; color: ghostwhite; border: 3px solid black;">
        Amrita Vishwa Vidyapeetham
    </h1>

    <pre style="text-align: center; color: burlywood; font-family: Georgia, 'Times New Roman', Times, serif; font-s

Amrita Vishwa Vidyapeetham, Amaravati is a premier, NAAC A++ accredited multi-disciplinary campus located in Andhra P

It offers research-intensive undergraduate and postgraduate programs, focusing on engineering, data science, and mana

The 200-acre lush green campus emphasizes sustainability and modern infrastructure, including advanced laboratories.

</pre>

    <table border="2" align="center" bgcolor="lightgray" style="text-align: center;">

        <caption style="font-size: x-large; font-weight: bold; color: black; background-color: gray;">
            All Amrita campus Details
        </caption>

        <tr style="color: red; font-style: bold; font-size: large;">
            <th>S.No</th>
            <th>Campus Name</th>
```

```html
            <th>Campus Name</th>
            <th>location</th>
            <th>Department</th>
            <th>faculty name</th>
            <th>Contact No,E mail</th>
        </tr>

        <tr style="color: rgb(10, 10, 10); font-style: italic; font-size: medium;">
            <td>1.</td>
            <td>AMRAVTHI</td>
            <td>amaravati</td>
            <td>UID-CSE</td>
            <td>DR.Raj kumar batchu</td>
            <td>94949 99944</td>
        </tr>

        <tr style="color: rgb(10, 10, 10); font-style: italic; font-size: medium;">
            <td>2.</td>
            <td>Amrithapuri</td>
            <td>amritapuri</td>
            <td>UID-CSE</td>
            <td>DR.Ajay kumar</td>
            <td>84848 44888</td>
        </tr>

        <tr style="color: rgb(10, 10, 10); font-style: italic; font-size: medium;">
            <td>3.</td>
            <td>Chennai</td>
            <td>chennai</td>
```

3 MB • 92%   Ln 7, Col 8   Spaces: 4   UTF-8   CRLF   { } HTML   Port: 3000   Go Live   Prettier

```html
    <table border="2" align="center" bgcolor="lightgray" style="text-align: center;">

        <tr style="color: rgb(10, 10, 10); font-style: italic; font-size: medium;">
            <td>3.</td>
            <td>Chennai</td>
            <td>chennai</td>
            <td>UID-CSE</td>
            <td>DR.Suresh babu</td>
            <td>96969 99969</td>
        </tr>

    </table>

    <button
        url="https://www.amrita.edu/campus/amaravati/"
        style="display: block; margin: auto; margin-top: 20px; padding: 10px 20px; font-size: large; background-color
        onclick="alert('Welcome to Amrita Vishwa Vidyapeetham!')">
        for more about amrita
    </button>

</body>
</html>
```

OUTPUT:



Explanation :

. <DOCTYPE html> : the <!DOCTYPE html> element tells the web browser that this an HTML documen

or this is the latest version of HTML

. <html> : the <html> element is the root element of an HTML web page

. <head> : the <head> element will contain meta data about the HTML web page

. <title> : the <title> element specifies a title for the HTML Page

. <body> : the <body> element contains all visible content all visible content like text , image , links ,

tables , etc

. <table>: the element <table> is used to create a table

. <tr>: the element <tr> creates a new row in the table

. <th>: the element <th> create a new heading inside the table

. <td>: the element <td> fill the columns in the row

. <button>: the element <button> is used for button .

The style attribute is used to add style the elements in the html webpage and url attribute for adding the url or link to the button

Week-3

3.A).Aim : to create a time table using html by using tags like

<table>,<td>,<th>,<tr>,and using style attribute in it

Program :

```html
Pranayk,UID-25209

<table border="2" width="100%" cellpadding="5">

<tr bgcolor="yellow">
    <th colspan="12">Mr Pranays time table </th>
</tr>

<tr>
<th>Time/Day</th>
<th>Slot 1<br>08.55-09.45am</th>
<th>Slot 2<br>09.45-10.35am</th>
<th>10.35-10.45am</th>
<th>Slot 3<br>10.45-11.35am</th>
<th>Slot 4<br>11.35am-12.25pm</th>
<th>Slot 5<br>12.25pm-01.15pm</th>
<th>01.15pm-02.05pm</th>
<th>Slot 6<br>02.10-03.00pm</th>
<th>03.00pm-03.10pm</th>
<th>Slot 7<br>03.10-4:00pm</th>
<th>Slot 8<br>4.00-4.50pm</th>
<tr>
    <td>Monday</td>
    <td colspan="2" bgcolor="orange">23MAT116 Discrete Mathematics Lab</td>
    <td style="background: yellow; writing-mode: sideways-lr;" rowspan="5">Short Break</td>
    <td>23PHY115 Modern Physics</td>
    <td>23CSE111 Object Oriented Programming</td>
    <td>22ADM111 Glimpses of Glorious India</td>
    <td style="background: yellow; writing-mode: sideways-lr;" rowspan="5">Lunch Break</td>
    <td>22ADM111 Glimpses of Glorious India (P)</td>
    <td style="background: yellow; writing-mode: sideways-lr;" rowspan="5">Short Break</td>
    <td colspan="2">Library/Sports</td>
</tr>
<tr>
```

```html
<tr>
<tr>
</tr>
<tr>
    <td>Tuesday</td>
    <td colspan="2" bgcolor="copper">23CSE113 User Interface Design LAB (BYOD)</td>
    <td>23MAT117 Linear Algebra</td>
    <td>23MAT116 Discrete Mathematics</td>
    <td>COUNSELLING</td>
    <td bgcolor="pink">23MEE115 Manufacturing Practice LAB</td>
    <td colspan="2" bgcolor="pink">23MEE115 Manufacturing Practice LAB</td>
</tr>

<tr>
    <td>Wednesday</td>
    <td>23PHY115 Modern Physics</td>
    <td>23CSE113 User Interface Design</td>
    <td colspan="2" bgcolor="Blue">23MAT117 Linear Algebra LAB (BYOD)</td>
    <td>COUNSELLING</td>
    <td>LIBRARY</td>
    <td>22ADM111 Glimpses of Glorious India</td>
    <td colspan="2">Library/Sports</td>
</tr>
<tr>
    <td>Thursday</td>
    <td>23CSE111 Object oriented programming</td>
    <td>23MAT116 Discrete Mathermatics</td>
    <td colspan="2" bgcolor="silver">23CSE111 Object oriented programming Lab(BYOD)</td>
    <td>23MAT117 Linear Algebra</td>
    <td>LIBRARY</td>
    <td colspan="2">Library/Sports</td>
</tr>
```

Ln 26, Col 37 Spaces: 4 UTF-8 CRLF { } HTML Port: 3000

```html
</tr>
<tr>
    <td>Friday</td>
    <td>23CSE11 Object oriented programming</td>
    <td>23MAT116 Discrete Mathematics </td>
    <td>23PHY Modern Physics(T)</td>
    <td>23CSE113 User Interface Design</td>
    <td>23MAT117 Linear Algebra</td>
    <td>LIBRARY</td>
    <td colspan="2">Library/Sports</td>

</tr>





</table>

Theory and lab




<table border="2" width="100%" cellpading="5">
<tr>
    <th>Course Code</th>
    <th>Course name</th>
    <th>     </th>
    <th>L-T-P</th>
```

```html
        <th>Credits</th>
        <th>Faculty Name</th>
        <th>    </th>
        <th>Assiting Faculty</th>
    </tr>

    <tr>
        <td>23MAT116</td>
        <td>Discrete Mathematics</td>
        <td> </td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Srinivas thota</td>
        <td> </td>
        <td>  </td>
    </tr>


    <tr>
        <td>23MAT117</td>
        <td>Linear Algebra</td>
        <td> </td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Sapavat Bixapati</td>
        <td> </td>
        <td>  </td>
    </tr>

    <tr>
        <td>23CSE111</td>
        <td>Object Oriented Programming</td>
```

The td element represents a data cell in a table.

Widely available across major browsers (Baseline since 2015)

MDN Reference

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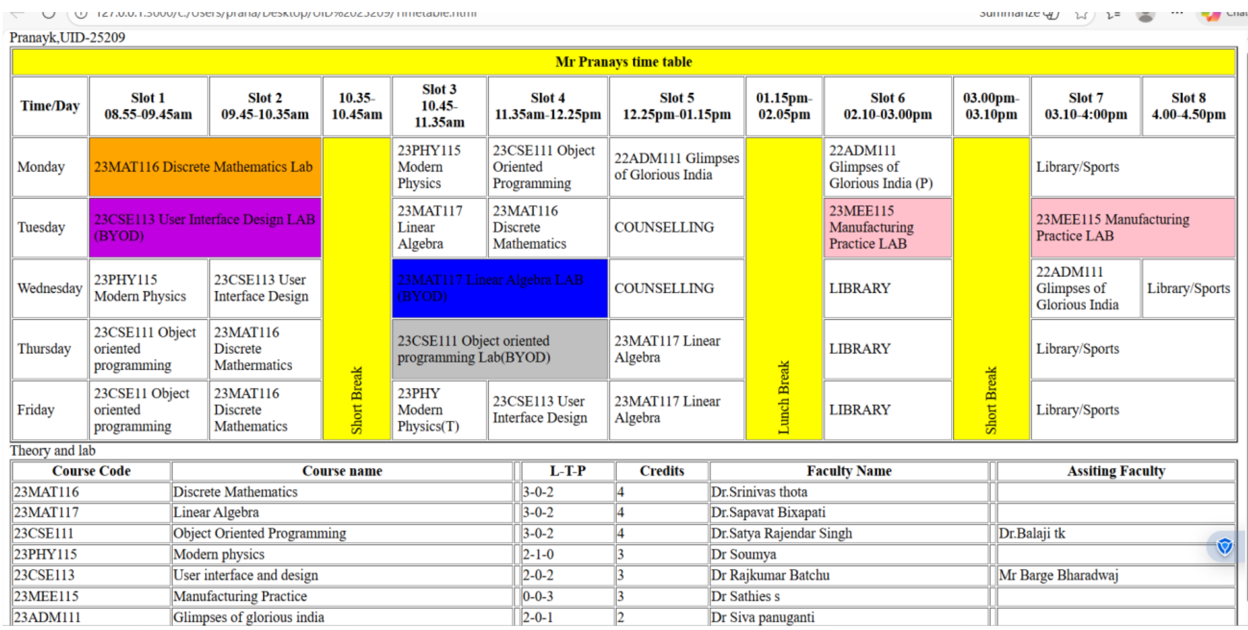
```html
        <td>23CSE111</td>
        <td>Object Oriented Programming</td>
        <td> </td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Satya Rajendar Singh</td>
        <td> </td>
        <td>Dr.Balaji tk </td>
    </tr>

    <tr>
        <td>23PHY115</td>
        <td>Modern physics</td>
        <td> </td>
        <td>2-1-0</td>
        <td>3</td>
        <td>Dr Soumya </td>
        <td> </td>
        <td>   </td>
    </tr>

    <tr>
        <td>23CSE113</td>
        <td>User interface and design</td>
        <td> </td>
        <td>2-0-2</td>
        <td>3</td>
        <td>Dr Rajkumar Batchu </td>
        <td> </td>
        <td> Mr Barge Bharadwaj  </td>
    </tr>
```

```
</tr>

<tr>
    <td>23MEE115</td>
    <td>Manufacturing Practice</td>
    <td> </td>
    <td>0-0-3</td>
    <td>3</td>
    <td>Dr Sathies s</td>
    <td> </td>
    <td>   </td>
</tr>
<tr>
    <td>23ADM111</td>
    <td>Glimpses of glorious india</td>
    <td> </td>
    <td>2-0-1</td>
    <td>2</td>
    <td>Dr Siva panuganti </td>
    <td> </td>
    <td>   </td>
</tr>

<tr>
    <td>  </td>
    <td>Total(15L+1Tut+12P=28hrs)</td>
    <td>  </td>
    <td>15-1-12 </td>
    <td>21</td>
    <td>  </td>
    <td> </td>
```


```
    <td>Glimpses of glorious india</td>
    <td> </td>
    <td>2-0-1</td>
    <td>2</td>
    <td>Dr Siva panuganti </td>
    <td> </td>
    <td>   </td>
</tr>

<tr>
    <td>  </td>
    <td>Total(15L+1Tut+12P=28hrs)</td>
    <td>  </td>
    <td>15-1-12 </td>
    <td>21</td>
    <td>  </td>
    <td> </td>
```


```

</tr>

```

OUTPUT



Pranayk,UID-25209

Mr Pranays time table											
Time/Day	Slot 1 08.55-09.45am	Slot 2 09.45-10.35am	10.35-10.45am	Slot 3 10.45-11.35am	Slot 4 11.35am-12.25pm	Slot 5 12.25pm-01.15pm	01.15pm-02.05pm	Slot 6 02.10-03.00pm	03.00pm-03.10pm	Slot 7 03.10-4:00pm	Slot 8 4.00-4.50pm
Monday	23MAT116 Discrete Mathematics Lab			23PHY115 Modern Physics	23CSE111 Object Oriented Programming	22ADM111 Glimpses of Glorious India		22ADM111 Glimpses of Glorious India (P)		Library/Sports	
Tuesday	23CSE113 User Interface Design LAB (BYOD)			23MAT117 Linear Algebra	23MAT116 Discrete Mathematics	COUNSELLING		23MEE115 Manufacturing Practice LAB		23MEE115 Manufacturing Practice LAB	
Wednesday	23PHY115 Modern Physics	23CSE113 User Interface Design		23MAT117 Linear Algebra LAB (BYOD)		COUNSELLING		LIBRARY		22ADM111 Glimpses of Glorious India	Library/Sports
Thursday	23CSE111 Object oriented programming	23MAT116 Discrete Mathermatics		23CSE111 Object oriented programming Lab(BYOD)		23MAT117 Linear Algebra		LIBRARY		Library/Sports	
Friday	23CSE11 Object oriented programming	23MAT116 Discrete Mathematics	Short Break	23PHY Modern Physics(T)	23CSE113 User Interface Design	23MAT117 Linear Algebra	Lunch Break	LIBRARY	Short Break	Library/Sports	

Theory and lab

Course Code	Course name	L-T-P	Credits	Faculty Name	Assiting Faculty
23MAT116	Discrete Mathematics	3-0-2	4	Dr.Srinivas thota	
23MAT117	Linear Algebra	3-0-2	4	Dr.Sapavat Bixapati	
23CSE111	Object Oriented Programming	3-0-2	4	Dr.Satya Rajendar Singh	Dr.Balaji tk
23PHY115	Modern physics	2-1-0	3	Dr Soumya	
23CSE113	User interface and design	2-0-2	3	Dr Rajkumar Batchu	Mr Barge Bharadwaj
23MEE115	Manufacturing Practice	0-0-3	3	Dr Sathies s	
23ADM111	Glimpses of glorious india	2-0-1	2	Dr Siva panuganti	

Explanation :
we use this tags in addition to the above tags that we used in other programs
<h> : we used <h> element for heading

<p> : we used <p> element for the paragraph

<table>: the element <table> is used to create a table

<tr>: the element <tr> creates a new row in the table

<th>: the element <th> create a new heading inside the table

<td>: the element <td> fill the columns in the row
And we used colspan and rowspan to combine two rows or colums in the table. And we used the style attribute in both the td and tr tags to get the text in vertical way and colour for the blocks and alignment of text.

2.Aim: to create the timetable with the respective syllabus at the end of the page.

In this we use the combined form of the week-1, 3rd one and week-2 1st one

Program :

C: > Users > prana > Desktop > UID 25209 > Timetable.html > ...

```html
Pranayk,UID-25209

<table border="2" width="100%" cellpadding="5">

<tr bgcolor="yellow">
    <th colspan="12">Mr Pranays time table </th>
</tr>

<tr>
<th>Time/Day</th>
<th>Slot 1<br>08.55-09.45am</th>
<th>Slot 2<br>09.45-10.35am</th>
<th>10.35-10.45am</th>
<th>Slot 3<br>10.45-11.35am</th>
<th>Slot 4<br>11.35am-12.25pm</th>
<th>Slot 5<br>12.25pm-01.15pm</th>
<th>01.15pm-02.05pm</th>
<th>Slot 6<br>02.10-03.00pm</th>
<th>03.00pm-03.10pm</th>
<th>Slot 7<br>03.10-4:00pm</th>
<th>Slot 8<br>4.00-4.50pm</th>
<tr>
    <td>Monday</td>
    <td colspan="2" bgcolor="orange">23MAT116 Discrete Mathematics Lab</td>
    <td style="background: yellow; writing-mode: sideways-lr;" rowspan="5">Short Break</td>
    <td>23PHY115 Modern Physics</td>
    <td>23CSE111 Object Oriented Programming</td>
    <td>22ADM111 Glimpses of Glorious India</td>
    <td style="background: yellow; writing-mode: sideways-lr;" rowspan="5">Lunch Break</td>
    <td>22ADM111 Glimpses of Glorious India (P)</td>
    <td style="background: yellow; writing-mode: sideways-lr;" rowspan="5">Short Break</td>
    <td colspan="2">Library/Sports</td>
</tr>
<tr>
```

Ln 1, Col 18 Spaces: 4 UTF-8 CRLF { } HTML Port: 3000

```
<table border="2" width="100%" cellpadding="5">
<tr>
<tr>
</tr>
<tr>
    <td>Tuesday</td>
    <td colspan="2" bgcolor="copper">23CSE113 User Interface Design LAB (BYOD)</td>
    <td>23MAT117 Linear Algebra</td>
    <td>23MAT116 Discrete Mathematics</td>
    <td>COUNSELLING</td>
    <td bgcolor="pink">23MEE115 Manufacturing Practice LAB</td>
    <td colspan="2" bgcolor="pink">23MEE115 Manufacturing Practice LAB</td>
</tr>

<tr>
    <td>Wednesday</td>
    <td>23PHY115 Modern Physics</td>
    <td>23CSE113 User Interface Design</td>
    <td colspan="2" bgcolor="Blue">23MAT117 Linear Algebra LAB (BYOD)</td>
    <td>COUNSELLING</td>
    <td>LIBRARY</td>
    <td>22ADM111 Glimpses of Glorious India</td>
    <td colspan="2">Library/Sports</td>
</tr>
<tr>
    <td>Thursday</td>
    <td>23CSE111 Object oriented programming</td>
    <td>23MAT116 Discrete Mathermatics</td>
    <td colspan="2" bgcolor="silver">23CSE111 Object oriented programming Lab(BYOD)</td>
    <td>23MAT117 Linear Algebra</td>
    <td>LIBRARY</td>
    <td colspan="2">Library/Sports</td>
</tr>
<tr>
```

Ln 26, Col 37 Spaces: 4 UTF-8 CRLF { } HTML Port: 3000

```
<tr>
</tr>
<tr>
    <td>Friday</td>
    <td>23CSE11 Object oriented programming</td>
    <td>23MAT116 Discrete Mathematics </td>
    <td>23PHY Modern Physics(T)</td>
    <td>23CSE113 User Interface Design</td>
    <td>23MAT117 Linear Algebra</td>
    <td>LIBRARY</td>
    <td colspan="2">Library/Sports</td>

</tr>






</table>

Theory and lab




<table border="2" width="100%" cellpading="5">
<tr>
    <th>Course Code</th>
    <th>Course name</th>
    <th>     </th>
    <th>L-T-P</th>
```

```html
        <th>Credits</th>
        <th>Faculty Name</th>
        <th>   </th>
        <th>Assiting Faculty</th>
    </tr>

    <tr>
        <td>23MAT116</td>
        <td>Discrete Mathematics</td>
        <td> </td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Srinivas thota</td>
        <td> </td>
        <td>  </td>
    </tr>


    <tr>
        <td>23MAT117</td>
        <td>Linear Algebra</td>
        <td> </td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Sapavat Bixapati</td>
        <td> </td>
        <td>  </td>
    </tr>

    <tr>
        <td>23CSE111</td>
```

The td element represents a data cell in a table.

Widely available across major browsers (Baseline since 2015)

MDN Reference

Ln 26, Col 37 Spaces: 4 UTF-8 CRLF { } HTML

```html
        <td>23CSE111</td>
        <td>Object Oriented Programming</td>
        <td> </td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.Satya Rajendar Singh</td>
        <td> </td>
        <td>Dr.Balaji tk </td>
    </tr>

    <tr>
        <td>23PHY115</td>
        <td>Modern physics</td>
        <td> </td>
        <td>2-1-0</td>
        <td>3</td>
        <td>Dr Soumya </td>
        <td> </td>
        <td>   </td>
    </tr>

    <tr>
        <td>23CSE113</td>
        <td>User interface and design</td>
        <td> </td>
        <td>2-0-2</td>
        <td>3</td>
        <td>Dr Rajkumar Batchu </td>
        <td> </td>
        <td> Mr Barge Bharadwaj  </td>
    </tr>
```

```html
</tr>

<tr>
    <td>23MEE115</td>
    <td>Manufacturing Practice</td>
    <td> </td>
    <td>0-0-3</td>
    <td>3</td>
    <td>Dr Sathies s</td>
    <td> </td>
    <td>   </td>
</tr>
<tr>
    <td>23ADM111</td>
    <td>Glimpses of glorious india</td>
    <td> </td>
    <td>2-0-1</td>
    <td>2</td>
    <td>Dr Siva panuganti </td>
    <td> </td>
    <td>   </td>
</tr>

<tr>
    <td>  </td>
    <td>Total(15L+1Tut+12P=28hrs)</td>
    <td>  </td>
    <td>15-1-12 </td>
    <td>21</td>
    <td>  </td>
    <td> </td>
```

```html
    <td>Glimpses of glorious india</td>
    <td> </td>
    <td>2-0-1</td>
    <td>2</td>
    <td>Dr Siva panuganti </td>
    <td> </td>
    <td>   </td>
</tr>

<tr>
    <td>  </td>
    <td>Total(15L+1Tut+12P=28hrs)</td>
    <td>  </td>
    <td>15-1-12 </td>
    <td>21</td>
    <td>  </td>
    <td> </td>

</tr>
```

```
<tr>
    <td>21</td>
    <td>  </td>
    <td> </td>

</tr>

</table>

<p style="background-color: yellow;">Discrete Mathematics</p>
<img src="discretemaths.jpeg">
<p style="background-color: yellow;">Linear Algebra</p>
<img src="Linearalgebra .jpeg">
<p style="background-color: yellow;">object oriented programing</p>
<img src="Java.jpeg">
<p style="background-color: yellow;">Modren Physics</p>
<img src="Moderphysics.jpeg">
<p style="background-color: yellow;">UI Design</p>
<img src="UID.jpeg">
<p style="background-color: yellow;">Manufacturing Practice</p>
<img src="Manufacturingprocess.jpeg">
<p style="background-color: yellow;">GGI</p>
<img src="Glimpsesofgloriousindia.jpeg">


</table>
```

Output:

Pranayk,UID-25209

Mr Pranays time table											
Time/Day	**Slot 1 08.55-09.45am**	**Slot 2 09.45-10.35am**	**10.35-10.45am**	**Slot 3 10.45-11.35am**	**Slot 4 11.35am-12.25pm**	**Slot 5 12.25pm-01.15pm**	**01.15pm-02.05pm**	**Slot 6 02.10-03.00pm**	**03.00pm-03.10pm**	**Slot 7 03.10-4:00pm**	**Slot 8 4.00-4.50pm**
Monday	23MAT116 Discrete Mathematics Lab		Short Break	23PHY115 Modern Physics	23CSE111 Object Oriented Programming	22ADM111 Glimpses of Glorious India	Lunch Break	22ADM111 Glimpses of Glorious India (P)	Short Break	Library/Sports	
Tuesday	23CSE113 User Interface Design LAB (BYOD)			23MAT117 Linear Algebra	23MAT116 Discrete Mathematics	COUNSELLING		23MEE115 Manufacturing Practice LAB		23MEE115 Manufacturing Practice LAB	
Wednesday	23PHY115 Modern Physics	23CSE113 User Interface Design		23MAT117 Linear Algebra LAB (BYOD)		COUNSELLING		LIBRARY		22ADM111 Glimpses of Glorious India	Library/Sports
Thursday	23CSE111 Object oriented programming	23MAT116 Discrete Mathermatics		23CSE111 Object oriented programming Lab(BYOD)		23MAT117 Linear Algebra		LIBRARY		Library/Sports	
Friday	23CSE11 Object oriented programming	23MAT116 Discrete Mathematics		23PHY Modern Physics(T)	23CSE113 User Interface Design	23MAT117 Linear Algebra		LIBRARY		Library/Sports	

Theory and lab

Course Code	Course name	L-T-P	Credits	Faculty Name	Assiting Faculty
23MAT116	Discrete Mathematics	3-0-2	4	Dr.Srinivas thota	
23MAT117	Linear Algebra	3-0-2	4	Dr.Sapavat Bixapati	
23CSE111	Object Oriented Programming	3-0-2	4	Dr.Satya Rajendar Singh	Dr.Balaji tk
23PHY115	Modern physics	2-1-0	3	Dr Soumya	
23CSE113	User interface and design	2-0-2	3	Dr Rajkumar Batchu	Mr Barge Bharadwaj
23MEE115	Manufacturing Practice	0-0-3	3	Dr Sathies s	
23ADM111	Glimpses of glorious india	2-0-1	2	Dr Siva panuganti	

Discrete Mathematics

Syllabus

Unit 1
Logic, Mathematical Reasoning and Counting: Logic, Prepositional Equivalence, Predicate and Quantifiers, Theorem Proving. Recursive Definitions, Recursive Algorithms, Basics of Counting, Pigeonhole Principle, Permutation and Combinations.

Unit 2
Relations and Their Properties: Representing Relations, Closure of Relations, Partial Ordering, Equivalence Relations and partitions.
Advanced Counting Techniques and Relations: Recurrence Relations, Solving Recurrence Relations, Generating Functions, Solutions of Homogeneous Recurrence Relations, Divide and Conquer Relations, Inclusion-Exclusion.

Unit 3
Graph Theory: Graphs and Sub graphs, isomorphism, matrices associated with graphs, degrees, walks, connected graphs, shortest path algorithm. Euler and Hamilton Graphs: Euler graphs, Euler's theorem. Fleury's algorithm for Eulerian trails. Hamilton cycles, Chinese-postman problem, approximate solutions of traveling salesman problem. Closest neighbour algorithm.

Linear Algebra

Unit 1
Vector Spaces: Vector spaces - Sub spaces - Linear independence - Basis – Dimension; Inner Product Spaces: Inner products - Orthogonality - Orthogonal basis - Gram Schmidt Process - Change of basis - Orthogonal complements - Projection on subspace - Least Square Principle. QR- Decomposition.

Unit 2
Linear Transformations: Linear transformation - Relation between matrices and linear transformations - Kernel and range of a linear transformation - Change of basis - Nilpotent transformations. Symmetric and Skew Symmetric Matrices, Adjoint and Hermitian Adjoint of a Matrix, Hermitian, Unitary and Normal Transformations, Self-Adjoint and Normal Transformations.

Unit 3
Eigen values and Eigen vectors: Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Diagonalization, Quadratic Forms, Diagonalizing Quadratic Forms, Conic Sections. Similarity of linear transformations - Diagonalization and its applications - Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

object oriented programing

Syllabus
Unit1
Structured to Object Oriented Approach by Examples - Object Oriented languages - Properties of Object Oriented system – UML and Object Oriented Software Development - Use case diagrams and documents as a functional model - Identifying Objects and classes - Representation of Objects and its state by Object Diagram - Simple Class using class diagram – Encapsulation - Data Hiding - Reading and Writing Objects - Class Level and Instance Level Attributes and Methods- JIVE environment for debugging.

Unit2
Aggregation and Composition using Class Diagram – Generalization using Class Diagram – Inheritance - `Constructor and Over Riding – Visibility – Attribute – Parameter – Package - Local and Global - Polymorphism – Overloading - Abstract Classes and Interfaces.

Unit3
Exception Handling - Inner Classes - Wrapper classes – String - and String Builder classes – Number – Math – Random - Array methods - File Streams - Serialization - Generics - Collection framework - Comparator and Comparable - Vector and Arraylist - Iterator and Iterable.

Modren Physics

Syllabus

Unit 1
Origin of quantum theory of radiation: Black body radiation, photo-electric effect, Compton Effect – pair production and annihilation, De-Broglie hypothesis, description of waves and wave packets,group velocities. Evidence for wave nature of particles: Davisson-Germer experiment, Heisenberguncertainty principle.

Unit 2
Atomic structure: Historical Development of atomic structures: Thomson's Model, Rutherford's Model: Scattering formula and its predictions, Atomic spectra - Bohr's Model, Sommerfield's Model, The correspondence principle, nuclear motion, and atomic excitation, Application: Lasers.

Unit 3
Quantum mechanics: Wave function, Probability density, expectation values - Schrodinger equation – time dependent and independent, Linearity and superposition, expectation values, operators, Eigen functions and Eigen values

Unit 4
Application of 1D Schrodinger Wave equation: Free particle, Particle in a box, Finite potential well,Tunnel effect, Harmonic oscillator.

Unit 5
Intro to Quantum computing- Q bits- II Quantum correlations: Bell inequalities and entanglement,Schmidt decomposition, super dense coding, teleportation. Module

UI Design

Syllabus

Unit 1
Introduction to Internet and Web design – Working of Web – Roles of Front-end, Back-end and Full stack development – Web Server – Internet protocols – Web Hosting – HTML – HTML Tags – Create a simple Web Page – Marking up Text – Adding Links – Adding Images – SVG – Table Markup – Embedded Media - Web Based Forms – HTML Forms - CSS for Presentation - Basic Style Sheet - A CSS Style Primer - Using Style Classes - Using Style IDs - Formatting Text - Advanced Typography with CSS3 - Working with Margins, Padding, Alignment, and Floating. Responsive web design, Introduction to version control systems.

Unit 2
CSS Box Model and Positioning - Creating Layouts Using Modern CSS - Backgrounds and Borders - CSS Transformations and Transitions - Animating with CSS and the Canvas - Dynamic Web Pages – Web Scripting - JavaScript programming for Web Applications – Including JavaScript in HTML – HTML5 Form Controls and Validation - Document Object Model

Unit 2
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- DOM manipulation and events - Event Handlers

Unit 3
Overview of UI Frameworks – User Interface Design Concepts in Web Development – Accessibility - Introduction to Electron - Overview of Electron - Setting up an Electron project - Creating a basic desktop application with Electron - Object communication - Introduction to other popular frameworks and libraries (e.g., JavaFX, WPF, GTK).

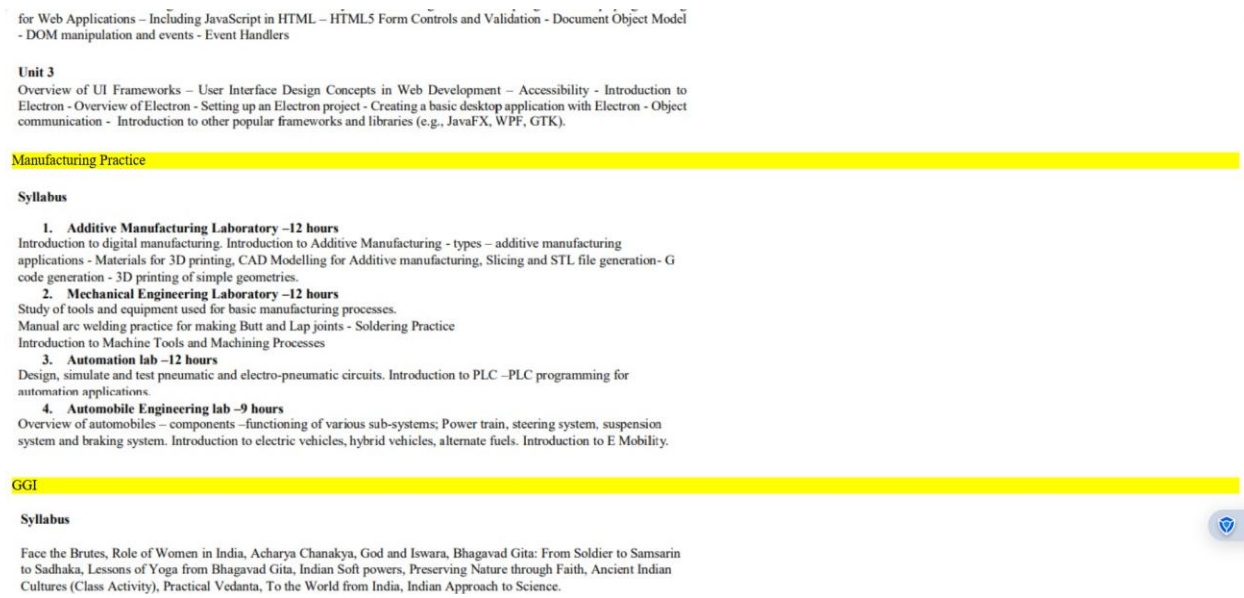
Manufacturing Practice

Syllabus

1. **Additive Manufacturing Laboratory –12 hours**
Introduction to digital manufacturing. Introduction to Additive Manufacturing - types – additive manufacturing applications - Materials for 3D printing, CAD Modelling for Additive manufacturing, Slicing and STL file generation- G code generation - 3D printing of simple geometries.
2. **Mechanical Engineering Laboratory –12 hours**
Study of tools and equipment used for basic manufacturing processes.
Manual arc welding practice for making Butt and Lap joints - Soldering Practice
Introduction to Machine Tools and Machining Processes
3. **Automation lab –12 hours**
Design, simulate and test pneumatic and electro-pneumatic circuits. Introduction to PLC –PLC programming for automation applications.
4. **Automobile Engineering lab –9 hours**
Overview of automobiles – components –functioning of various sub-systems; Power train, steering system, suspension system and braking system. Introduction to electric vehicles, hybrid vehicles, alternate fuels. Introduction to E Mobility.

GGI

Syllabus


for Web Applications – Including JavaScript in HTML – HTML5 Form Controls and Validation - Document Object Model - DOM manipulation and events - Event Handlers

Unit 3

Overview of UI Frameworks – User Interface Design Concepts in Web Development – Accessibility - Introduction to Electron - Overview of Electron - Setting up an Electron project - Creating a basic desktop application with Electron - Object communication - Introduction to other popular frameworks and libraries (e.g., JavaFX, WPF, GTK).

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GGI

Syllabus

Face the Brutes, Role of Women in India, Acharya Chanakya, God and Iswara, Bhagavad Gita: From Soldier to Samsarin to Sadhaka, Lessons of Yoga from Bhagavad Gita, Indian Soft powers, Preserving Nature through Faith, Ancient Indian Cultures (Class Activity), Practical Vedanta, To the World from India, Indian Approach to Science.

Explanation:

In this we use the combined form of the week-1, 3rd one and week-2 1st one . In detail we use the table tags that we used in week-2, 1st program and the image elements from week-1 3rd program by combining both of those tags we get this .

<h> : we used <h> element for heading

<p> : we used <p> element for the paragraph

<table> : we used <table> element for the creation of the table

<tr> : we used <tr> element for creation of row in the table

<td> : we used <td> element for creation of table details

<th> : we used <th> element for creation of table headings

<img> : we used <img> element for adding the images into the web page

And we used colspan and rowspan for combining rows and coloums and scr attribute for adding the image link and we also used style attribute for the colurs and the vertical text and alignment of text.

Week – 4:

4.A)Aim: Design a registration form using only HTML tags (no CSS) with proper labels, table

alignment, background color, and borders.

Program :

CollegeAdmission.html > html > body > center > table > tr > td

```
UID-AV.SC.U4CSE25209
<!DOCTYPE html>
<html>

<head>
    <title>College Admission Registration</title>
</head>

<body bgcolor="#e6e6fa">

<center>
    <h2>College Admission Registration</h2>

    <table border="2" cellpadding="8" cellspacing="0" bgcolor="#ffffff">

        <tr>
            <td><label for="name">Full Name:</label></td>
            <td><input type="text" id="name" name="name" size="40"></td>
        </tr>

        <tr>
            <td><label for="email">Email:</label></td>
            <td><input type="email" id="email" name="email" size="40"></td>
        </tr>

        <tr>
            <td><label for="phone">Phone Number:</label></td>
            <td><input type="tel" id="phone" name="phone" size="40"></td>
        </tr>

        <tr>
            <td>Gender:</td>
            <td>
```

```html
            <td>
                <input type="radio" name="gender" value="male"> Male
                <input type="radio" name="gender" value="female"> Female
                <input type="radio" name="gender" value="other"> Other
            </td>
        </tr>

        <tr>
            <td><label for="course">Course Applying For:</label></td>
            <td>
                <select id="course" name="course">
                    <option>B.Tech</option>
                    <option>B.Sc</option>
                    <option>B.Com</option>
                    <option>BA</option>
                </select>
            </td>
        </tr>

        <tr>
            <td>Hobbies:</td>
            <td>
                <input type="checkbox" name="hobby" value="sports"> Sports
                <input type="checkbox" name="hobby" value="music"> Music
                <input type="checkbox" name="hobby" value="reading"> Reading
            </td>
        </tr>
```

```html
        </tr>

        <tr>
            <td><label for="dob">Date of Birth:</label></td>
            <td><input type="date" id="dob" name="dob"></td>
        </tr>

        <tr>
            <td><label for="file">Upload Certificates:</label></td>
            <td><input type="file" id="file" name="file"></td>
        </tr>

        <tr>
            <td><label for="address">Address:</label></td>
            <td>
                <textarea id="address" name="address" rows="4" cols="35"></textarea>
            </td>
        </tr>

        <tr align="center">
            <td colspan="2">
                <input type="submit" value="Submit">
                <input type="reset" value="Reset">
            </td>
        </tr>

    </table>
</center>
```

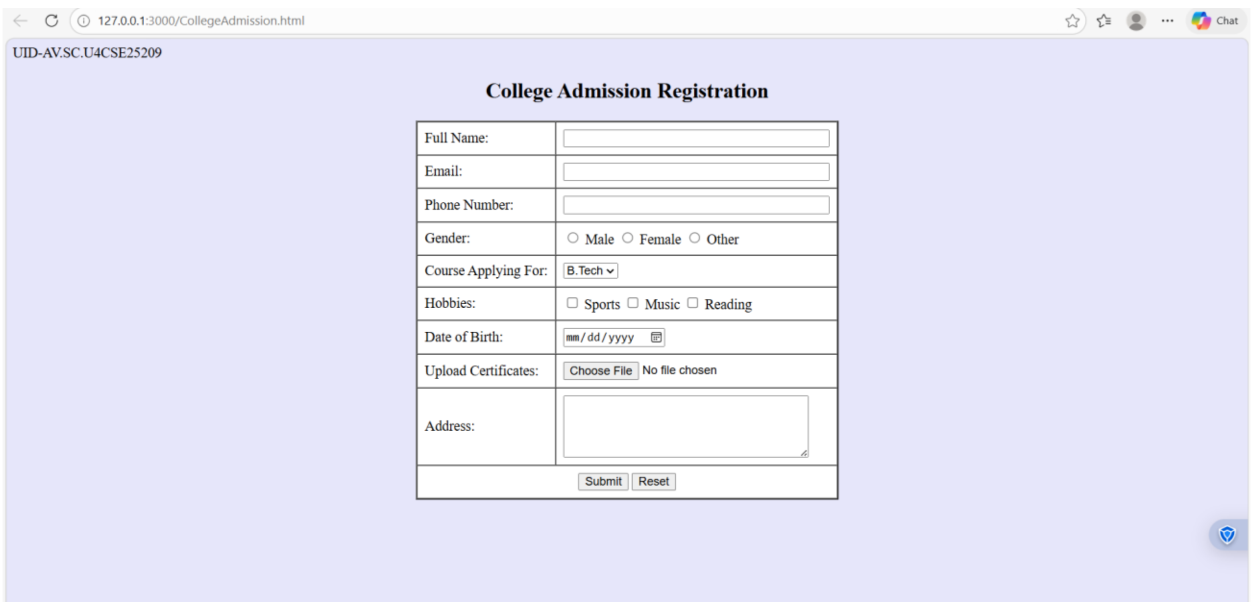
```html
            </tr>

            <tr align="center">
                <td colspan="2">
                    <input type="submit" value="Submit">
                    <input type="reset" value="Reset">
                </td>
            </tr>

        </table>
    </center>

</body>
</html>
```

OUTPUT:



Explanation :

<DOCTYPE html> : the <!DOCTYPE html> element tells the web browser that this an HTML document

or this is the latest version of HTML

<html> : the <html> element is the root element of an HTML web page

<head> : the <head> element will contain meta data about the HTML web page

<title> : the <title> element specifies a title for the HTML Page

<body> : the <body> element contains all visible content all visible content like text , image , links ,

tables , etc

<h> : we used <h> element for heading

<p> : we used <p> element for the paragraph

<table>: the element <table> is used to create a table

<tr>: the element <tr> creates a new row in the table

<th>: the element <th> create a new heading inside the table

<td>: the element <td> fill the columns in the row

<label>:we used <label> element for linking the input to specific id name and we use for attribute for it

<input>:we use <input> element for the input elements in the forms

<select>:we use <select> element for creation of drop box

<option>:we use <option> element for creating the options in the drop box

And we use style attribute for styling and designing the webpage and type for selecting the type of input that is going to In it and for and id tag is used to link the input to label.

4.B)Aim: Two forms displayed side by side using iframes .one is login and another registration

Program :

Main code with Iframes:

```html
UID-AV.SC.U4CSE25209

<!DOCTYPE html>
<html>
<head>

<title>Two Forms Using Iframes</title>

</head>

<body bgcolor="#e6e6cc">

<h2 align="center">Two Forms Displayed Using Iframes</h2>

<table align="center" cellpadding="20">

<tr>

<td>
<iframe src="login.html" width="350" height="300"></iframe>
</td>

<td>
<iframe src="register.html" width="350" height="300"></iframe>
</td>

</tr>

</table>

</body>

</html>
```

Login.html:

```html
UID-AV.SC.U4CSE25209
<!DOCTYPE html>

<html>
<head>

<title>Login Form</title>

</head>
<body>

<h3 align="center">User Login</h3>

<form>

<table border="1" align="center" cellpadding="8">

<tr>
<td><label for="username">Username:</label></td>
<td><input type="text" id="username" name="username"></td>
</tr>

<tr>
<td><label for="password">Password:</label></td>
<td><input type="password" id="password" name="password"></td>
</tr>

<tr align="center">
<td colspan="2">
<input type="submit" value="Login">
</td>
</tr>
</table>
</form>
```

```
</tr>

<tr>
<td><label for="password">Password:</label></td>
<td><input type="password" id="password" name="password"></td>
</tr>

<tr align="center">
<td colspan="2">
<input type="submit" value="Login">
</td>
</tr>
</table>
</form>
</body>
</html>
```

Register.html:

```
UID-AV.SC.U4CSE25209
<!DOCTYPE html>
<html>

<head>
    <title>Registration Form</title>
</head>

<body>

<h3 align="center">User Registration</h3>

<form>

<table border="1" align="center" cellpadding="8">

<tr>
<td><label for="name">Full Name:</label></td>
<td><input type="text" id="name" name="name"></td>
</tr>

<tr>
<td><label for="email">Email:</label></td>
<td><input type="email" id="email" name="email"></td>
</tr>

<tr>
<td><label for="phone">Phone:</label></td>
<td><input type="tel" id="phone" name="phone"></td>
</tr>

<tr>
<td>Password:</td>
<td><input type="password" name="password"></td>
```

```
<tr>
<td>Password:</td>
<td><input type="password" name="password"></td>
</tr>

<tr align="center">
<td colspan="2">
<input type="submit" value="Register">
<input type="reset" value="Reset">
</td>
</tr>

</table>

</form>

</body>
</html>
```

Output :

Explanation :

<DOCTYPE html> : the <!DOCTYPE html> element tells the web browser that this an HTML document

or this is the latest version of HTML

<html> : the <html> element is the root element of an HTML web page

<head> : the <head> element will contain meta data about the HTML web page

<title> : the <title> element specifies a title for the HTML Page

<body> : the <body> element contains all visible content all visible content like text , image , links ,

tables , etc

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<p> : we used <p> element for the paragraph

<table>: the element <table> is used to create a table

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<th>: the element <th> create a new heading inside the table

<td>: the element <td> fill the columns in the row

<label>:we used <label> element for linking the input to specific id name and we use for attribute for it

<input>:we use <input> element for the input elements in the forms

<select>:we use <select> element for creation of drop box

<option>:we use <option> element for creating the options in the drop box

<iframes>:we use <iframes> element for adding anoter html file to main html file

And we use style attribute for styling and designing the webpage and type for selecting the type of input that is going to In it and for and id tag is used to link the input to label and use the iframe for use the table and add two iframes in two boxes

Week-5:

5.A)Aim: creating a card layout using <div> tag consider three cards and a button using html and css

Program :

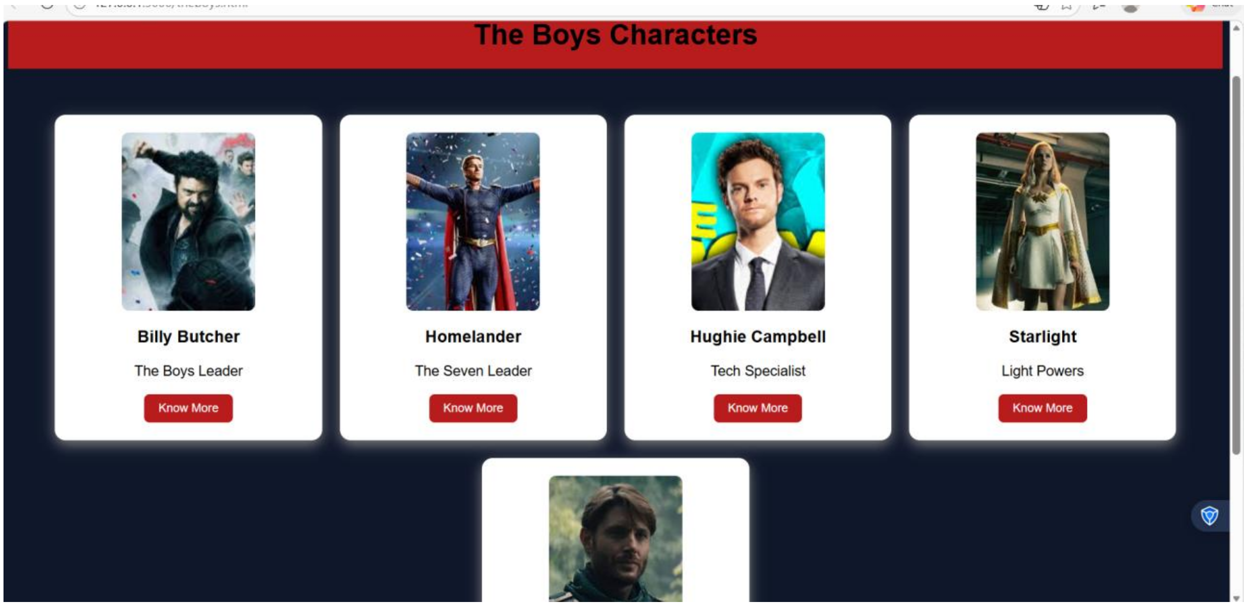
```
UID-AV.SC.U4CSE25209
<!DOCTYPE html>
<html>
<head>
<title>The Boys</title>
<style>
body{
    font-family:Arial;
    background:#0f172a;
    text-align:center;
}
h1{
    background:#b91c1c;
    padding:20px;
}
.container{
    display:flex;
    flex-wrap:wrap;
    justify-content:center;
    gap:20px;
    padding:30px;
}
.card{
    background:white;
    color:black;
    width:260px;
    padding:20px;
    border-radius:12px;
    box-shadow:0 5px 15px gray;
}
.hero-img{
    width:150px;
    height:200px;
```

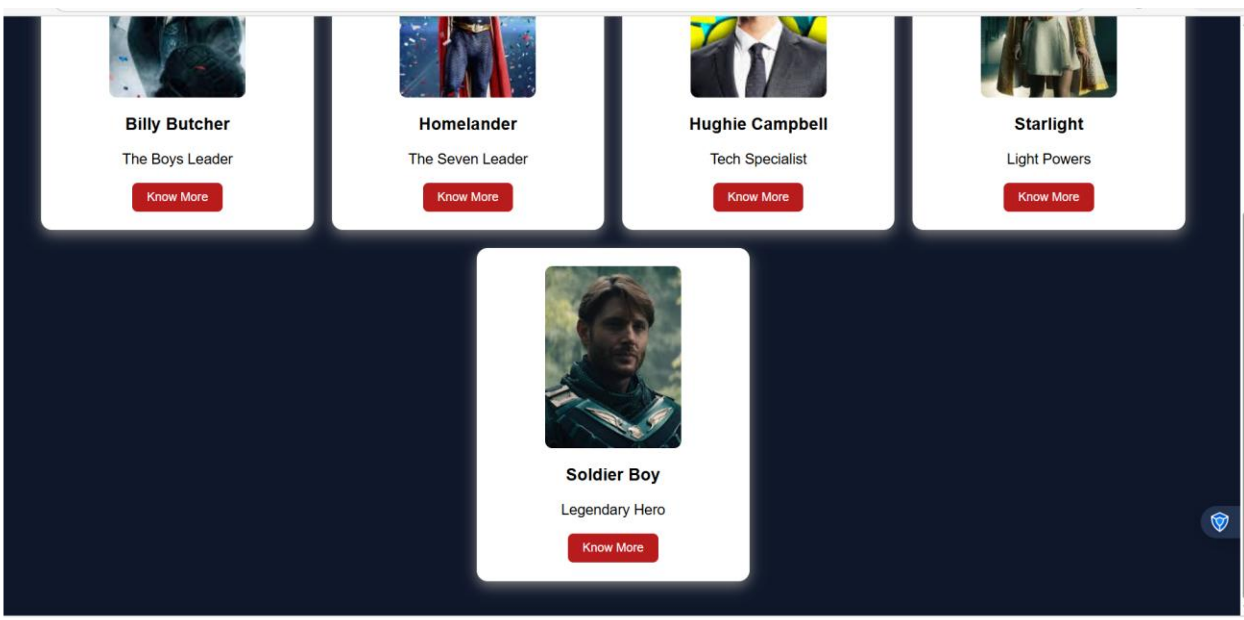
```
    height:200px;
    object-fit:cover;
    border-radius:8px;
}
button{
    padding:8px 16px;
    border:none;
    background:#b91c1c;
    color:white;
    border-radius:6px;
}
button:hover{
    background:black;
}
</style>
</head>
<body>

<h1>The Boys Characters</h1>

<div class="container">

<div class="card">
<img class="hero-img" src="billybutcher.jpg">
<h3>Billy Butcher</h3>
<p>The Boys Leader</p>
<button>Know More</button>
</div>
```

```html

<div class="card">
<img class="hero-img" src="homelander.jpg">
<h3>Homelander</h3>
<p>The Seven Leader</p>
<button>Know More</button>
</div>

<div class="card">
<img class="hero-img" src="hughie.jpg">
<h3>Hughie Campbell</h3>
<p>Tech Specialist</p>
<button>Know More</button>
</div>

<div class="card">
<img class="hero-img" src="starlight.jpg">
<h3>Starlight</h3>
<p>Light Powers</p>
<button>Know More</button>
</div>

<div class="card">
<img class="hero-img" src="soldierboy.jpg">
<h3>Soldier Boy</h3>
<p>Legendary Hero</p>
<button>Know More</button>
</div>

</div>
```

```html
</div>

<div class="card">
<img class="hero-img" src="soldierboy.jpg">
<h3>Soldier Boy</h3>
<p>Legendary Hero</p>
<button>Know More</button>
</div>

</div>

</body>
</html>
```

Output :





Explanation :

<DOCTYPE html> : the <!DOCTYPE html> element tells the web browser that this an HTML document

or this is the latest version of HTML

<html> : the <html> element is the root element of an HTML web page

<head> : the <head> element will contain meta data about the HTML web page

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<body> : the <body> element contains all visible content all visible content like text , image , links ,

tables , etc

<h> : we used <h> element for heading

<p> : we used <p> element for the paragraph

<div>:we use <div> element for encapsulating the the divisions and easy to modify and

<button>: we use <button> element for button

<img>: we use <img> tag for displaying the image

And we internal css for the html page and use element and class selector to apply the css to the div elements and src attribute for image path ,by using the class and internal css we created this webpage

5.B)Aim: Navigation using iframe (mini website) upon clicking each buttons the details about that

button should be display on the same page

program:

vicecity.html:(main)

```html
UID-AV.SC.U4CSE25209
<!DOCTYPE html>
<html>
<head>
<title>GTA Vice City</title>
<style>
body{margin:0;font-family:Arial;background-color: gray;}
.menu{
background: #ff6600;
padding:15px;
text-align:center;
}
.menu a{
color: white;
margin:0 15px;
text-decoration:none;
background: black;
padding:8px 12px;
}
iframe{
width:100%;
height:400px;
border:none;
}
</style>
</head>

<body>

<h1 style="text-align:center;background: #ff6600;color: white;padding:20px;">
GTA Vice City
</h1>

<div class="menu">
```

```html

<body>

<h1 style="text-align:center;background: #ff6600;color: white;padding:20px;">
GTA Vice City
</h1>

<div class="menu">
<a href="features.html" target="frame">Game Features</a>
<a href="character.html" target="frame">Main Character</a>
<a href="missions.html" target="frame">Famous Missions</a>
</div>

<iframe name="frame"></iframe>

</body>
</html>
```

FEATURE.html:

vice city page > features.html > ...

```html

<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center;}
.box{
border:2px solid black;
padding:20px;
display:inline-block;}
h2{color:#ff6600;}
</style>
</head>

<body>

<div class="box">
<h2>Game Features</h2>

<p>Open world gameplay.</p>
<p>Many vehicles and weapons.</p>
<p>80s themed Vice City map.</p>

<img src="vicecity1.jpg" style="width:200px;height:200px;">

</div>

</body>
</html>
```

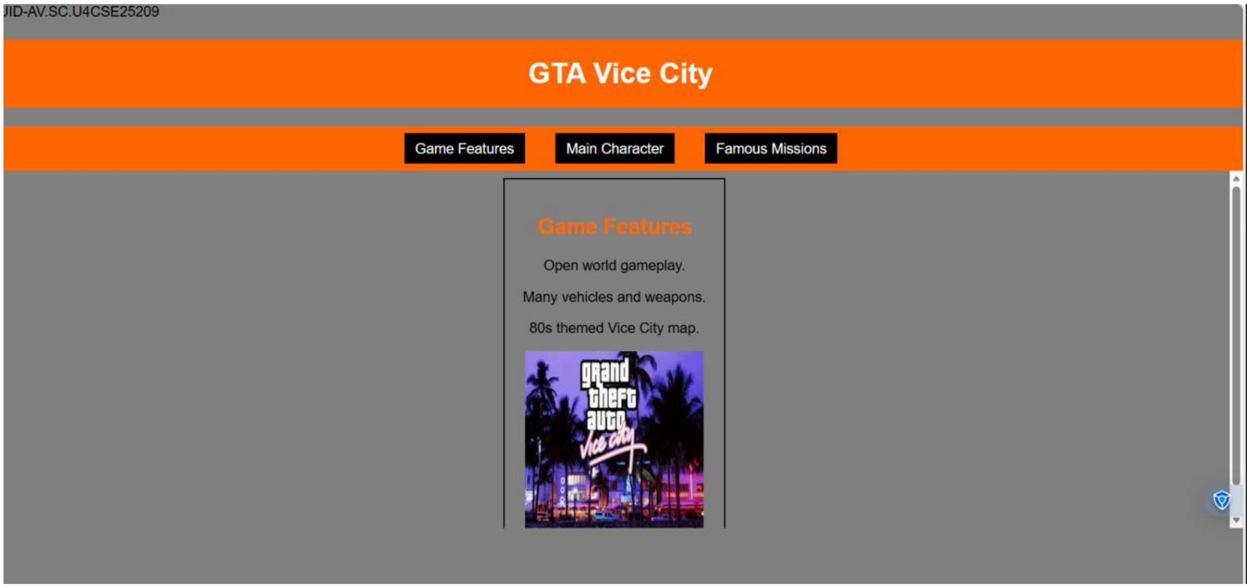
CHARACTER.html:

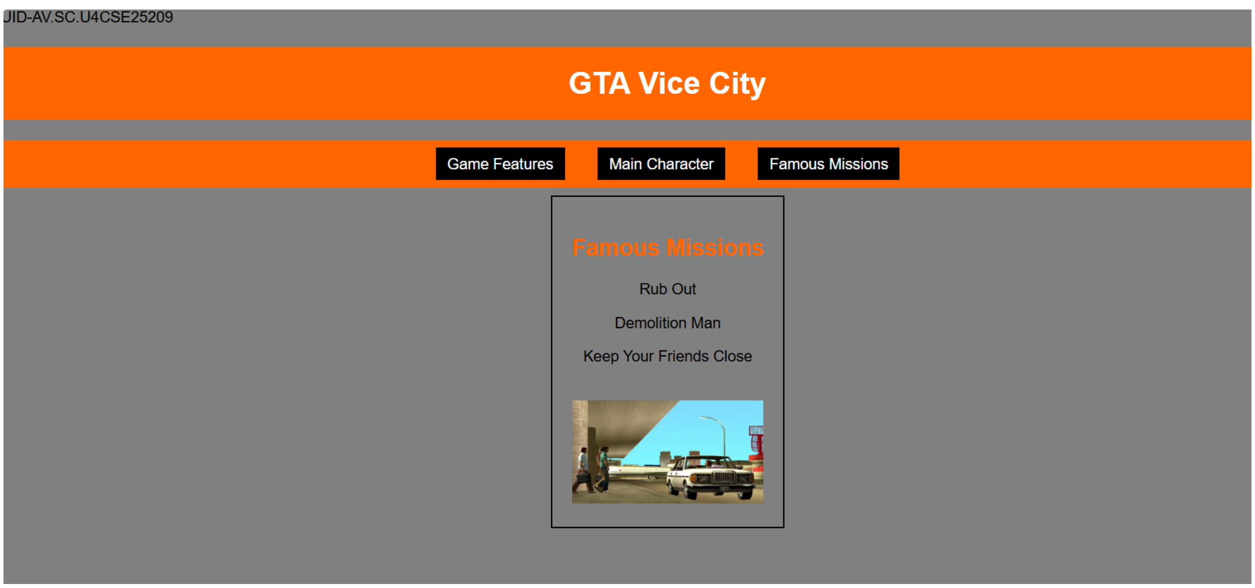
```html
<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center}
.box{
border:2px solid black;
padding:20px;
display:inline-block;}
h2{color:#ff6600;}
</style>
</head>

<body>

<div class="box">

<h2>Main Character</h2>

<p>Name: Tommy Vercetti</p>
<p>Gang: Forelli Family</p>
<p>City: Vice City</p>

<img src="tommy.jpg" style="width:200px;margin-top:20px;">

</div>

</body>
</html>
```

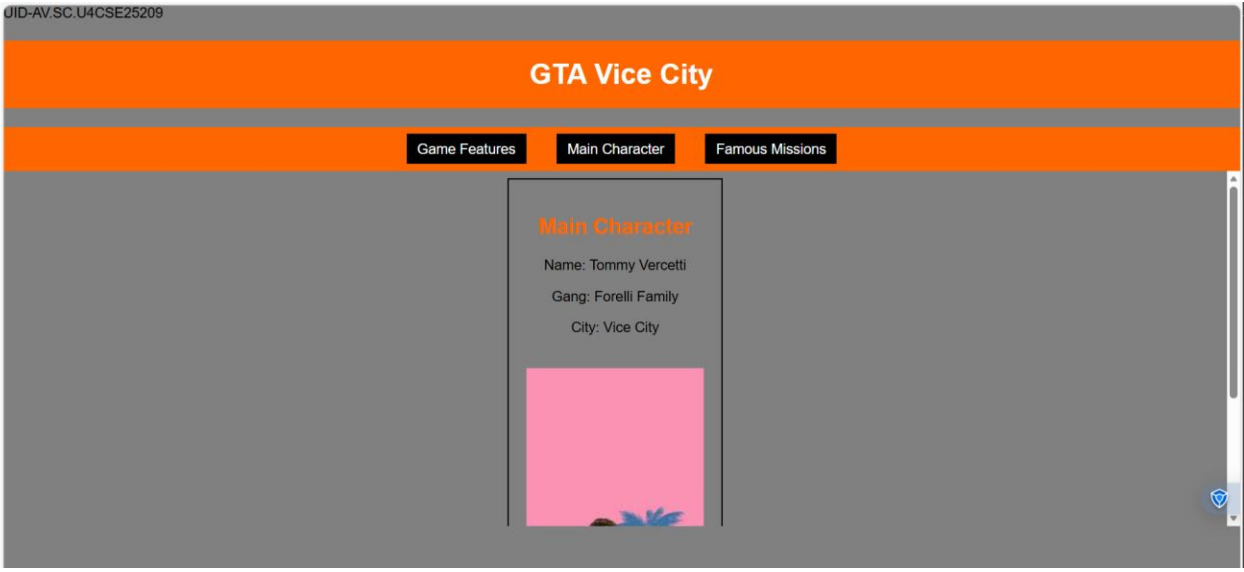
MISSIONS.html:

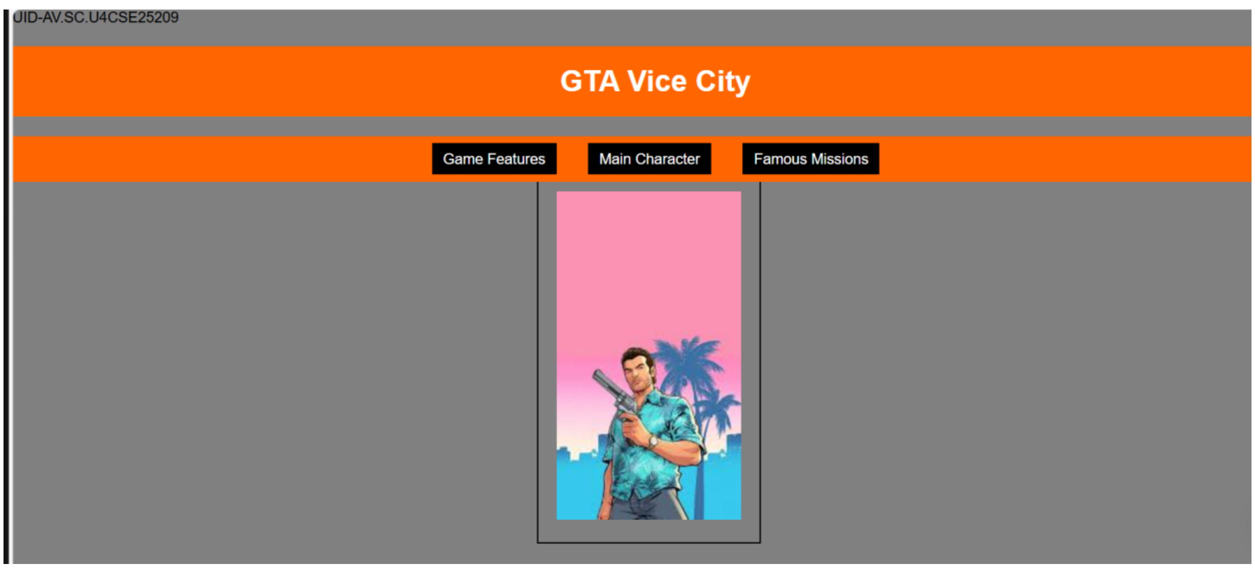
```html

<!DOCTYPE html>
<html>
<head>
<style>
body{font-family:Arial;text-align:center}
.box{
border:2px solid black;
padding:20px;
display:inline-block;}
h2{color:#ff6600;}
</style>
</head>

<body>

<div class="box">

<h2>Famous Missions</h2>

<p>Rub Out</p>
<p>Demolition Man</p>
<p>Keep Your Friends Close</p>

<img src="mission.jpg" style="width:200px;margin-top:20px;">

</div>

</body>
</html>
```

OUTPUT:


UID-AV.SC.U4CSE25209
GTA Vice City
Game Features
Main Character
Famous Missions
Game Features
Open world gameplay.
Many vehicles and weapons.
80s themed Vice City map.
grand theft auto vice city


UID-AV.SC.U4CSE25209
GTA Vice City
Game Features
Main Character
Famous Missions
Famous Missions
Rub Out
Demolition Man
Keep Your Friends Close





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<title> : the <title> element specifies a title for the HTML Page

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<p> : we used <p> element for the paragraph

<div>:we use <div> element for encapsulating the the divisions and easy to modify and

<button>: we use <button> element for button

<img>: we use <img> tag for displaying the image

And we internal css for the html page and use element and class selector to apply the css to the div elements and src attribute for image path ,by using the class and internal css we created this webpage and href attribute is used to link the html pages to the main webpage and use target attribute to link the anchor element to href attribute . and the three webpages are linked to main webpage .and inline attribute for the boder to wrap around it .

5.c)Aim: Design amrita login page by showcampus on home page.upon clicking on each campus

the details about each campus should be display on the same page use html and css to

design a stylish webpage

program:

Main file:

```html
<!DOCTYPE html>
<html>
<head>
<title>Amrita Vishwa Vidhyapeetam</title>
<style>
body{margin:0;font-family:Arial;background:rgb(101, 101, 139);}
header{background:#a00000;color:#fff;text-align:center;




    padding:20px;font-size:36px;}
nav{background:#0b5ea6;text-align:center;padding:10px;}
nav a{color:yellow;margin:12px;text-decoration:none;
    font-weight:bold;padding:5px;}
nav a:hover{background:#fff;color:#0b5ea6;
    border-radius:5px;}
main{display:flex;padding:20px;}
.login{
    width:350px;
    background:#2ecc71;
    padding:15px;
    border-radius:10px;
    color:white;
}
.login input[type="email"],
.login input[type="password"]{
    width:90%;
    padding:6px;
    margin:6px 0;
}
.remember{display:flex;align-items:center;gap:6px;margin:8px 0;}
iframe{
```

```html
iframe{
    flex:1;
    margin-left:20px;
    height:520px;
    border:none;
    background:lightskyblue;
    border-radius:10px;
}
</style>
</head>
<body>
<header>AMRITA VISHWA VIDHYAPEETAM</header>
<nav>
<a href="amaravati.html" target="campus">AMARAVATI</a>
<a href="amritapuri.html" target="campus">AMRITAPURI</a>
<a href="bangalore.html" target="campus">BENGALURU</a>
<a href="chennai.html" target="campus">CHENNAI</a>
<a href="coimbatore.html" target="campus">COIMBATORE</a>
<a href="faridabhad.html" target="campus">FARIDABAD</a>
<a href="kochi.html" target="campus">KOCHI</a>
<a href="mysore.html" target="campus">MYSURU</a>




</nav>
<main>
<div class="login">
<h2>Login</h2>
Email
```

```html
<html>
<body>
<main>
<div class="login">
<h2>Login</h2>
Email
<input type="email">
Password
<input type="password">
<div class="remember">
<input type="checkbox"><label>Save me</label>
</div>
<input type="button" value="Login" style="width:100%;padding:6px;">
<br><br>
<a href="#">Forgot Password?</a><br>
No account? <a href="#">Register</a>
</div>
<iframe name="campus" src="amaravati.html"></iframe>
</main>
</body>
</html>
```

Amaravathi:

amaravati.html > html > body > img

```html
<!DOCTYPE html>
<html>
<body style="font-family:Arial; text-align:center; padding:30px;">
<h1 style="color:#a00000;">Amaravati Campus</h1>
<p>
Amrita Amaravati campus is located near the Krishna River in Andhra Pradesh.
It provides modern classrooms, research labs, and excellent hostel facilities.
</p>
<p>
Popular Courses: CSE, AI, Data Science<br>
Facilities: Library, Smart Classrooms, Hostel<br>
Focus: Innovation and Research
</p>
<img src="amaravati.jpg"
width="450">
</body>
</html>
```

Bangalore:

```html
<!DOCTYPE html>
<html>

<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Bengaluru Campus</h1>
<p>
Located in India's tech hub, Bengaluru campus offers advanced
engineering programs with strong industry connections.
</p>
<p>
Popular Courses: Robotics, CSE, ECE<br>
Facilities: Coding Labs, Innovation Centres<br>
Focus: Industry Collaboration
</p>
<img src="bangalore.jpg"
width="450">
</body>
</html>
```

Amritapuri:

```html
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Amritapuri Campus</h1>
<p>
Amritapuri campus in Kerala is the main campus of Amrita University,
located near the Arabian Sea with a peaceful academic atmosphere.
</p>
<p>
Popular Courses: Engineering, Arts, Sciences<br>
Facilities: Research Labs, Beach Campus, Hostel<br>
Focus: Academic Excellence with Values
</p>
<img src="amritapuri.jpg"
width="450">
</body>
</html>
```

Chennai:

```html
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Chennai Campus</h1>
<p>
Chennai campus provides top-quality computing and management education
with modern infrastructure and placement support.
</p>
<p>
Popular Courses: IT, MBA, Electronics<br>
Facilities: Digital Library, Placement Cell<br>
Focus: Career Development
</p>
<img src="chennai.jpg"
width="450">
</body>
</html>
```

Coimbatore:

```html
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Coimbatore Campus</h1>
<p>
Coimbatore campus is well known for its engineering excellence
and research-oriented education.
</p>
<p>
Popular Courses: Mechanical, Civil, CSE<br>
Facilities: Workshops, Labs, Hostel<br>
Focus: Core Engineering Skills
</p>
<img src="coimbatore.jpg"
width="450">
</body>
</html>
```

Faridabhad:

```html
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Faridabad Campus</h1>
<p>
Faridabad campus focuses on medical and healthcare education
with advanced hospital training facilities.
</p>
<p>
Popular Courses: Medical, Nursing, Allied Health<br>
Facilities: Teaching Hospital, Labs<br>
Focus: Healthcare Research

</p>
<img src="faridabhad.jpg" >
</body>
</html>
```

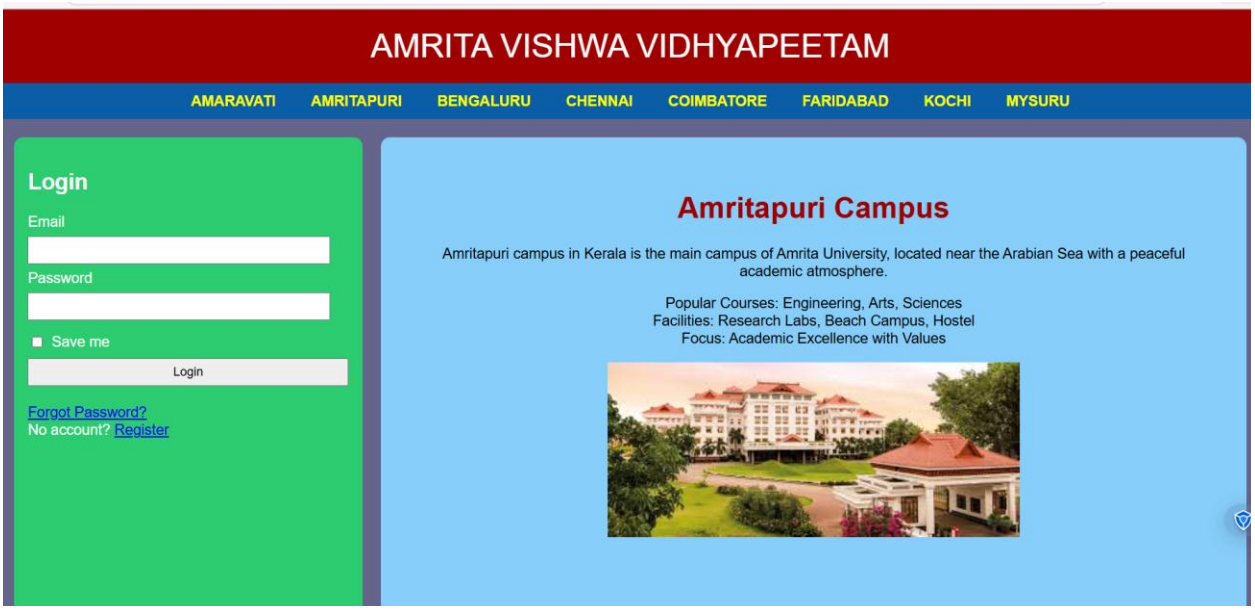
Kochi:

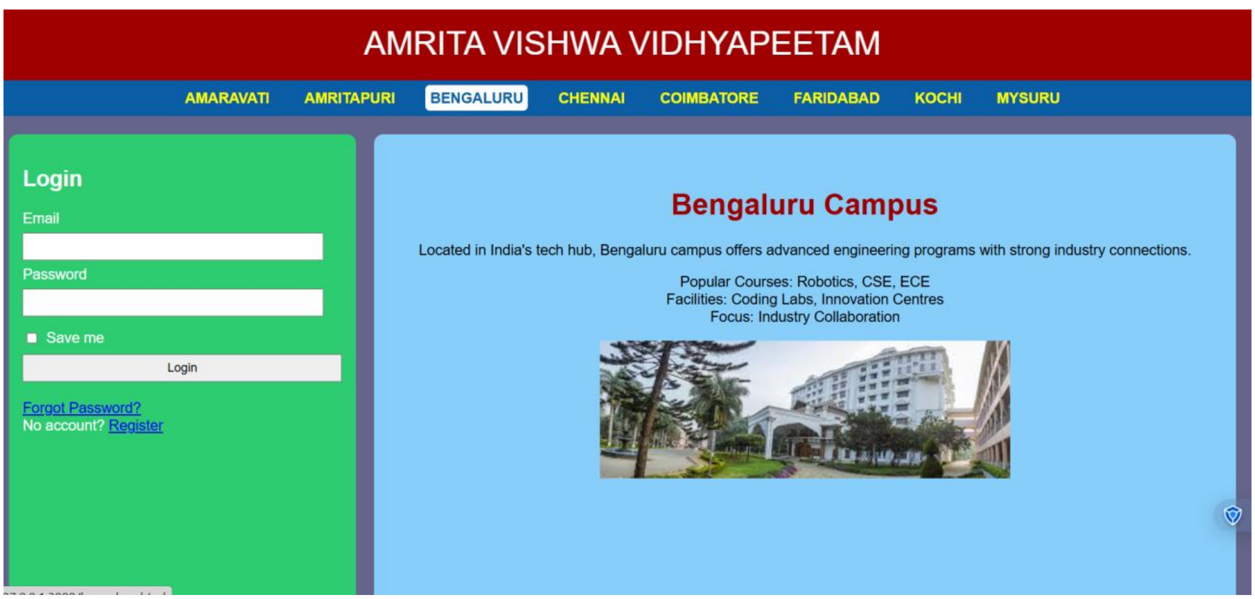
```html
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Kochi Campus</h1>
<p>
Kochi campus offers programs in business, commerce,
and applied sciences with modern facilities.
</p>
<p>
Popular Courses: MBA, Commerce, Finance<br>
Facilities: Seminar Halls, Library<br>
Focus: Industry Training
</p>
<img src="kochi.jpg" >
</body>
</html>
```

Mysore:

```html
<!DOCTYPE html>
<html>
<body style="font-family:Arial;text-align:center;padding:30px;">
<h1 style="color:#a00000;">Mysuru Campus</h1>
<p>
Mysuru campus is known for its peaceful environment,
green campus, and quality education.
</p>
<p>
Popular Courses: Arts, Sciences, Engineering<br>
Facilities: Hostel, Library, Sports Complex<br>
Focus: Calm Learning Atmosphere
</p>
<img src="mysore.jpg" width="450">
</body>
</html>
```

Output:


AMRITA VISHWA VIDHYAPEETAM
AMARAVATI AMRITAPURI BENGALURU CHENNAI COIMBATORE FARIDABAD KOCHI MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Amritapuri Campus
Amritapuri campus in Kerala is the main campus of Amrita University, located near the Arabian Sea with a peaceful academic atmosphere.
Popular Courses: Engineering, Arts, Sciences
Facilities: Research Labs, Beach Campus, Hostel
Focus: Academic Excellence with Values


AMRITA VISHWA VIDHYAPEETAM
AMARAVATI AMRITAPURI BENGALURU CHENNAI COIMBATORE FARIDABAD KOCHI MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Bengaluru Campus
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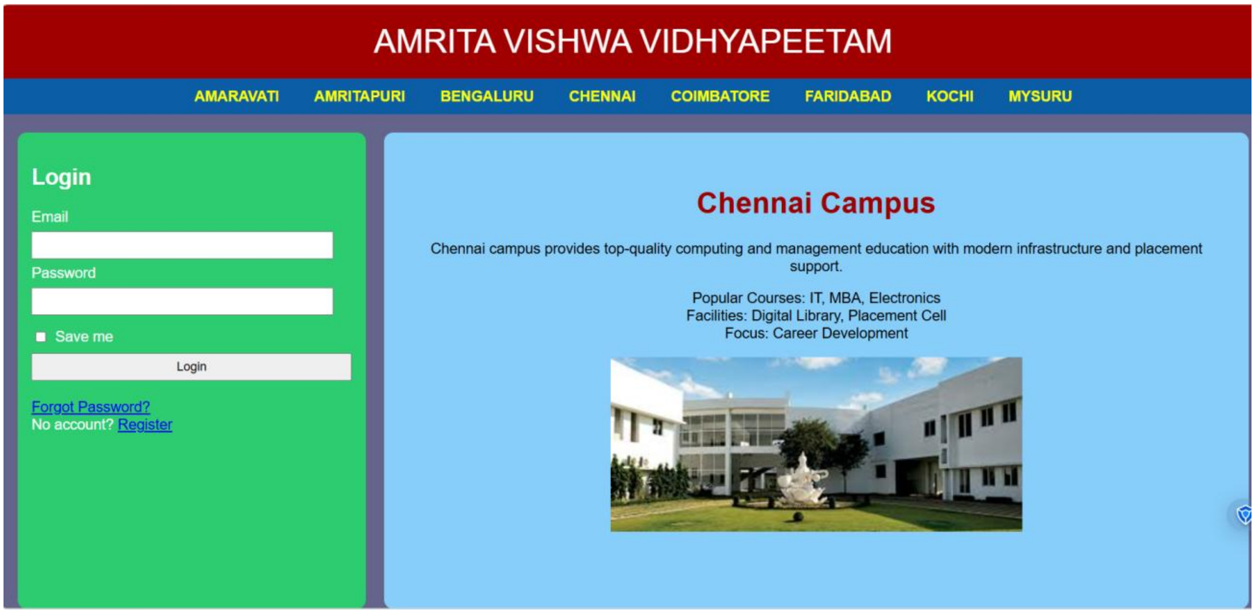
Facilities: Coding Labs, Innovation Centres
Focus: Industry Collaboration


AMRITA VISHWA VIDHYAPEETAM
AMARAVATI AMRITAPURI BENGALURU CHENNAI COIMBATORE FARIDABAD KOCHI MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
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Chennai campus provides top-quality computing and management education with modern infrastructure and placement support.
Popular Courses: IT, MBA, Electronics
Facilities: Digital Library, Placement Cell
Focus: Career Development


AMRITA VISHWA VIDHYAPEETAM
AMARAVATI AMRITAPURI BENGALURU CHENNAI COIMBATORE FARIDABAD KOCHI MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Coimbatore Campus
Coimbatore campus is well known for its engineering excellence and research-oriented education.
Popular Courses: Mechanical, Civil, CSE
Facilities: Workshops, Labs, Hostel
Focus: Core Engineering Skills
27.0.0.1:3000/bangalore.html


AMRITA VISHWA VIDHYAPEETAM
AMARAVATI
AMRITAPURI
BENGALURU
CHENNAI
COIMBATORE
FARIDABAD
KOCHI
MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Faridabad Campus
Faridabad campus focuses on medical and healthcare education with advanced hospital training facilities.
Popular Courses: Medical, Nursing, Allied Health
Facilities: Teaching Hospital, Labs
Focus: Healthcare Research


AMRITA VISHWA VIDHYAPEETAM
AMARAVATI
AMRITAPURI
BENGALURU
CHENNAI
COIMBATORE
FARIDABAD
KOCHI
MYSURU
Login
Email
Password
Save me
Login
Forgot Password?
No account? Register
Kochi Campus
Kochi campus offers programs in business, commerce, and applied sciences with modern facilities.
Popular Courses: MBA, Commerce, Finance
Facilities: Seminar Halls, Library
Focus: Industry Training

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<body> : the <body> element contains all visible content all visible content like text , image , links ,

tables , etc

<h> : we used <h> element for heading

<p> : we used <p> element for the paragraph

<div>:we use <div> element for encapsulating the the divisions and easy to modify and

<button>: we use <button> element for button

<img>: we use <img> tag for displaying the image

We used different types of input attributes for different inputs

And we internal css for the html page and use element and class selector to apply the css to the div elements and src attribute for image path ,by using the class and internal css we created this webpage and href attribute is used to link the html pages to the main webpage and use target attribute to link the anchor element to href attribute . and the three webpages are linked to main webpage .and inline attribute for the boder to

WEEK-6

6.A. <u>Aim</u>: Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books**.**

Code:-

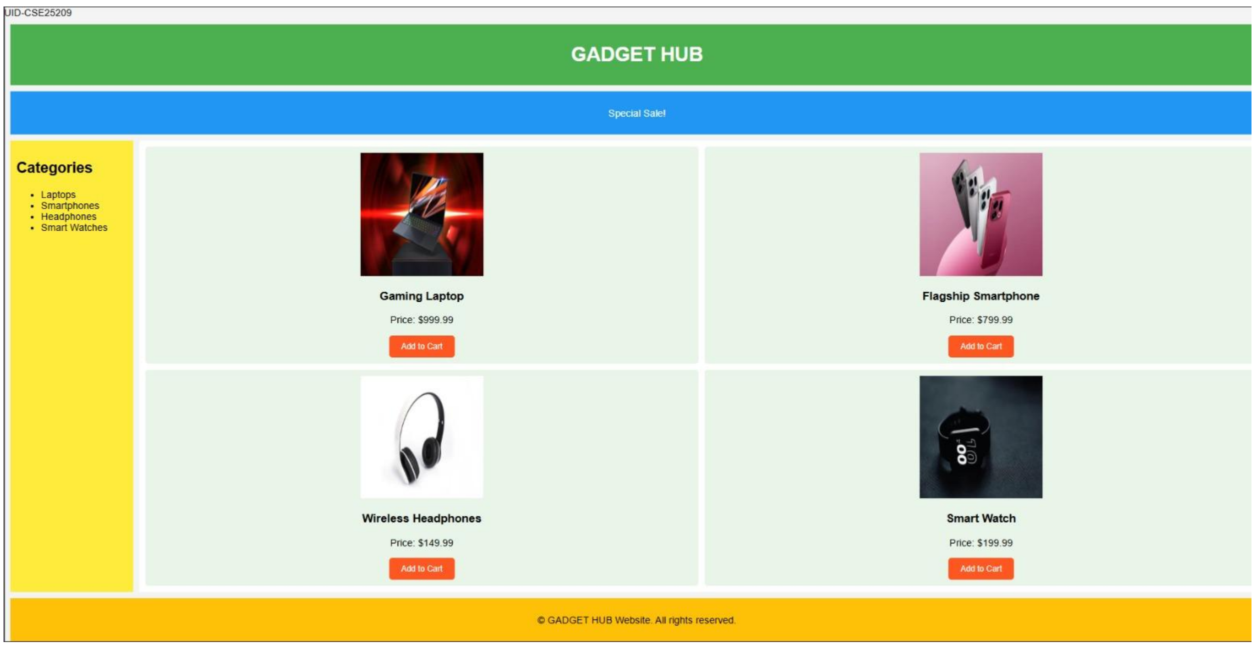
```
<!DOCTYPE html>
UID-CSE25209
<html>
<head>
  <title>Gadget Hub</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #f4f4f4;
    }
    .container {
      display: grid;
      grid-template-rows: auto auto 1fr auto;
      grid-template-columns: 200px 1fr;
      grid-template-areas:
        "header header"
        "sale sale"
        "sidebar main"
        "footer footer";
      gap: 10px;
      padding: 10px;
    }
    .header {
      grid-area: header;
      background-color: #4CAF50;
      padding: 10px;
      text-align: center;
      color: white;
    }
    .sale {
      grid-area: sale;
      background-color: #2196F3;
```

```
        padding: 10px;
        text-align: center;
        color: white;
    }
    .sidebar {
        grid-area: sidebar;
        background-color: #FFEB3B;
        padding: 10px;
    }
    .main {
        grid-area: main;
        background-color: white;
        padding: 10px;
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 10px;
    }
    .footer {
        grid-area: footer;
        background-color: #FFC107;
        padding: 10px;
        text-align: center;
    }
    .product {
        background-color: #e8f5e8;
        padding: 10px;
        border-radius: 5px;
        text-align: center;
    }
    .product-image {
        width: 200px;
        height: 200px;
    }
    .button {
        background-color: #FF5722;
        color: white;
        border: none;
        padding: 10px 20px;
        border-radius: 5px;
    }
    .button:hover {
        background-color: #E64A19;
    }
  </style>
</head>
```

```html
<body>
  <div class="container">
    <div class="header">
      <h1>GADGET HUB</h1>
    </div>
    <div class="sale">
      <p>Special Sale!</p>
    </div>
    <div class="sidebar">
      <h2>Categories</h2>
      <ul>
        <li>Laptops</li>
        <li>Smartphones</li>
        <li>Headphones</li>
        <li>Smart Watches</li>
      </ul>
    </div>
    <div class="main">
      <div class="product">
        <img class="product-image" src="game.webp" alt="Gaming Laptop">
        <h3>Gaming Laptop</h3>
        <p>Price: $999.99</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product">
        <img class="product-image" src="phone.webp" alt="Smartphone">
        <h3>Flagship Smartphone</h3>
        <p>Price: $799.99</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product">
        <img class="product-image" src="wireless.webp" alt="Headphones">
        <h3>Wireless Headphones</h3>
        <p>Price: $149.99</p>
        <button class="button">Add to Cart</button>
      </div>
      <div class="product">
        <img class="product-image" src="watch.webp" alt="Smart Watch">
        <h3>Smart Watch</h3>
        <p>Price: $199.99</p>
        <button class="button">Add to Cart</button>
      </div>
    </div>
    <div class="footer">
      <p>&copy; GADGET HUB Website. All rights reserved.</p>
```

```
      </div>
    </div>
</body>
</html>
```

Output:-



Explanation of the tags:

- display: grid: Enables CSS Grid layout system
- grid-template-areas: Defines 5 regions (header/sale/sidebar/main/footer)
- grid-area: header: Assigns element to header grid area
- grid-template-columns: 200px 1fr: Fixed 200px sidebar + flexible main
- gap: 10px: Spacing between all grid items
- padding: 10px: Inner spacing for all sections
- background-color: #4CAF50: Green header background

- text-align: center: Centers content in all sections
- repeat(2, 1fr): Creates 2-column product grid in main
- border-radius: 5px: Rounded corners on products/buttons
- .product-image { width: 200px; height: 200px }: Fixed-size product images
- background-color: #FF5722: Orange Add to Cart buttons
- :hover: Button darkens on mouseover (#E64A19)
- border: none: Removes button borders
- 1fr: Flexible units fill available space

6.B. Aim: Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.

Code:-

Main.html

```
<!DOCTYPE html>
<html>
<head>
  <title>GADGET HUB</title>
  <style>
    body {
      margin: 0;
      font-family: Arial;
      color: #333;
      background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
    }
    .container {
      display: grid;
      grid-template:
        "header header" auto
        "sale sale" auto
        "sidebar main" 1fr
        "footer footer" auto / 220px 1fr;
      gap: 15px;
      padding: 15px;
    }
    .header, .sale, .sidebar, .main, .footer {
      background: rgba(255,255,255,.95);
      border-radius: 12px;
      padding: 15px;
      box-shadow: 0 4px 20px rgba(0,0,0,0.1);
    }
    .header {
      grid-area: header;
      text-align: center;
      font-size: 28px;
      letter-spacing: 3px;
      background: linear-gradient(45deg, #4CAF50, #45a049);
      color: white;
    }
    .sale {
      grid-area: sale;
      overflow: hidden;
      white-space: nowrap;
      height: 40px;
      position: relative;
      background: linear-gradient(45deg, #2196F3, #21CBF3);
```

```
    color: white;
    font-weight: bold;
  }
  .sale p {
    position: absolute;
    font-size: 20px;
    animation: move 10s linear infinite;
  }
  @keyframes move {
    from { left: -100%; }
    to { left: 100%; }
  }
  .sidebar {
    grid-area: sidebar;
    background: linear-gradient(135deg, #FFEB3B, #FF9800);
  }
  .sidebar ul { list-style: none; padding: 0; }
  .sidebar li { margin: 15px 0; }
  .sidebar a {
    color: #333;
    text-decoration: none;
    font-size: 18px;
    transition: .3s;
    display: block;
  }
  .sidebar a:hover {
    color: #2196F3;
    padding-left: 8px;
    background: rgba(255,255,255,0.5);
    border-radius: 6px;
  }
  .main {
    grid-area: main;
    background: white;
  }
  iframe {
    width: 100%;
    height: 520px;
    border: 0;
    border-radius: 10px;
    box-shadow: 0 4px 15px rgba(0,0,0,0.1);
  }
  .footer {
    grid-area: footer;
    text-align: center;
```

```
        font-size: 14px;
        background: linear-gradient(45deg, #FFC107, #FF9800);
        color: #333;
      }
  </style>
</head>
<body>
  <div class="container">
    <div class="header">GADGET HUB</div>
    <div class="sale">
      <p>🔥 SPECIAL SALE – UP TO 50% OFF ON GADGETS 🔥</p>
    </div>
    <div class="sidebar">
      <h2>Categories</h2>
      <ul>
        <li><a href="laptops.html" target="contentFrame">LAPTOPS</a></li>
        <li><a href="smartphones.html" target="contentFrame">SMARTPHONES</a></li>
        <li><a href="headphones.html" target="contentFrame">HEADPHONES</a></li>
        <li><a href="smartwatches.html" target="contentFrame">SMART WATCHES</a></li>
      </ul>
    </div>
    <div class="main">
      <iframe name="contentFrame"></iframe>
    </div>
    <div class="footer">© GADGET HUB Website</div>
  </div>
</body>
</html>
```

laptops.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Laptops</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 20px;
      background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
    }
    .container {
```

```
            display: grid;
            grid-template-columns: 1fr 1fr;
            gap: 20px;
        }
        .card {
            background: white;
            padding: 20px;
            border-radius: 15px;
            box-shadow: 0 8px 25px rgba(0,0,0,0.1);
            text-align: center;
        }
        .card-img {
            width: 200px;
            height: 200px;
            border-radius: 12px;
            margin-bottom: 15px;
        }
        .price {
            font-size: 20px;
            color: #4CAF50;
            font-weight: bold;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="card">
            <img class="card-img" src="game.webp" alt="Gaming Laptop">
            <h2>GAMING PRO</h2>
            <p>RTX 4080 gaming laptop</p>
            <p class="price">$999.99</p>
        </div>
        <div class="card">
            <img class="card-img" src="ultrabook.webp" alt="Ultrabook">
            <h2>ULTRABOOK</h2>
            <p>Lightweight productivity laptop</p>
            <p class="price">$799.99</p>
        </div>
        <div class="card">
            <img class="card-img" src="creator.webp" alt="Creator Laptop">
            <h2>CREATOR STUDIO</h2>
            <p>4K video editing laptop</p>
            <p class="price">$1299.99</p>
        </div>
        <div class="card">
```

```
        <img class="card-img" src="business.webp" alt="Business Laptop">
        <h2>BUSINESS ELITE</h2>
        <p>Enterprise grade laptop</p>
        <p class="price">$899.99</p>
      </div>
    </div>
</body>
</html>
```

smartphones.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Smartphones</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 20px;
        background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
      }
      .container {
        display: grid;
        grid-template-columns: 1fr 1fr;
        gap: 20px;
      }
      .card {
        background: white;
        padding: 20px;
        border-radius: 15px;
        box-shadow: 0 8px 25px rgba(0,0,0,0.1);
        text-align: center;
      }
      .card-img {
        width: 200px;
        height: 200px;
        border-radius: 12px;
        margin-bottom: 15px;
      }
      .price {
        font-size: 20px;
        color: #4CAF50;
```

```
            font-weight: bold;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="card">
            <img class="card-img" src="ultra.webp" alt="Flagship Phone">
            <h2>ULTRA PRO</h2>
            <p>Latest flagship smartphone</p>
            <p class="price">$799.99</p>
        </div>
        <div class="card">
            <img class="card-img" src="phone.webp" alt="Midrange Phone">
            <h2>MIDRANGE X</h2>
            <p>Best value smartphone</p>
            <p class="price">$499.99</p>
        </div>
        <div class="card">
            <img class="card-img" src="gaming.webp" alt="Gaming Phone">
            <h2>GAMER PHONE</h2>
            <p>120Hz gaming smartphone</p>
            <p class="price">$649.99</p>
        </div>
        <div class="card">
            <img class="card-img" src="fold.webp" alt="Foldable Phone">
            <h2>FOLD PRO</h2>
            <p>Foldable display smartphone</p>
            <p class="price">$999.99</p>
        </div>
    </div>
</body>
</html>
```

smartwatches.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Smart Watches</title>
    <style>
```

```
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 20px;
            background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
        }
        .container {
            display: grid;
            grid-template-columns: 1fr 1fr;
            gap: 20px;
        }
        .card {
            background: white;
            padding: 20px;
            border-radius: 15px;
            box-shadow: 0 8px 25px rgba(0,0,0,0.1);
            text-align: center;
        }
        .card-img {
            width: 200px;
            height: 200px;
            border-radius: 12px;
            margin-bottom: 15px;
        }
        .price {
            font-size: 20px;
            color: #4CAF50;
            font-weight: bold;
        }
    </style>
</head>
<body>
    <div class="container">
        <div class="card">
            <img class="card-img" src="s9.webp" alt="Premium Smartwatch">
            <h2>PREMIUM S9</h2>
            <p>Advanced health tracking watch</p>
            <p class="price">$199.99</p>
        </div>
        <div class="card">
            <img class="card-img" src="fit.webp" alt="Fitness Tracker">
            <h2>FIT TRACKER</h2>
            <p>24/7 fitness monitoring</p>
            <p class="price">$129.99</p>
        </div>
```

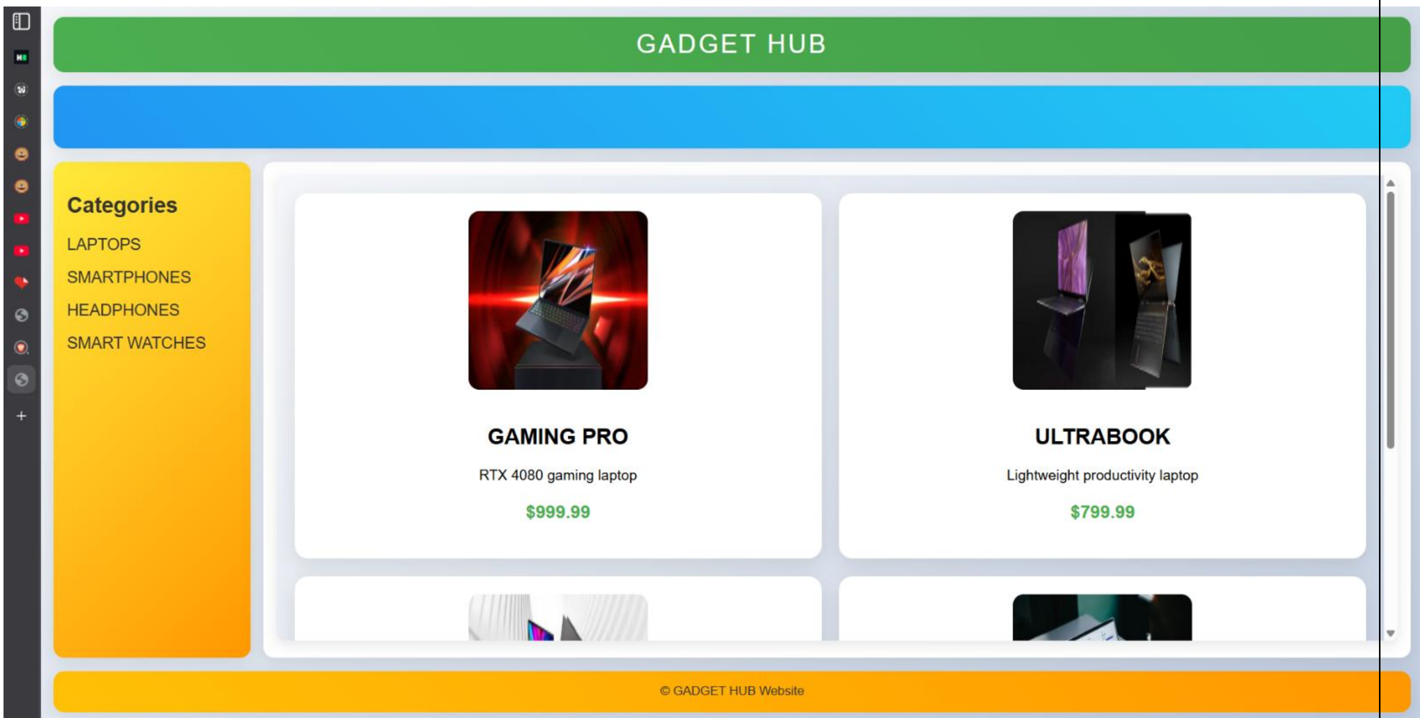
```
        <div class="card">
            <img class="card-img" src="sports.webp" alt="Sports Watch">
            <h2>SPORTS PRO</h2>
            <p>GPS running and cycling watch</p>
            <p class="price">$249.99</p>
        </div>
        <div class="card">
            <img class="card-img" src="watch.webp" alt="Classic Smartwatch">
            <h2>CLASSIC SERIES</h2>
            <p>Traditional smartwatch design</p>
            <p class="price">$179.99</p>
        </div>
    </div>
</body>
</html>
```

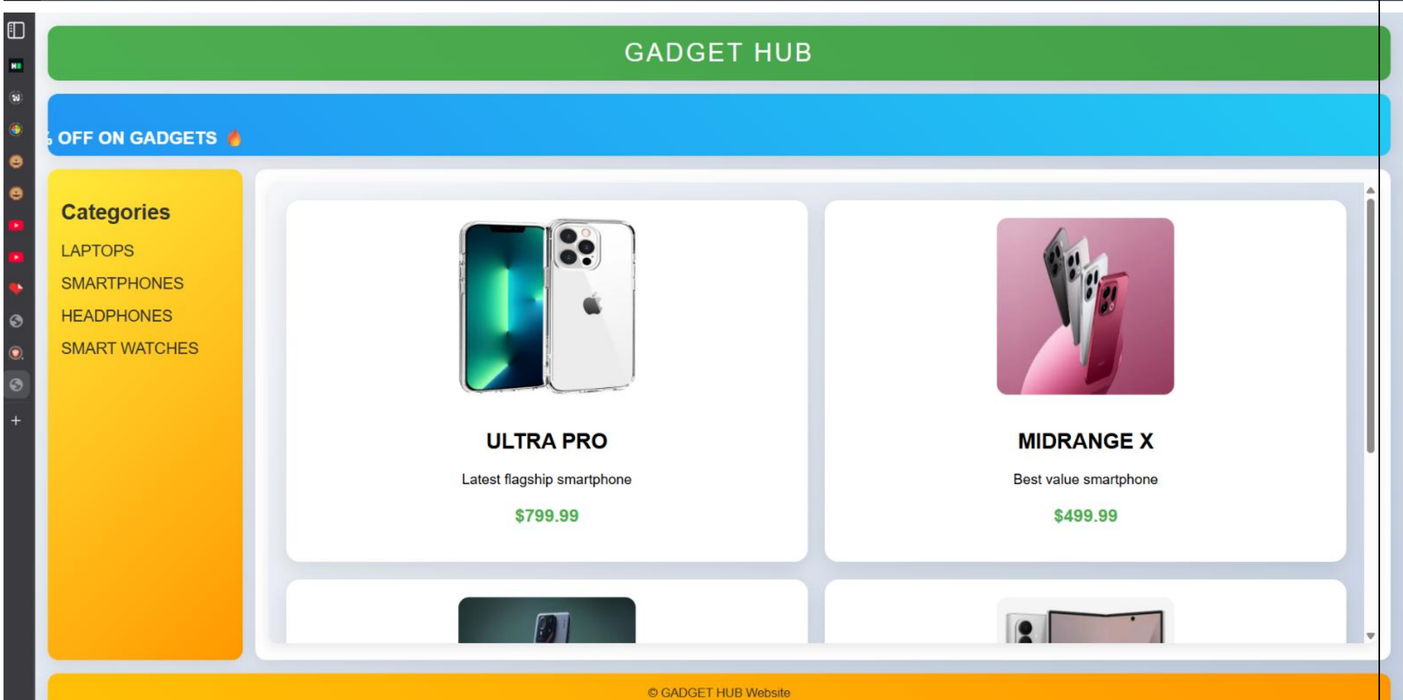
headphones.html

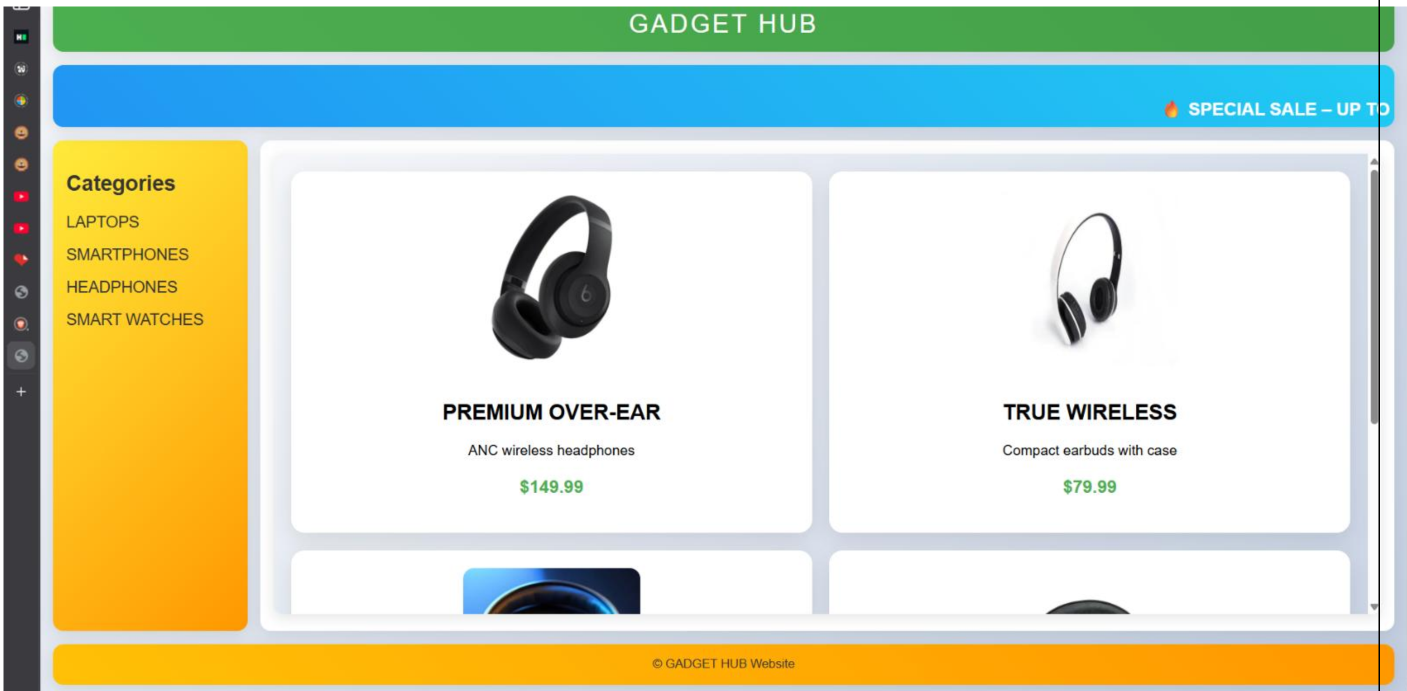
```
<!DOCTYPE html>
<html>
<head>
    <title>Headphones</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 20px;
            background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%);
        }
        .container {
            display: grid;
            grid-template-columns: 1fr 1fr;
            gap: 20px;
        }
        .card {
            background: white;
            padding: 20px;
            border-radius: 15px;
            box-shadow: 0 8px 25px rgba(0,0,0,0.1);
            text-align: center;
        }
        .card-img {
            width: 200px;
            height: 200px;
```

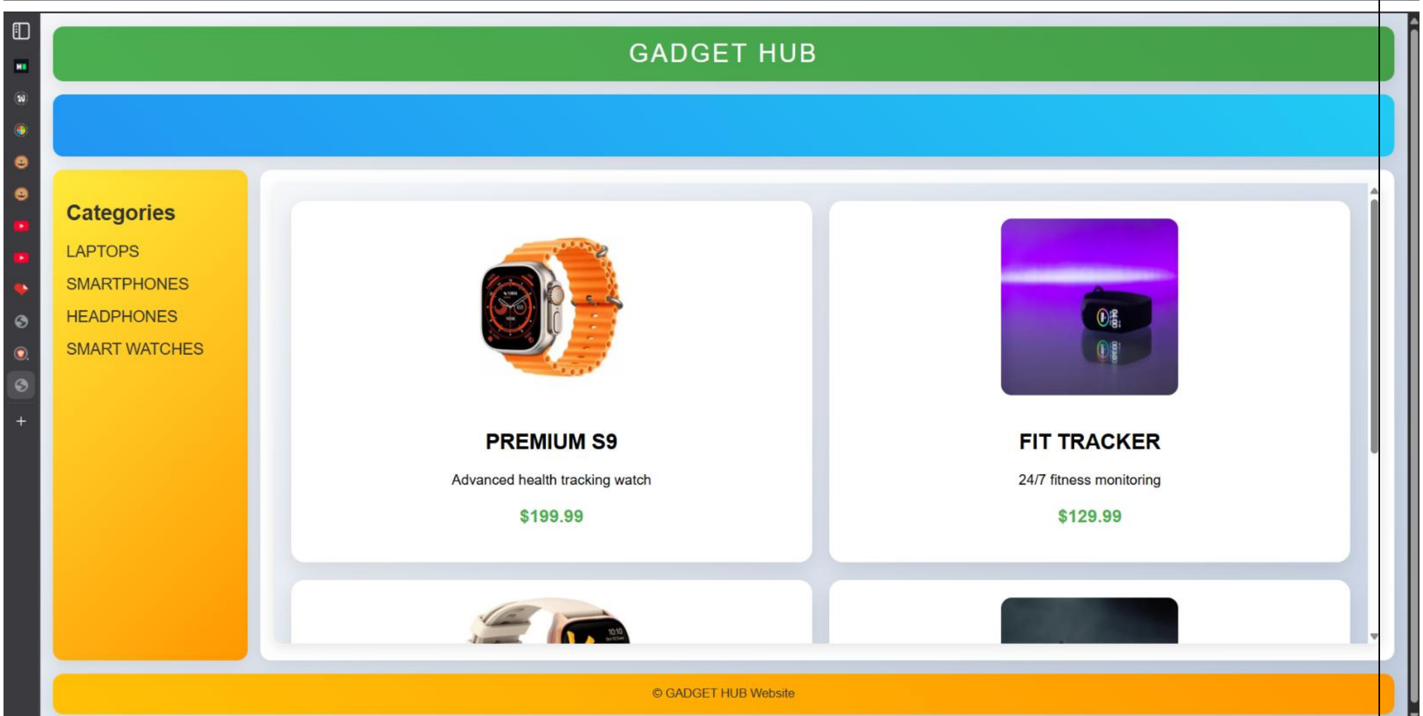
```
        border-radius: 12px;
        margin-bottom: 15px;
      }
      .price {
        font-size: 20px;
        color: #4CAF50;
        font-weight: bold;
      }
  </style>
</head>
<body>
  <div class="container">
    <div class="card">
      <img class="card-img" src="premium.webp" alt="Wireless Over-ear">
      <h2>PREMIUM OVER-EAR</h2>
      <p>ANC wireless headphones</p>
      <p class="price">$149.99</p>
    </div>
    <div class="card">
      <img class="card-img" src="wireless.webp" alt="True Wireless">
      <h2>TRUE WIRELESS</h2>
      <p>Compact earbuds with case</p>
      <p class="price">$79.99</p>
    </div>
    <div class="card">
      <img class="card-img" src="gami.webp" alt="Gaming Headset">
      <h2>GAMING HEADSET</h2>
      <p>7.1 surround sound headset</p>
      <p class="price">$99.99</p>
    </div>
    <div class="card">
      <img class="card-img" src="studio.webp" alt="Studio Headphones">
      <h2>STUDIO PRO</h2>
      <p>Professional monitoring headphones</p>
      <p class="price">$199.99</p>
    </div>
  </div>
</body>
</html>
```

Output:


GADGET HUB
Categories
LAPTOPS
SMARTPHONES
HEADPHONES
SMART WATCHES
GAMING PRO
RTX 4080 gaming laptop
$999.99
ULTRABOOK
Lightweight productivity laptop
$799.99
© GADGET HUB Website


GADGET HUB
OFF ON GADGETS
Categories
LAPTOPS
SMARTPHONES
HEADPHONES
SMART WATCHES
ULTRA PRO
Latest flagship smartphone
$799.99
MIDRANGE X
Best value smartphone
$499.99
© GADGET HUB Website





Explanation of the tags:

- **target="contentFrame": Loads linked pages inside specific iframe by name**
- **name="contentFrame": Names iframe so menu links can target it**

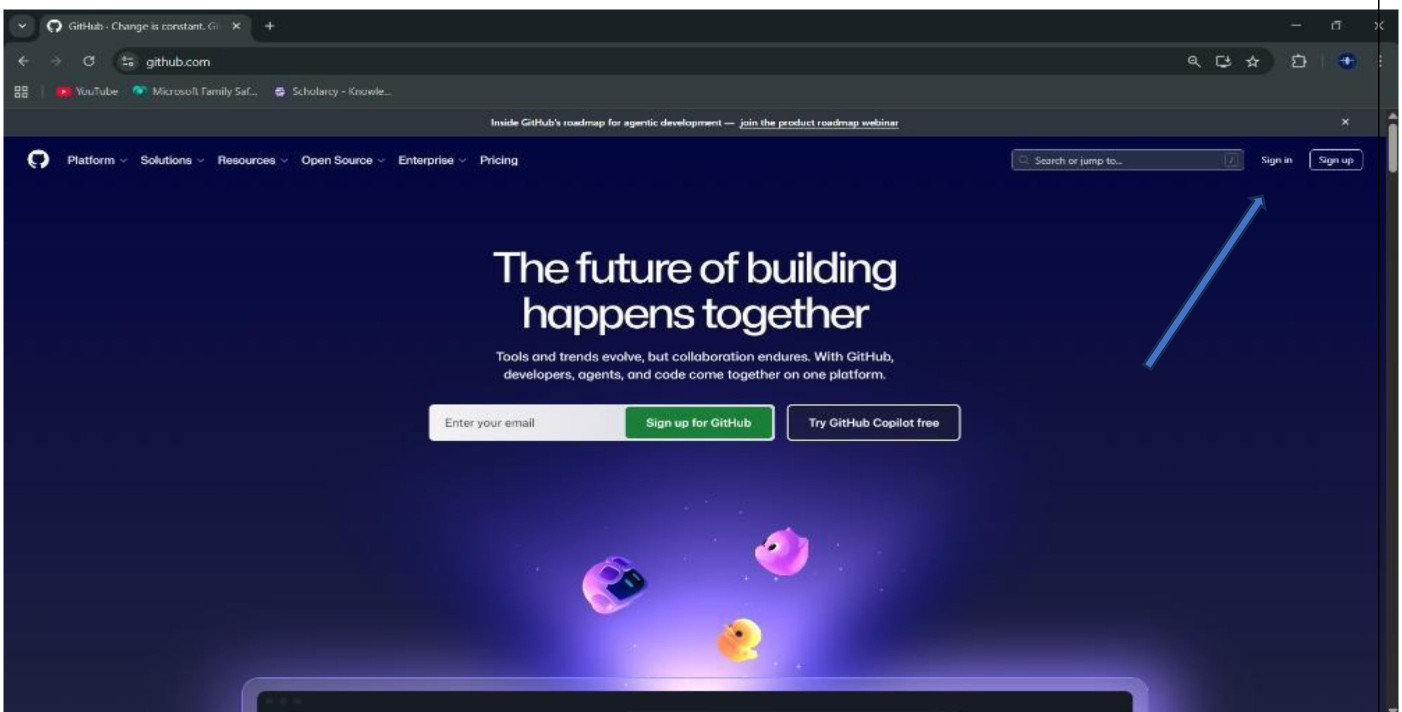
- **grid-template: CSS Grid shorthand for rows, columns, and areas**
- **grid-area: header: Assigns element to named grid area**
- **linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%): Light theme background gradient**
- **@keyframes move: Creates scrolling sale text animation**
- **animation: move 10s linear infinite: Continuous movement of sale banner text**
- **box-shadow: 0 4px 20px rgba(0,0,0,0.1): Adds soft shadow around sections/cards**
- **transition: .3s: Smooth hover effect on sidebar links**
- **display: block: Makes sidebar links take full line width**
- **border-radius: 10px: Rounds iframe corners**
- **grid-template-columns: 1fr 1fr: Creates 2-column product layout in sub-pages**
- **rgba(255,255,255,.95): Semi-transparent white section background**
- **font-weight: bold: Makes text thicker for emphasis**
- **letter-spacing: 3px: Adds spacing between header letters**

WEEK-7

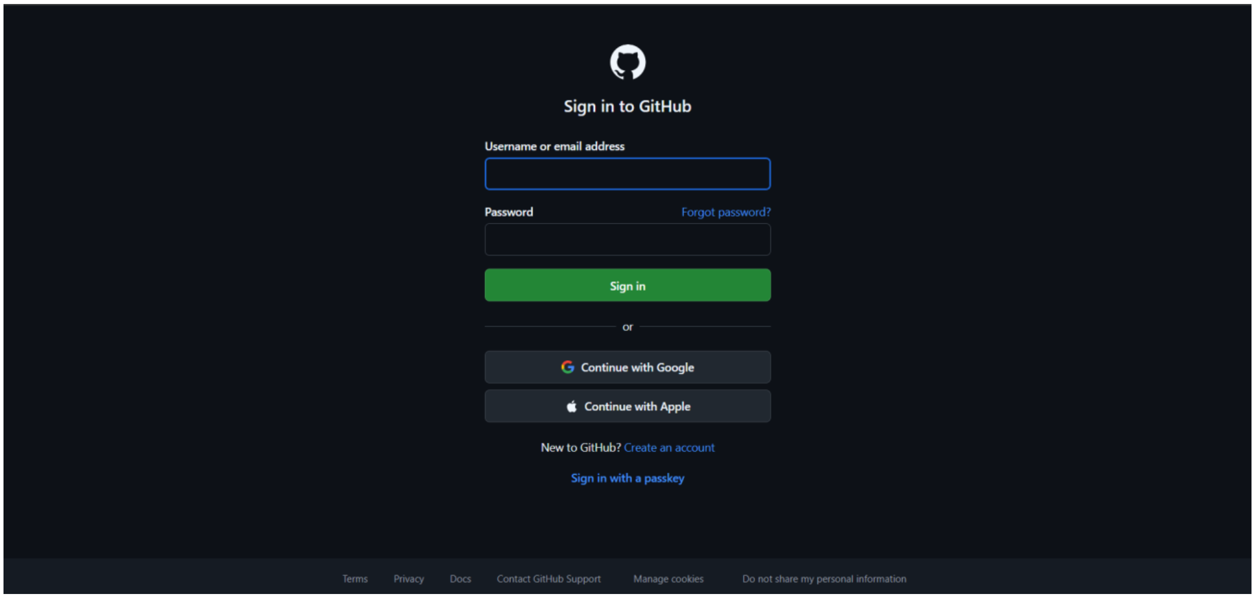
7.Aim: Create a Github Repository and upload all related files of lab manual.

Procedure:

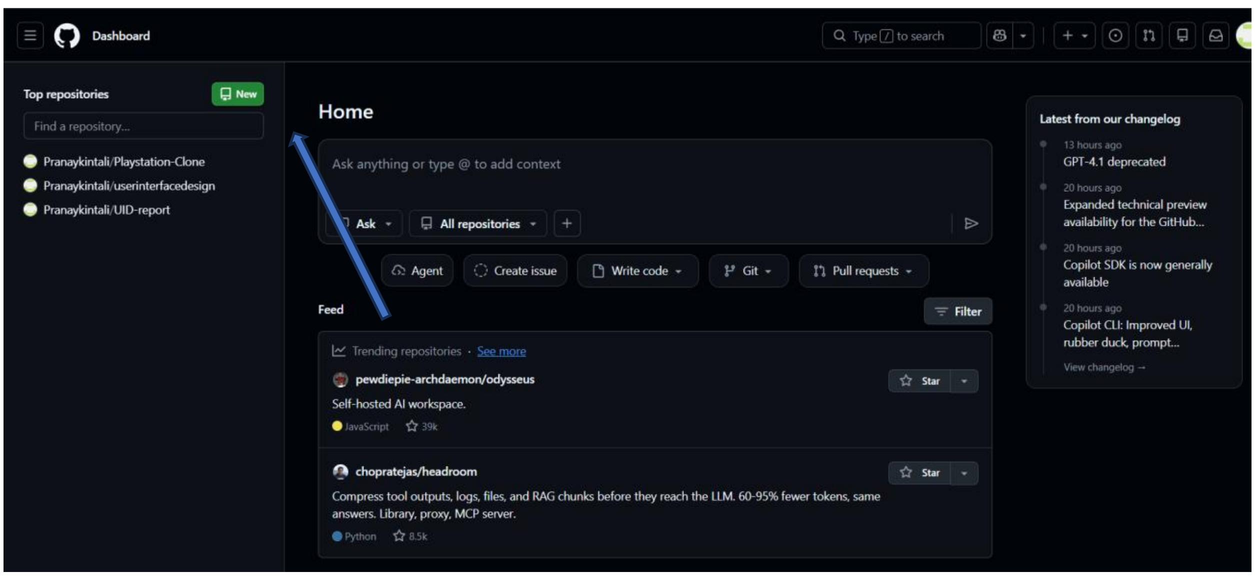
Step-1: Browse to Github and create an Account.



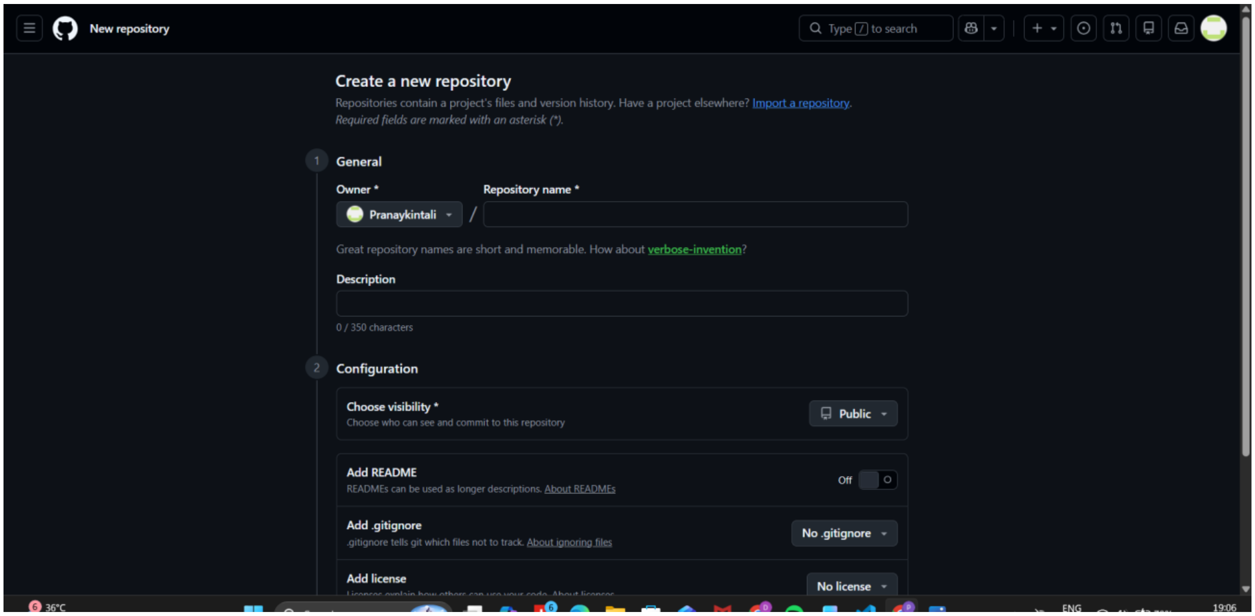
Step-2:Login to Github and Create a Repository



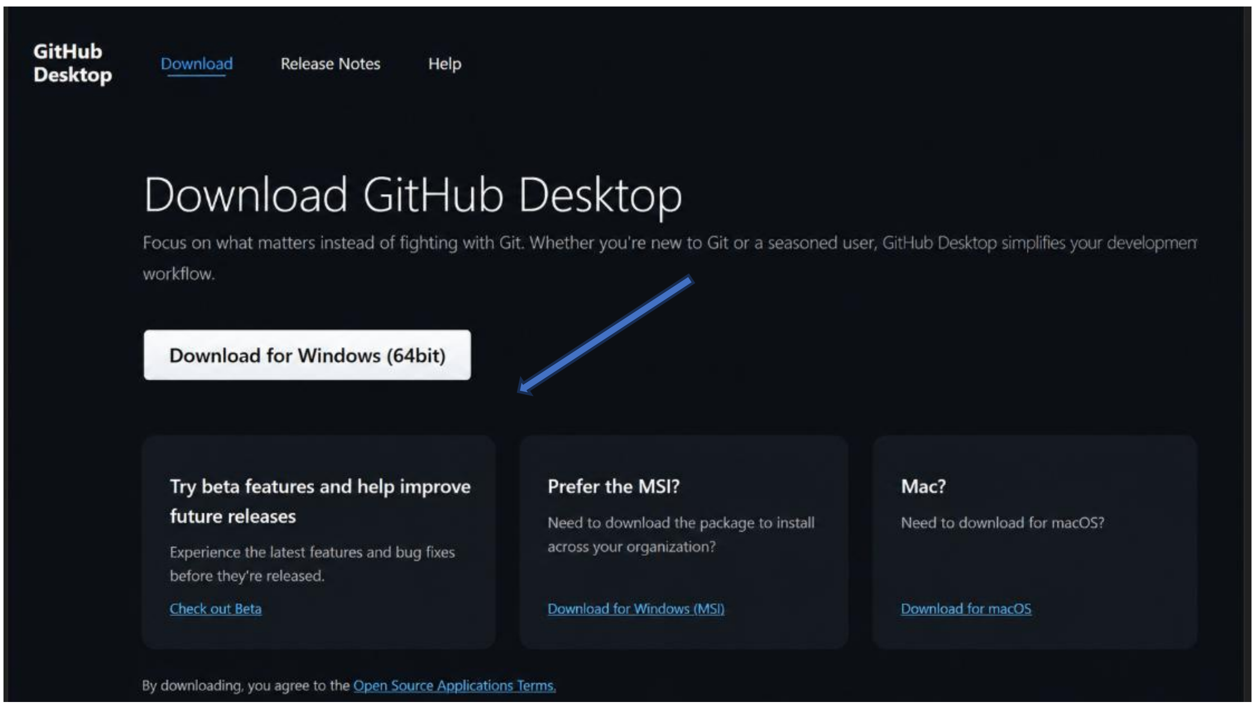
Step-3: Create a repository after login by clicking new.



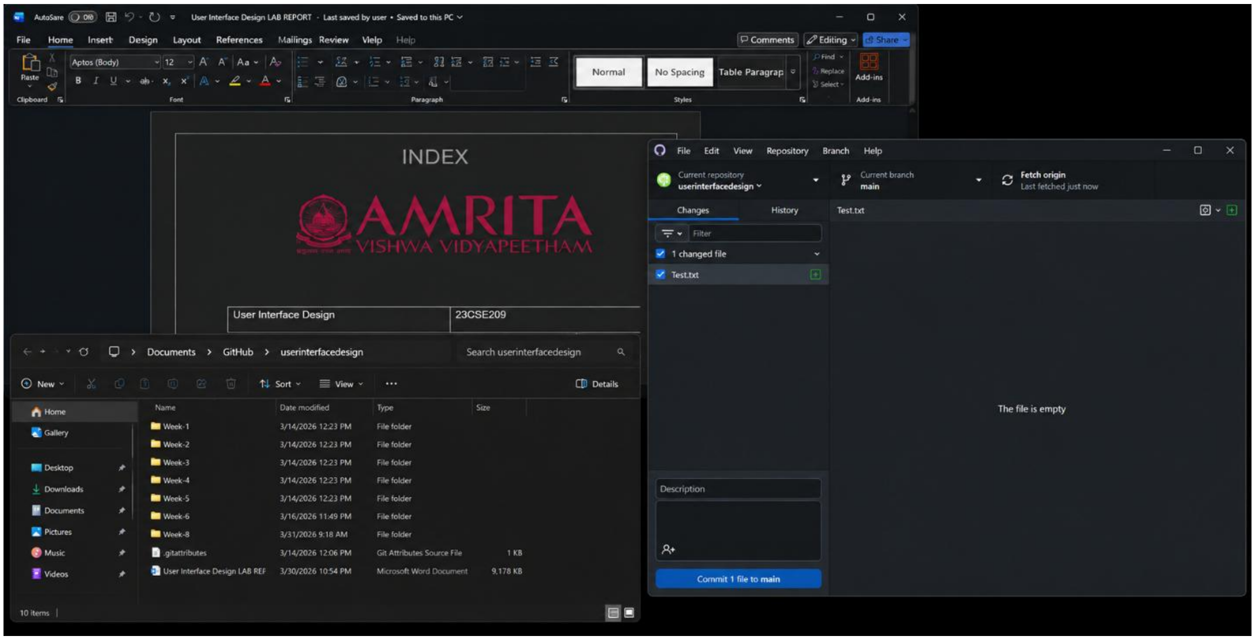
Step-4: Specify the Repository specifications



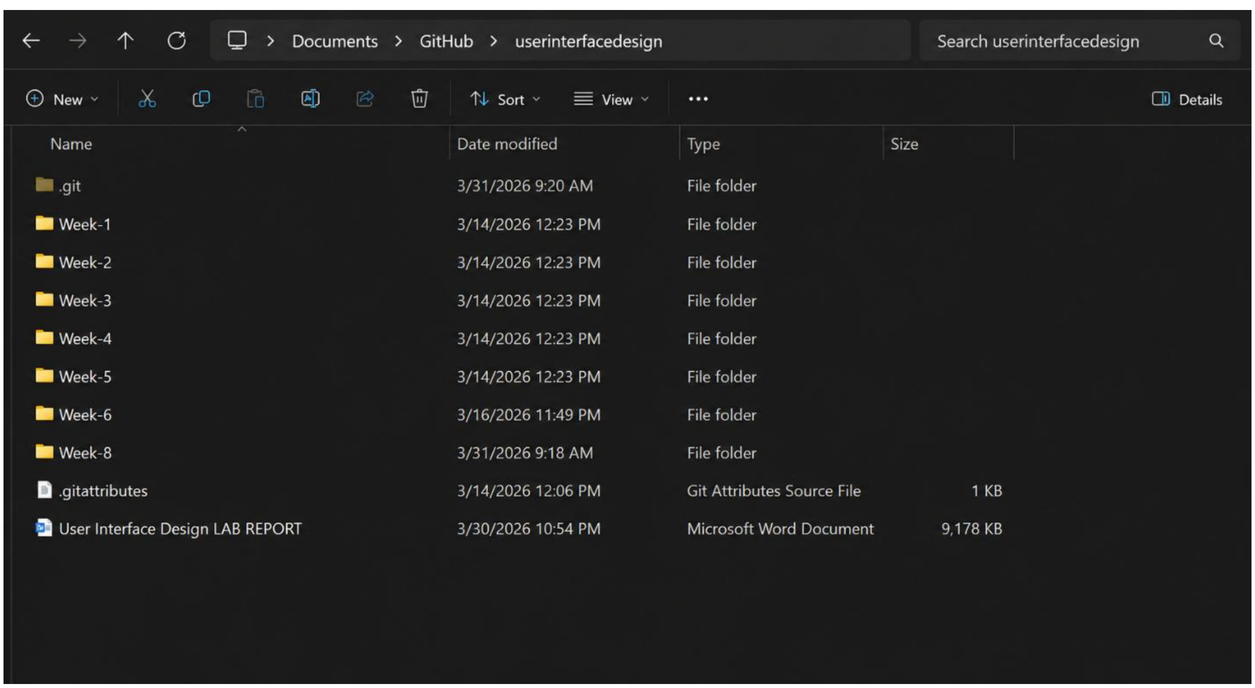
Step-5: Download Github Desktop and Launch it



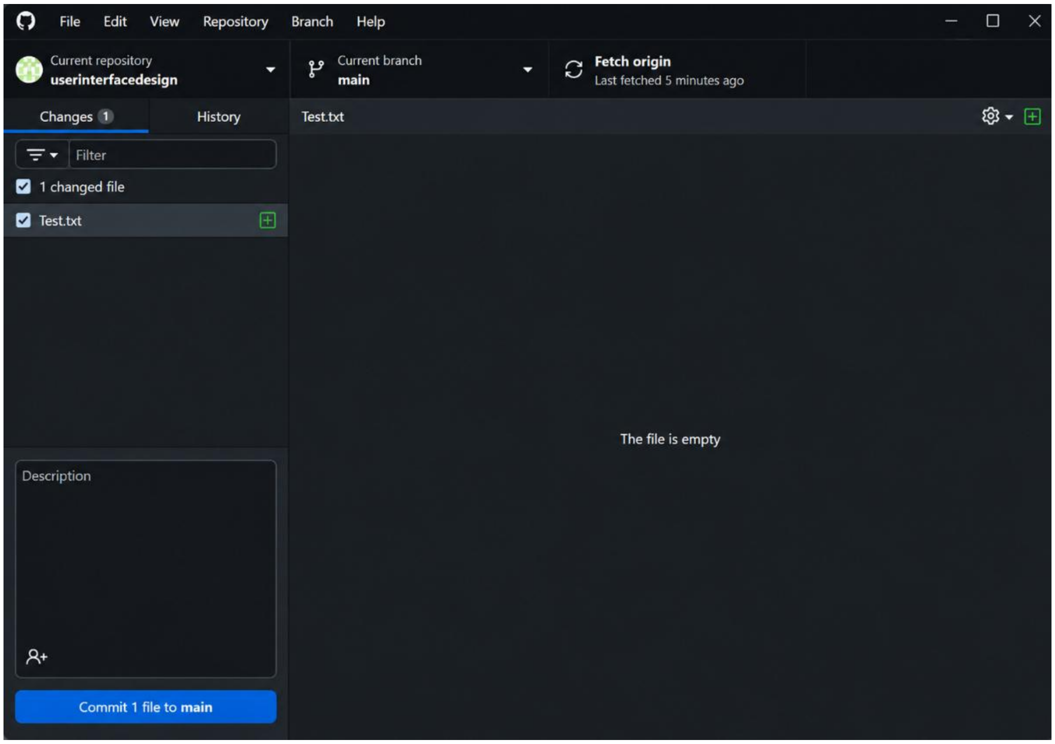
Step-6:In Github Desktop app setup a folder from your explorer.

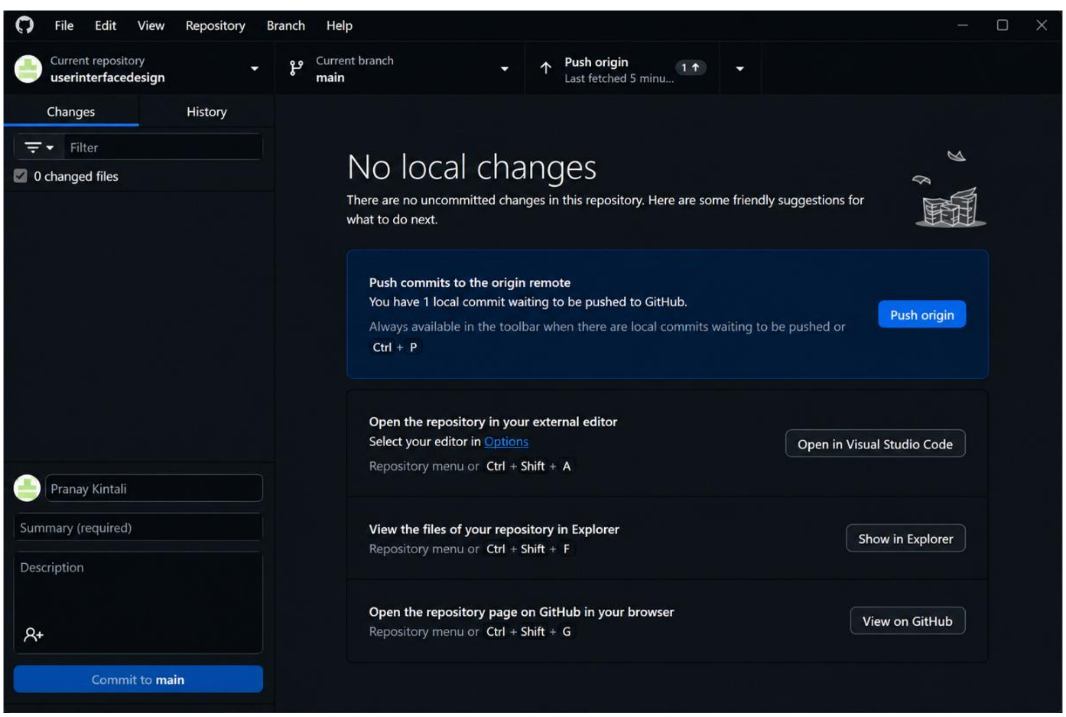


Step-7:Add all Week files to the folder linked to Github.

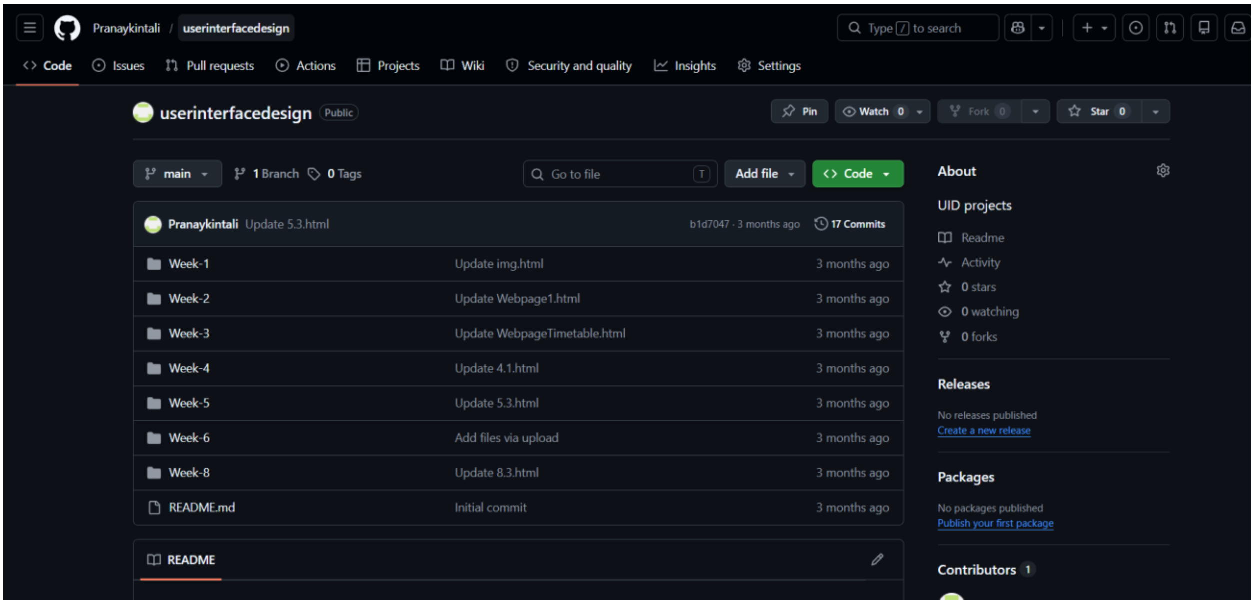


Step-8:Now click on Commit changes and Push the files in the GithubDesktop Applications.





Step-9: Now go to browser and check it. All the new added files in the repository .



Link:https://github.com/Pranaykintali/userinterfacedesign

WEEK-8

8.A.<u>Aim:</u> Create a webpage that displays a square that continuously rotates 360
degree using CSS Animation.

Code:

```html
<!DOCTYPE html>
<head>
   AV.SC.U4CSE25209
<style>
div {
  width: 100px;
  height: 100px;
  background: whitesmoke;
  margin: 100px auto;
  animation: mymove 5s infinite linear;
}

@keyframes mymove {
  from {transform: rotate(0deg);}
  to {transform: rotate(360deg);}
}
</style>
</head>

<body style="background-color: greenyellow;">
<div></div>
</body>
```

Output:

<u>Explanation of the tags:</u>

• animation: mymove 5s infinite linear: Runs animation for 5s, repeats forever, linear speed.

• @keyframes mymove: Defines animation steps from start to end

• transform: rotate(0deg): Rotates element 0 degrees (start position)

• transform: rotate(360deg): Rotates element full circle (end position)

• width: 100px; height: 100px: Creates square shape

• margin: 100px auto: Centers square horizontally, adds top spacing

• background: whitesmoke: Light gray square color

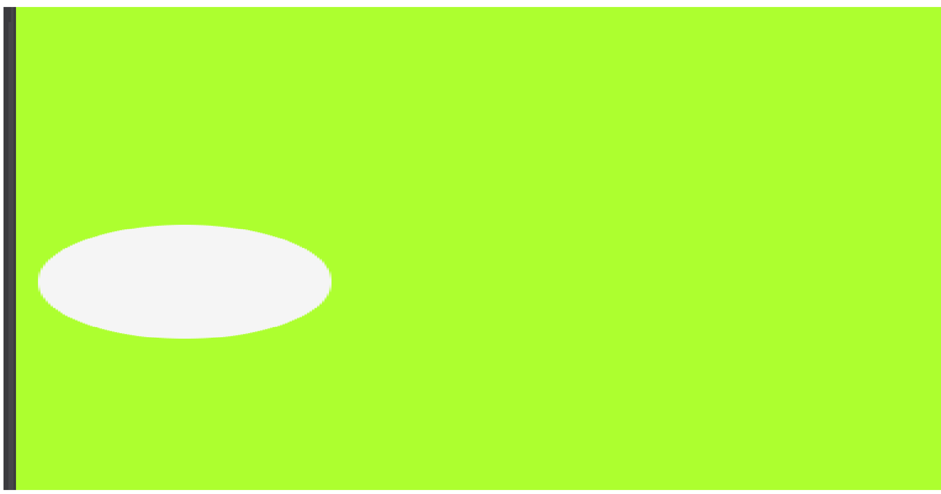
8.B.<u>Aim:</u> Create a ball that moves up and down continuously using CSS Animation.

Code:

```
<!DOCTYPE html>
<head>
   AV.SC.U4CSE25209
<style>
div {
  width: 100px;
  height: 100px;
  background: whitesmoke;
```

```
  border-radius: 50%;
  position: relative;
  animation: mymove 2s infinite alternate;
}

@keyframes mymove {
  from {top: 0px;}
  to {top: 200px;}
}
</style>
</head>

<body style="background-color: greenyellow;">
<div></div>
</body>
```

Output:



<u>Explanation of the tags:</u>

• border-radius: 50%: Makes square into perfect circle

• position: relative: Allows top positioning from original spot

• animation: mymove 2s infinite alternate: Bounces up/down every 2s, reverses direction

• top: 0px: Starting position at top

• top: 200px: Ending position 200px down

• @keyframes mymove: Defines bounce path from top to bottom

• width/height: Creates circular ball shape

• infinite alternate: Keeps bouncing back and forth

8.C.<u>Aim:</u> Create a circle that moves horizontally and rotates at the same time
using CSS Animation.

Code:

```
<!DOCTYPE html>
<head>
  AV.SC.U4CSE25209
<style>
div {
 width: 100px;
 height: 100px;
 background: whitesmoke;
 border-radius: 50%;
 position: relative;
 animation: mymove 5s infinite linear;
}
@keyframes mymove {
 from {
  left: 0px;
  transform: rotate(0deg);
 }
 to {
  left: 200px;
  transform: rotate(360deg);
 }
}
</style>
</head>
<body style="background-color: greenyellow;">
<div></div>
</body>
```

Output:

Explanation of the tags:

• border-radius: 50%: Makes square into perfect circle

• position: relative: Allows top positioning from original spot

• animation: mymove 2s infinite alternate: Bounces up/down every 2s, reverses direction

• top: 0px: Starting position at top

• top: 200px: Ending position 200px down

• @keyframes mymove: Defines bounce path from top to bottom

• width/height: Creates circular ball shape

• infinite alternate: Keeps bouncing back and forth

WEEK-9

9.Aim: Create a webpage using HTML, CSS, JS based on the give conditions:

a) Create a button for each program, based on the button, clicked the corresponding program has to be executed and should display the output.

b)The logics for each button includes finding an even or odd number, prime or not, factorial of a number, Armstrong or not.

c)Use appropriate design to display the output.

Code:

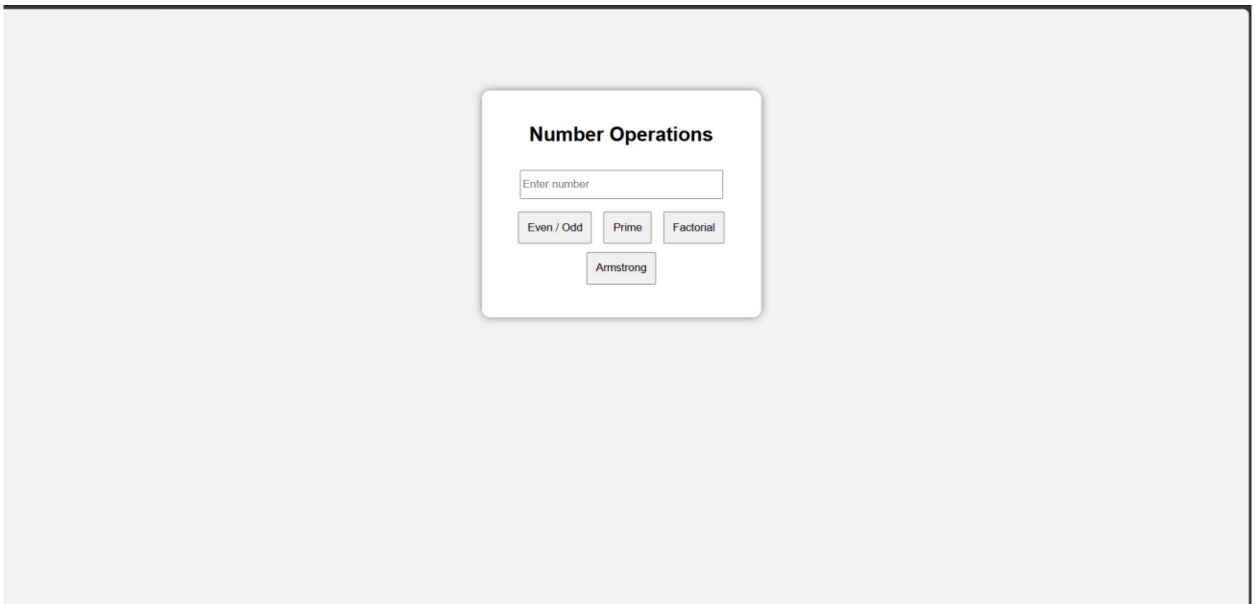
```
<!DOCTYPE html>

AV.SC.U4CSE25209

<html>
<head>
   <title>Number Checker</title>

   <style>
      body {
         font-family: Arial;
         text-align: center;
         background-color: #f2f2f2;
      }

      .box {
         width: 300px;
         margin: 100px auto;
         padding: 20px;
         background: white;
         border-radius: 10px;
         box-shadow: 0 0 10px gray;
      }

      input {
         width: 80%;
         height: 30px;
         margin: 10px;
      }

      button {
         margin: 5px;
```

```
            padding: 10px;
            cursor: pointer;
        }

        #output {
            margin-top: 15px;
            font-weight: bold;
        }
    </style>
</head>

<body>

<div class="box">
    <h2>Number Operations</h2>

    <input type="number" id="num" placeholder="Enter number">

    <br>

    <button onclick="checkEvenOdd()">Even / Odd</button>
    <button onclick="checkPrime()">Prime</button>
    <button onclick="findFactorial()">Factorial</button>
    <button onclick="checkArmstrong()">Armstrong</button>

    <div id="output"></div>
</div>

<script>

function getNumber() {
    return parseInt(document.getElementById("num").value);
}

// Even / Odd
function checkEvenOdd() {
    let n = getNumber();
    if (n % 2 === 0)
        document.getElementById("output").innerText = n + " is EVEN";
    else
        document.getElementById("output").innerText = n + " is ODD";
}

// Prime
function checkPrime() {
```

```
    let n = getNumber();
    let isPrime = true;

    if (n <= 1) isPrime = false;

    for (let i = 2; i <= Math.sqrt(n); i++) {
        if (n % i === 0) {
            isPrime = false;
            break;
        }
    }

    document.getElementById("output").innerText =
        isPrime ? n + " is PRIME" : n + " is NOT PRIME";
}

// Factorial
function findFactorial() {
    let n = getNumber();
    let fact = 1;

    if (n < 0) {
        document.getElementById("output").innerText = "Factorial not possible";
        return;
    }

    for (let i = 1; i <= n; i++) {
        fact *= i;
    }

    document.getElementById("output").innerText =
        "Factorial of " + n + " = " + fact;
}

// Armstrong
function checkArmstrong() {
    let n = getNumber();
    let sum = 0;
    let temp = n;
    let digits = n.toString().length;

    while (temp > 0) {
        let rem = temp % 10;
        sum += Math.pow(rem, digits);
        temp = Math.floor(temp / 10);
```

```
    }

    document.getElementById("output").innerText =
      (sum === n) ? n + " is ARMSTRONG" : n + " is NOT ARMSTRONG";
}

</script>

</body>
</html>
```

Output:



Explanation of the tags:

- <html> – Top-level container for the entire HTML page.
- <head> – Holds metadata, title, styles, and scripts.
- <title> – Sets the page title shown in the browser tab.
- <body> – Contains all visible content of the page.
- <div class="container"> – Groups the main UI into a styled card.
- <h2> – Second-level heading for "Logic Programs".
- <input type="number"> – Text box that accepts only numeric input.
- id="numInput" – Unique name used in JavaScript to access the input.
- onclick="checkEvenOdd()" – Runs the checkEvenOdd function when the button is clicked.

WEEK-10

- **Aim:**
- **10(A): write JavaScript conditional statement to find the sign of product of three numbers using functions. Display an alert box with specified sign.**
- **10(B): write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result.**
- **10(C): write a java script conditional statement to find largest of find largest of five numbers using function and onclick event. display an alert box to show the result**
- **10(D): write a java script for loop that will iterate from 0 to 15. For each iteration, it will**
- **check if the current number odd or even, and display a message on screen**
- **10(E) write a java script program to read 6 subjects marks from the student then computes the average marks of students, this average is used to determine the corresponding grade**
-
- **<u>Code:</u>**

```
<!DOCTYPE html>
<html>
<head>
    <title>JavaScript Lab Programs</title>

    <style>
        *{
            margin:0;
            padding:0;
            box-sizing:border-box;
            font-family:Arial, sans-serif;
        }
```

```
body{
  background:#f2f2f2;
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
}

.container{
  width:950px;
  background:white;
  display:flex;
  border-radius:8px;
  overflow:hidden;
  box-shadow:0 0 15px rgba(0,0,0,0.2);
}

.left{
  width:45%;
  padding:30px;
}

.right{
  width:55%;
  padding:30px;
  border-left:1px solid #ddd;
}

h2{
  margin-bottom:20px;
}

button{
  width:100%;
  padding:15px;
  margin-bottom:18px;
  border:none;
  border-radius:6px;
  background:#4a89dc;
  color:white;
  font-size:18px;
  cursor:pointer;
}
```

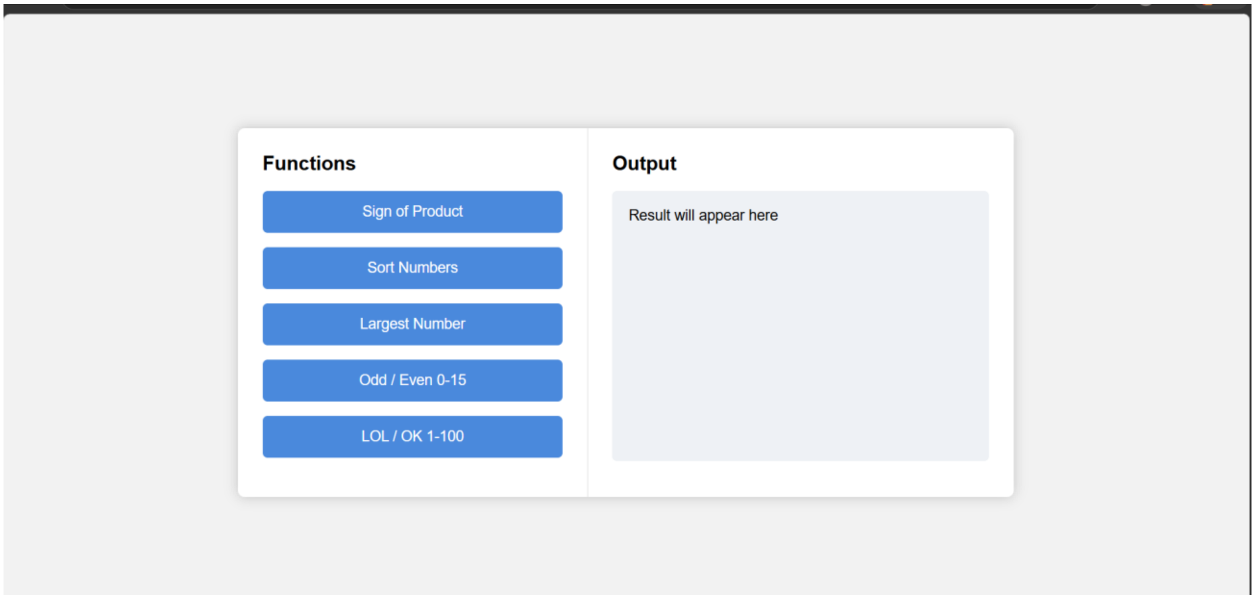
```
        button:hover{
            background:#3575cc;
        }

        #output{
            background:#eef1f5;
            height:330px;
            padding:20px;
            border-radius:6px;
            overflow:auto;
            font-size:18px;
        }
    </style>
</head>

<body>

<div class="container">

    <div class="left">
        <h2>Functions</h2>

        <button onclick="signProduct()">
            Sign of Product
        </button>

        <button onclick="sortNumbers()">
            Sort Numbers
        </button>

        <button onclick="largestNumber()">
            Largest Number
        </button>

        <button onclick="oddEven()">
            Odd / Even 0-15
        </button>

        <button onclick="lolOk()">
            LOL / OK 1-100
        </button>

    </div>

    <div class="right">
```

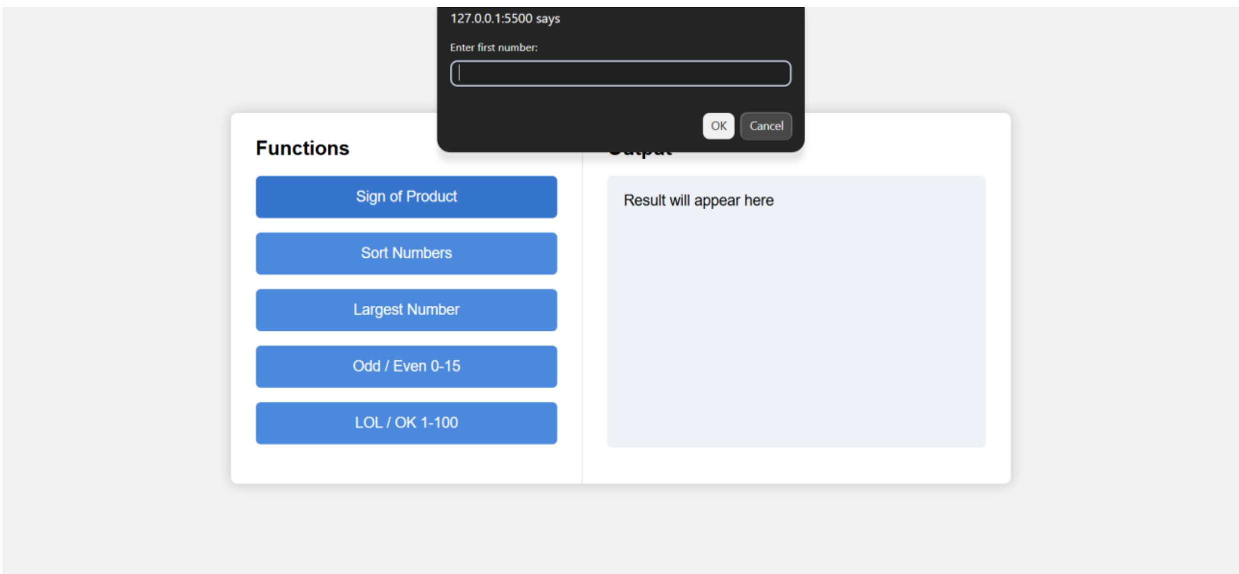
```
        <h2>Output</h2>

        <div id="output">
            Result will appear here
        </div>

    </div>

</div>

<script>

// 1. Sign of Product

function signProduct(){

    let a = Number(prompt("Enter first number:"));
    let b = Number(prompt("Enter second number:"));
    let c = Number(prompt("Enter third number:"));

    let result="";

    if(a==0 || b==0 || c==0){
        result="The product is 0";
    }
    else if(a*b*c > 0){
        result="The sign is Positive (+)";
    }
    else{
        result="The sign is Negative (-)";
    }

    document.getElementById("output").innerHTML=result;
}

// 2. Sort Numbers

function sortNumbers(){

    let a = Number(prompt("Enter first number:"));
    let b = Number(prompt("Enter second number:"));
    let c = Number(prompt("Enter third number:"));

    let arr=[a,b,c];
```

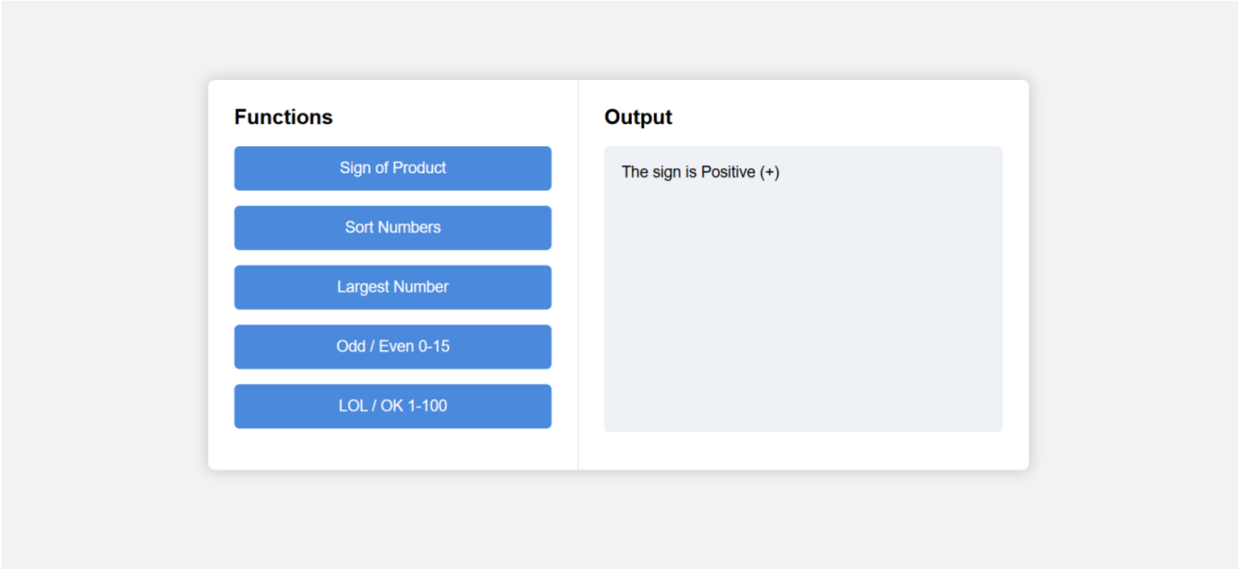
```
    arr.sort(function(x,y){
        return x-y;
    });

    document.getElementById("output").innerHTML=
    "Sorted Numbers:<br><br>" +
    arr.join(" , ");
}

// 3. Largest Number

function largestNumber(){

    let a = Number(prompt("Enter number 1:"));
    let b = Number(prompt("Enter number 2:"));
    let c = Number(prompt("Enter number 3:"));
    let d = Number(prompt("Enter number 4:"));
    let e = Number(prompt("Enter number 5:"));

    let max=a;

    if(b>max) max=b;
    if(c>max) max=c;
    if(d>max) max=d;
    if(e>max) max=e;

    document.getElementById("output").innerHTML=
    "Largest Number = <b>"+max+"</b>";
}

// 4. Odd Even

function oddEven(){

    let result="";

    for(let i=0;i<=15;i++){

        if(i%2==0){
            result += i + " is EVEN<br>";
        }
        else{
            result += i + " is ODD<br>";
        }
```

```
        }

        document.getElementById("output").innerHTML=result;
    }

    // 5. LOL OK

    function lolOk(){

        let result="";

        for(let i=1;i<=100;i++){

            if(i%3==0 && i%5==0){
                result += "LOL-OK<br>";
            }
            else if(i%3==0){
                result += "LOL<br>";
            }
            else if(i%5==0){
                result += "OK<br>";
            }
            else{
                result += i + "<br>";
            }
        }

        document.getElementById("output").innerHTML=result;
    }

    </script>

    </body>
    </html>
```

Output:


Functions
Sign of Product
Sort Numbers
Largest Number
Odd / Even 0-15
LOL / OK 1-100
Output
Result will appear here


127.0.0.1:5500 says
Enter first number:
OK
Cancel
Functions
Sign of Product
Sort Numbers
Largest Number
Odd / Even 0-15
LOL / OK 1-100
Result will appear here



Explanation:

- <!DOCTYPE html> → Declares the document as HTML5.
- <html> → Root element containing entire webpage.
- <head> → Holds metadata like title.
- <title> → Sets the browser tab name.
- <body> → Contains all visible content.
- <h3> → Displays a heading.
- <input> → Takes user input.
- type="password" → Hides typed characters.
- id → Uniquely identifies an element.
- onkeyup → Triggers function when user types.
- <p> → Displays text/output.
- <script> → Contains JavaScript code.
- function → Defines a reusable block of code.
- document. getElementById() → Selects an HTML element using its ID.
- .value → Gets input field value.
- let → Declares a variable.
- if → Executes code if condition is true.

- **else if → Checks another condition if previous is false.**
- **else → Runs if all conditions are false.**
- **. length → Returns number of characters in a string.**
- **match () → Checks if a pattern exists in a string.**
- **/[A-Z]/ → Checks for uppercase letters.**
- **/[0-9]/ → Checks for digits.**
- **/[@$!%*? &]/ → Checks for special characters.**
- **&& → Logical AND (all conditions must be true).**
- **= → Assigns value to a variable.**
- **inner Text → Updates text inside an HTML element.**

- # WEEK-11

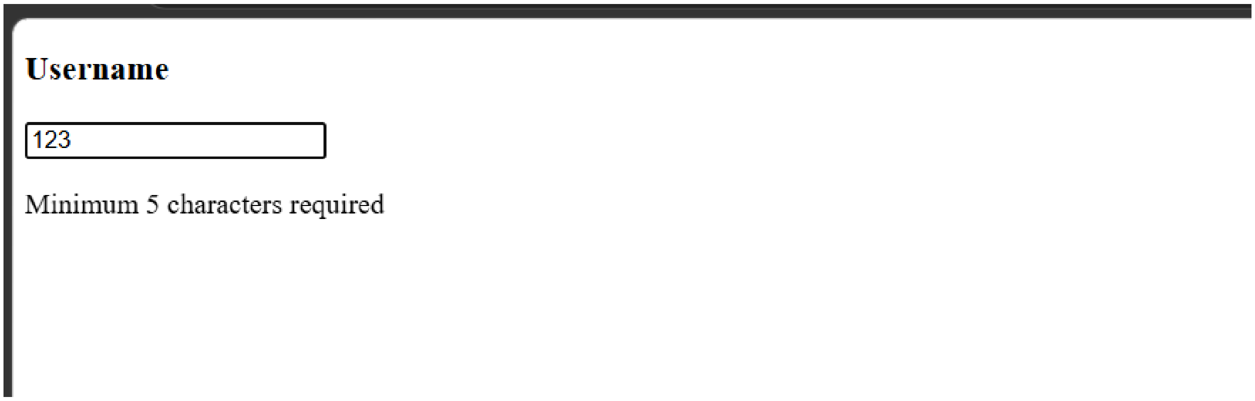
AIM:

- **1. A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters.**
- **2. A registration form should validate email format before submission. If invalid, the form should not be submitted.**
- **3. A system should check password strength while typing and display whether it is weak or strong.**
- **4. A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits.**
- **5. Design an HTML page, such that the registration form must validate:**
- **Name (not empty)**
- **Email (valid format)**
- **Password (minimum 6 characters).**

- ## CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Username Validation</title>
</head>
<body>
  <h3>Username</h3>
  <input type="text" id="username" onkeyup="checkUsername()" />
  <p id="msg"></p>

  <script>
    function checkUsername() {
      let name = document.getElementById("username").value;

      document.getElementById("msg").innerText =
        name.length < 5 ? "Minimum 5 characters required" : "";
    }
  </script>
</body>
</html>
```

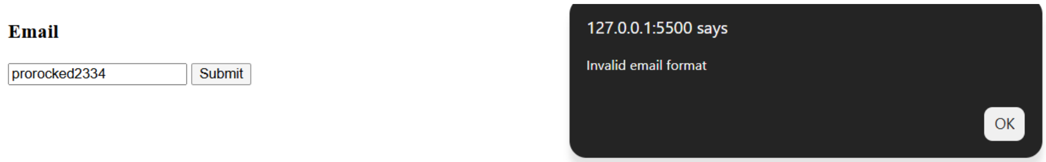
OUTPUT:



CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Email Validation</title>
</head>
<body>
   <h3>Email</h3>

   <form onsubmit="return validateEmail();">
      <input type="text" id="email" />
      <button type="submit">Submit</button>
   </form>

   <script>
      function validateEmail() {
         let email = document.getElementById("email").value;

         let pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

         if (!pattern.test(email)) {
            alert("Invalid email format");
            return false;
         }

         return true;
      }
   </script>
</body>
</html>
```

OUTPUT:

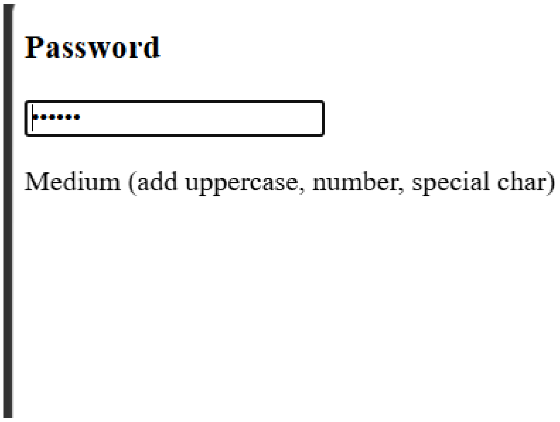


CODE:

```html
<!DOCTYPE html>

<html>
<head>
  <title>Password Strength</title>
</head>
<body>
  <h3>Password</h3>

  <input type="password" id="password" onkeyup="checkPassword()" />
  <p id="strength"></p>

  <script>
    function checkPassword() {
      let pass = document.getElementById("password").value;
      let msg = "";

      if (pass.length < 6) {
        msg = "Weak (too short)";
      }
      else if (
        pass.match(/[A-Z]/) &&   // uppercase
        pass.match(/[0-9]/) &&   // number
        pass.match(/[@$!%*?&]/)  // special char
      ) {
        msg = "Strong Password";
      }
      else {
        msg = "Medium (add uppercase, number, special char)";
      }

      document.getElementById("strength").innerText = msg;
    }
```

```
    </script>
</body>
</html>
```

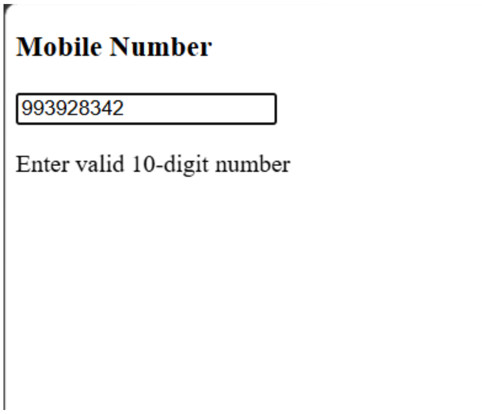
OUTPUT:



CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Mobile Validation</title>
</head>
<body>
    <h3>Mobile Number</h3>

    <input type="text" id="mobile" onblur="checkMobile()" />
    <p id="msg"></p>

    <script>
        function checkMobile() {
            let num = document.getElementById("mobile").value;

            if (num.length !== 10 || isNaN(num)) {
                document.getElementById("msg").innerText =
                    "Enter valid 10-digit number";
            } else {
                document.getElementById("msg").innerText = "";
            }
        }
    </script>
</body>
</html>
```

OUTPUT:

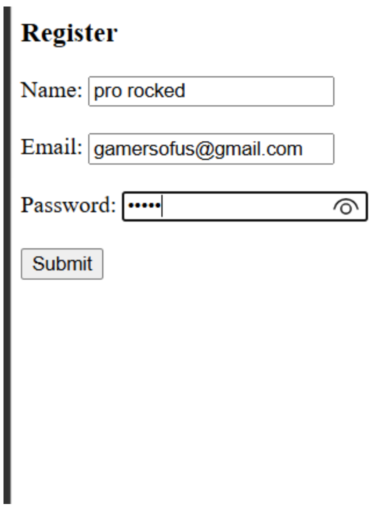


CODE:

```
<!DOCTYPE html>

<html>
<head>
  <title>Registration Form</title>
</head>
<body>
  <h3>Register</h3>

  <form onsubmit="return validateForm();">
    Name: <input type="text" id="name" /><br /><br />
    Email: <input type="text" id="email" /><br /><br />
    Password: <input type="password" id="password" /><br /><br />

    <button type="submit">Submit</button>
  </form>

  <script>
    function validateForm() {
      let name = document.getElementById("name").value;
      let email = document.getElementById("email").value;
      let password = document.getElementById("password").value;

      let pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

      if (name === "") {
        alert("Name required");
        return false;
      }

      if (!pattern.test(email)) {
```

```
            alert("Invalid email");
            return false;
        }

        if (password.length < 6) {
            alert("Password must be at least 6 characters");
            return false;
        }

        return true;
    }
  </script>
</body>
</html>
```

OUTPUT:



Explanation:

<!DOCTYPE html> → Declares HTML5 document type.

<html> → Root of the webpage.

<head> → Contains metadata.

<title> → Sets tab title.

<body> → Holds visible content.

<h3> → Displays heading text.

<form> → Collects user input.

on submit → Runs function before form submission.

<input> → Takes user input.

type="text" → Plain text input.

type="password" → Hides typed characters.

id → Unique identifier for element.

onkeyup → Triggers function while typing.

onblur → Triggers when input loses focus.

<p> → Displays messages.

<button> → Submits form.

<script> → Contains JavaScript code.

function → Defines reusable block of code.

document.getElementById() → Selects element using ID.

.value → Gets input value.

let → Declares variable.

if → Executes code if condition is true.

else if → Checks another condition.

else → Executes if all conditions fail.

.length → Returns number of characters.

match() → Checks pattern in string.

test() → Tests regex pattern (true/false).

/regex/ → Defines pattern for validation.

[A-Z] → Matches uppercase letters.

[0-9] → Matches digits.

[@$!%*?&] → Matches special characters.

&& → Logical AND operator.

! → Logical NOT operator.

=== → Strict equality check.

!== → Strict inequality check.

= → Assigns value.

innerText → Updates text inside element.

alert() → Displays popup message.

return false → Stops form submission.

isNaN() → Checks if value is not a number.

WEEK-12

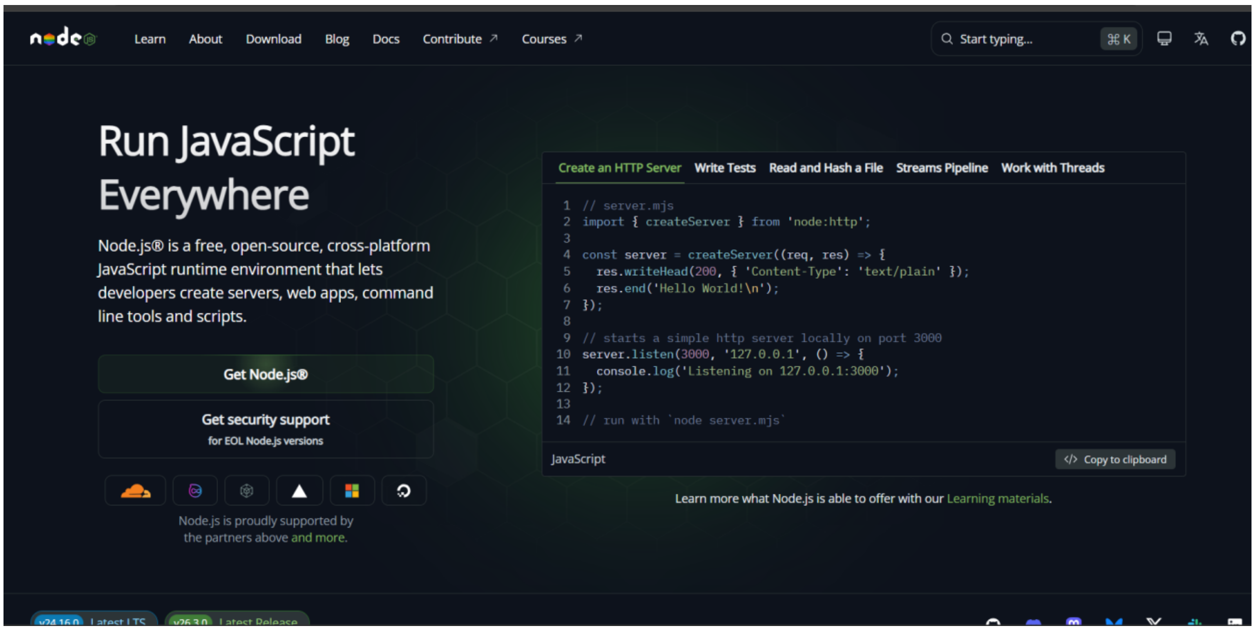
Aim: Installation of Electron

Steps for installation of electron:

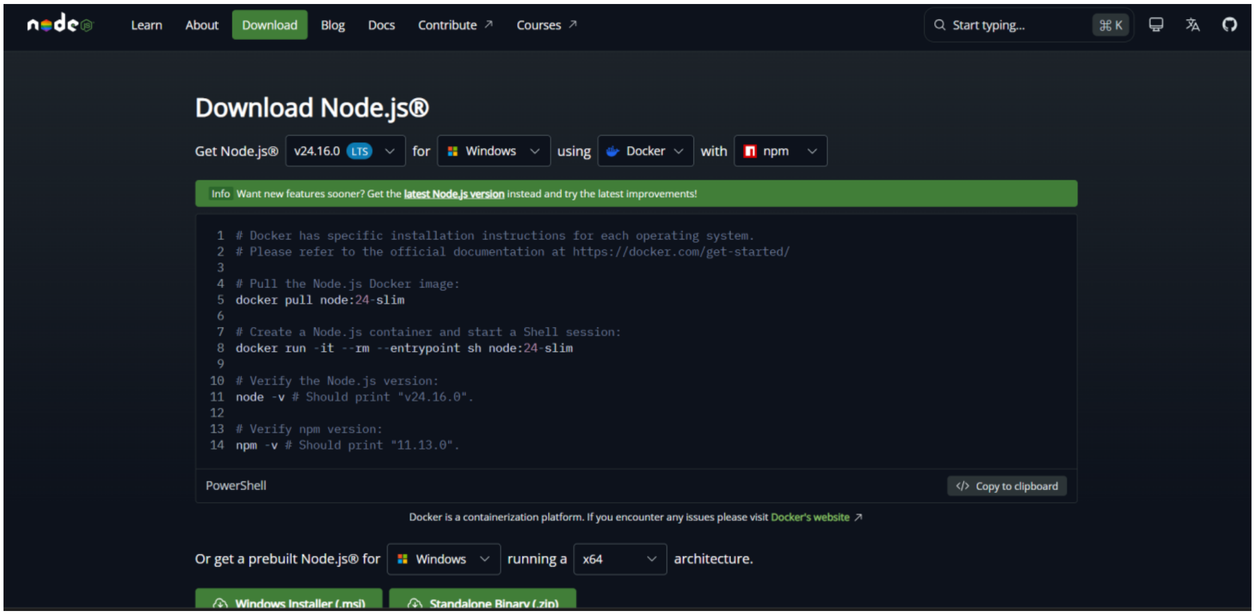
Step1: Open Google Chrome browser.

open:

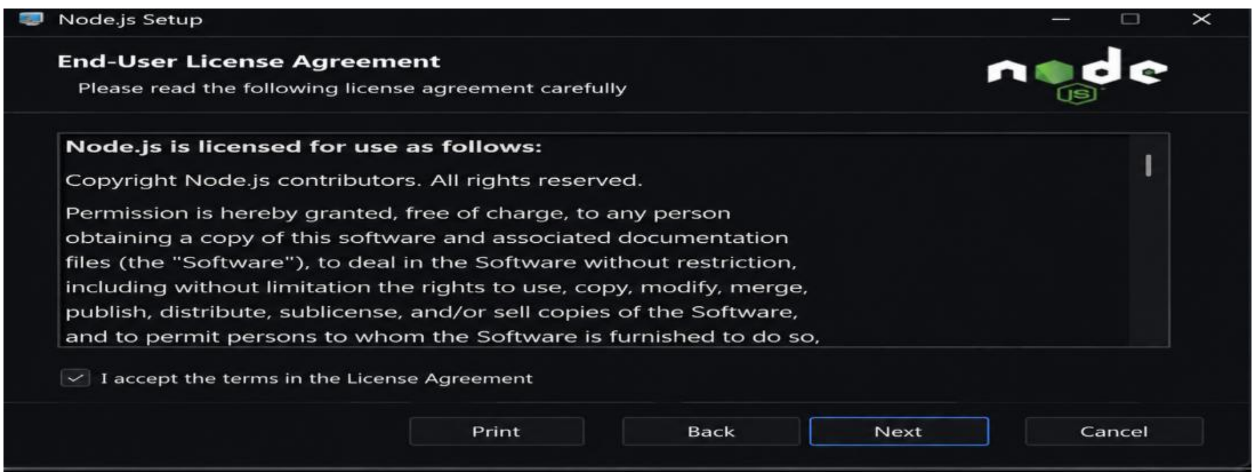
[Node.js Official Website]



- **Download the LTS version**

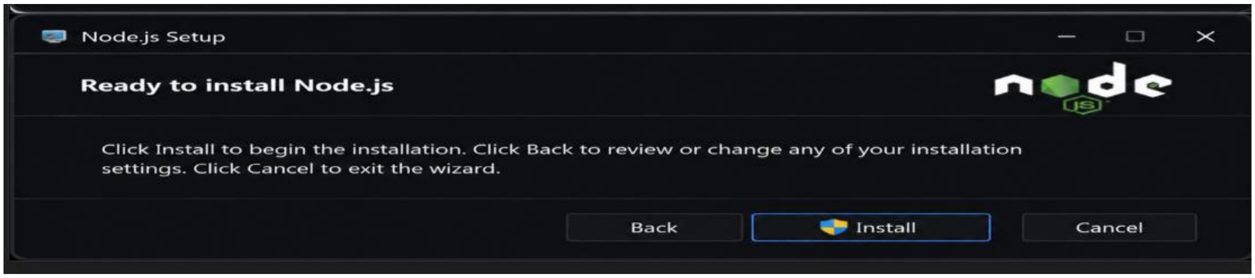


- **Install it**
- **Keep clicking Next → Next → Install**



Node.js automatically installs:

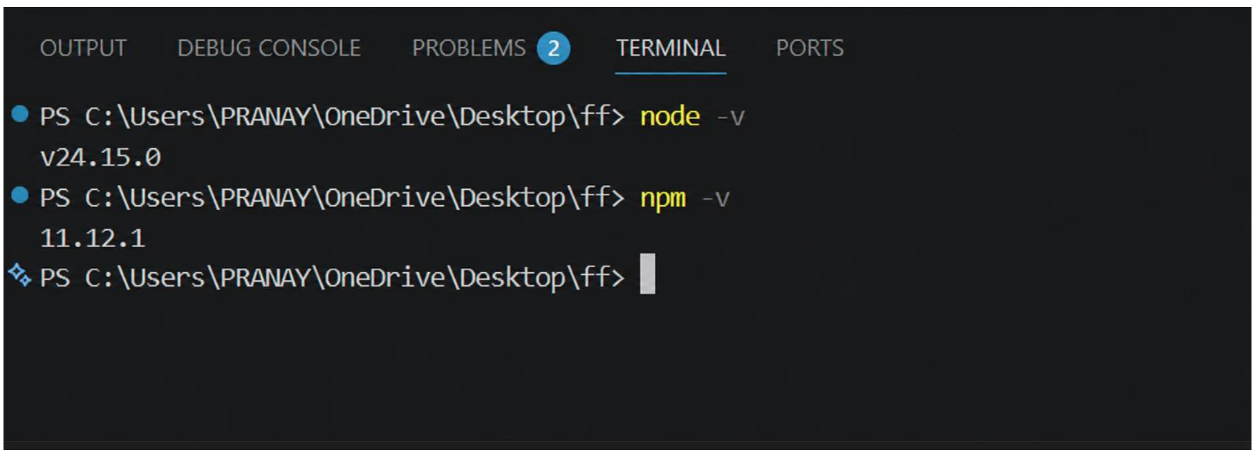
- **Node**
- **npm (Node Package Manager)**



Step2:

Open VS Code.

Open terminal:

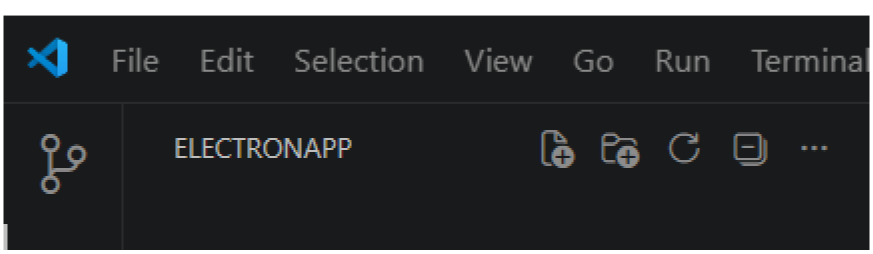


Step3:

Create New Empty Folder

- **Create a new folder anywhere**
  Example:

ElectronApp



Step4 — Initialize Node Project

Run:

npm init -y

This creates:

package.json

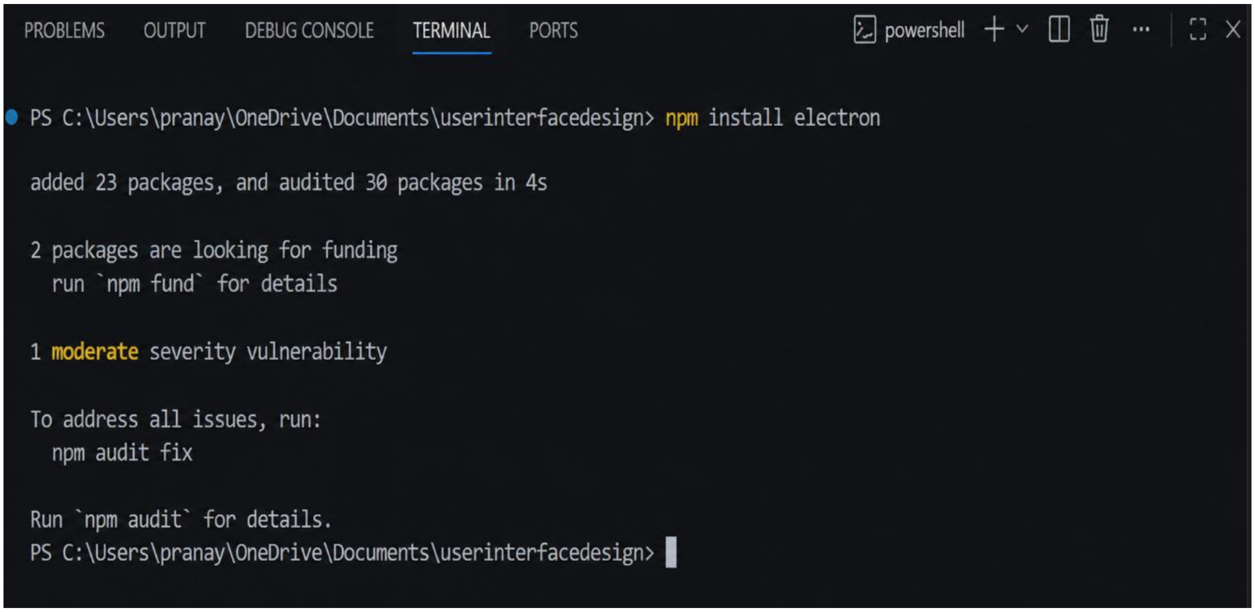

```
PS C:\Users\pranay\OneDrive\Documents\userinterfacedesign> npm init -y
Wrote to C:\Users\pranay\OneDrive\Documents\userinterfacedesign\package.json:

{
  "name": "userinterfacedesign",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "repository": {
    "type": "git",
    "url": "git+https://github.com/pranaySaitb-UI/userinterfacedesign.git"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "bugs": {
    "url": "https://github.com/pranaySaitb-UI/userinterfacedesign/issues"
  },
  "homepage": "https://github.com/pranaySaitb-UI/userinterfacedesign#readme"
}
PS C:\Users\pranay\OneDrive\Documents\userinterfaceedesign>
```

Step5 — Install Electron

Run:

npm install electron --save-dev

Wait until installation finishes.



Step6 — Modify package.json

Open package.json

Add:

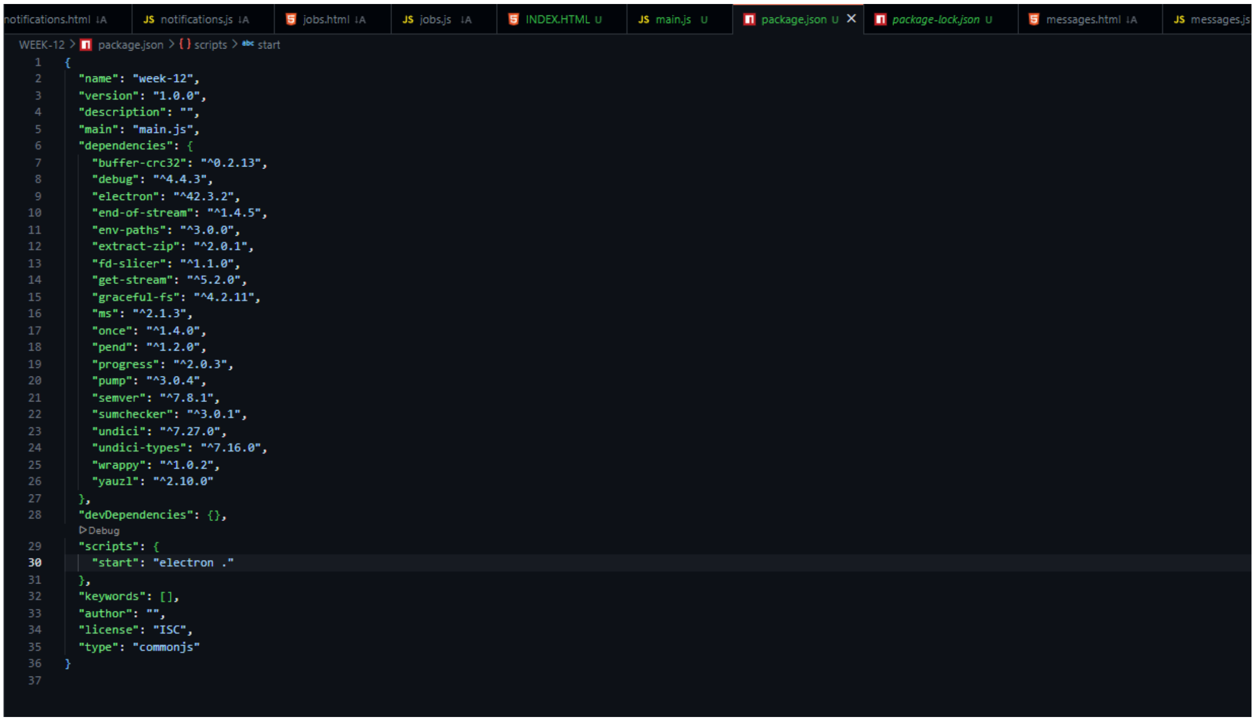
"main": "main.js",

Inside "scripts" add:

"start": "electron ."

Final example:

```
{
  "name": "electronapp",
  "version": "1.0.0",
  "main": "main.js",
  "scripts": {
    "start": "electron ."
  }
}
```



```json
{
  "name": "week-12",
  "version": "1.0.0",
  "description": "",
  "main": "main.js",
  "dependencies": {
    "buffer-crc32": "^0.2.13",
    "debug": "^4.4.3",
    "electron": "^42.3.2",
    "end-of-stream": "^1.4.5",
    "env-paths": "^3.0.0",
    "extract-zip": "^2.0.1",
    "fd-slicer": "^1.1.0",
    "get-stream": "^5.2.0",
    "graceful-fs": "^4.2.11",
    "ms": "^2.1.3",
    "once": "^1.4.0",
    "pend": "^1.2.0",
    "progress": "^2.0.3",
    "pump": "^3.0.4",
    "semver": "^7.8.1",
    "sumchecker": "^3.0.1",
    "undici": "^7.27.0",
    "undici-types": "^7.16.0",
    "wrappy": "^1.0.2",
    "yauzl": "^2.10.0"
  },
  "devDependencies": {},
  "scripts": {
    "start": "electron ."
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

Step7 — Open Electron Official Website

Open Chrome.

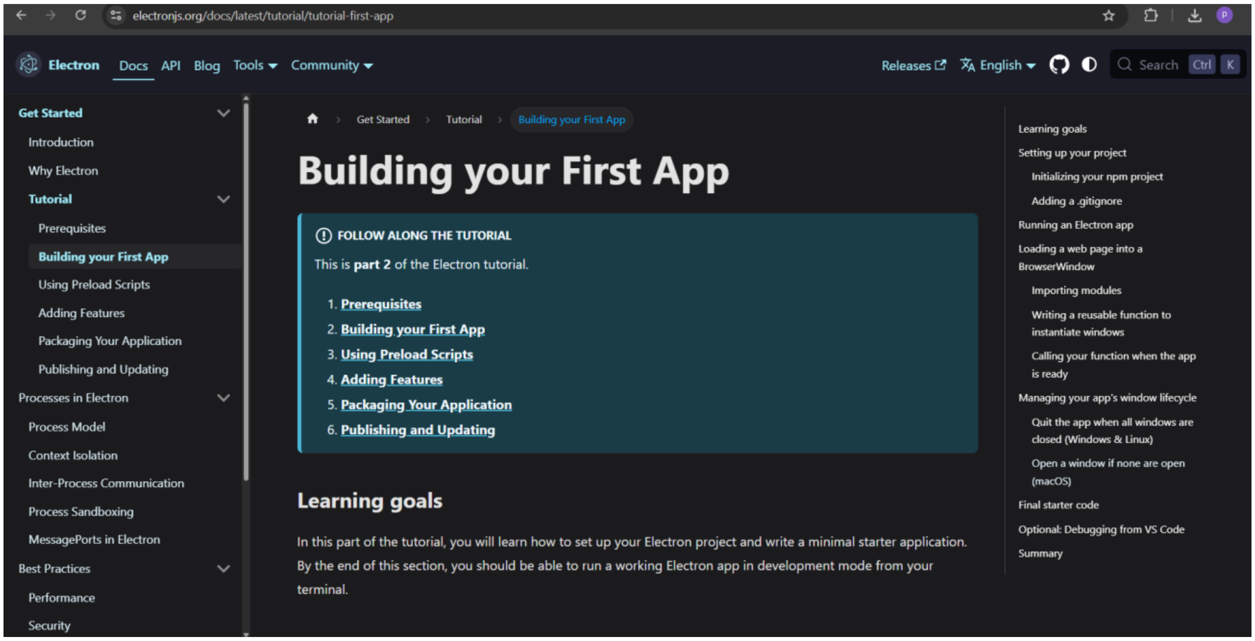
Go to:

[Electron.js Official Website]

Navigate:

Docs → Tutorial → Building Your First App

Copy the Final Starter Code.



Step8 — Create Required Files

Inside your project folder create:

main.js
index.html

Step9 — Paste Electron Code in main.js

```javascript
const { app, BrowserWindow } = require("electron/main");
const path = require("node:path");

function createWindow() {
  const win = new BrowserWindow({
    width: 800,
    height: 600,
    webPreferences: {
      preload: path.join(__dirname, "preload.js"),
    },
  });

  win.loadFile("index.html");
}

app.whenReady().then(() => {
  createWindow();

  app.on("activate", () => {
    if (BrowserWindow.getAllWindows().length === 0) {
      createWindow();
    }
  });
});

app.on("window-all-closed", () => {
  if (process.platform !== "darwin") {
    app.quit();
  }
});
```

Step10 — Add HTML File

```html
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>Hello World!</title>
    <meta
      http-equiv="Content-Security-Policy"
      content="script-src 'self' 'unsafe-inline';"
    />
  </head>
  <body>
    <h1>Hello World!</h1>
    <p>
      We are using Node.js <span id="node-version"></span>, Chromium
      <span id="chrome-version"></span>, and Electron
      <span id="electron-version"></span>.
    </p>
  </body>
</html>
```

notifications.html ↓A | notifications.js ↓A | jobs.html ↓A | jobs.js ↓A | main.js U | index.html WEEK-12 U

WEEK-12 > index.html > html > body > div.card

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Electron Test</title>

    <style>
      * {
        margin: 0;
        padding: 0;
        box-sizing: border-box;
        font-family:
          Inter,
          Segoe UI,
          sans-serif;
      }

      body {
        background: #f8fafc;
        color: #1e293b;
        min-height: 100vh;
        display: flex;
        justify-content: center;
        align-items: center;
        padding: 2rem;
      }

      .card {
        width: 100%;
        max-width: 500px;
        background: white;
        padding: 2rem;
        border-radius: 16px;
        box-shadow: 0 10px 30px rgba(0, 0, 0, 0.08);
      }

      h1 {
        font-size: 2rem;
        margin-bottom: 0.5rem;
      }
```

```
    h1 {
      font-size: 2rem;
      margin-bottom: 0.5rem;
    }

    p {
      color: #64748b;
      line-height: 1.6;
      margin-bottom: 1.5rem;
    }

    button {
      background: #2563eb;
      color: white;
      border: none;
      padding: 12px 20px;
      border-radius: 8px;
      cursor: pointer;
      font-size: 0.95rem;
      transition: 0.2s;
    }

    button:hover {
      background: #1d4ed8;
    }

    .status {
      margin-top: 1rem;
      font-size: 0.9rem;
      color: #16a34a;
    }
  </style>
</head>
<body>
  <div class="card">
    <h1>Electron Test</h1>
```

```
  </style>
</head>
<body>
  <div class="card">
    <h1>Electron Test</h1>
    <p>
      Simple UI for testing Electron rendering, JavaScript execution, and CSS
      styling.
    </p>

    <button onclick="testElectron()">Run Test</button>

    <div class="status" id="status">Ready</div>
  </div>

  <script>
    function testElectron() {
      document.getElementById("status").textContent =
        "JavaScript is working ✓";
    }
  </script>
</body>
</html>
```

Step11 — Verify File Names

Make sure:

win.loadFile('index.html')

matches your actual HTML filename.

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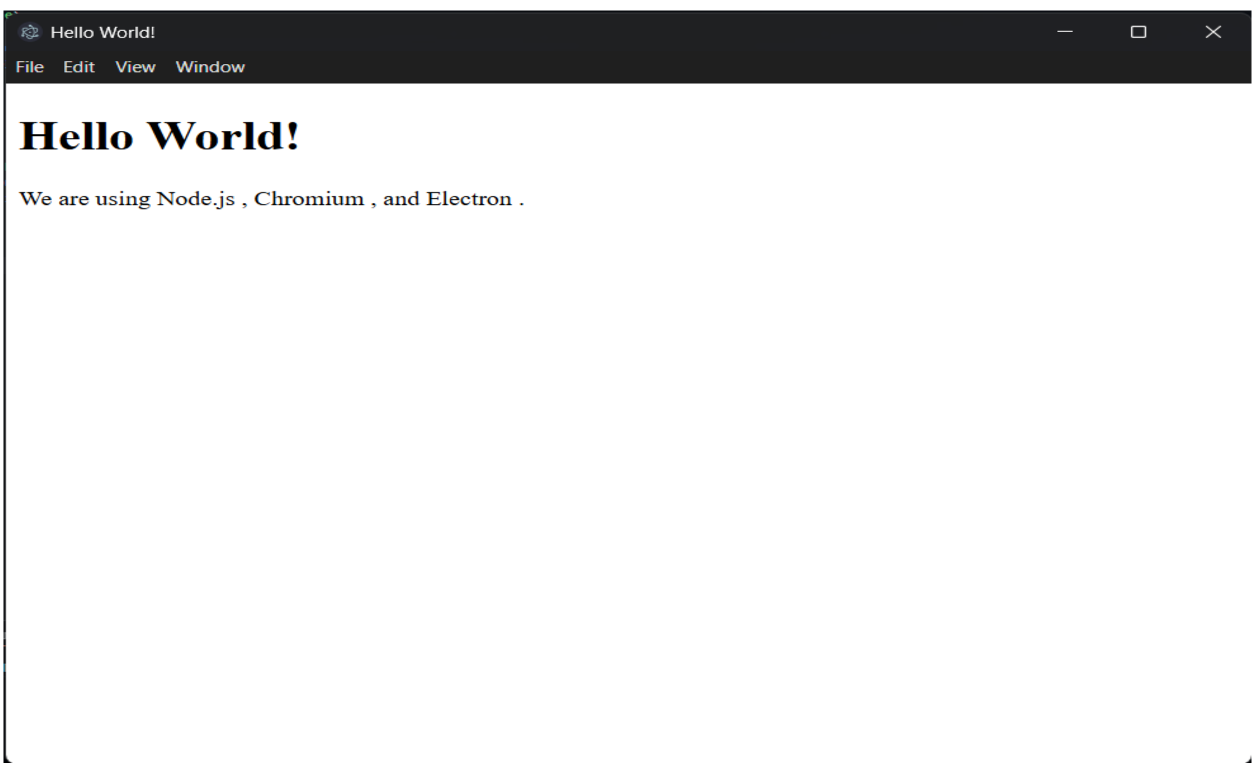
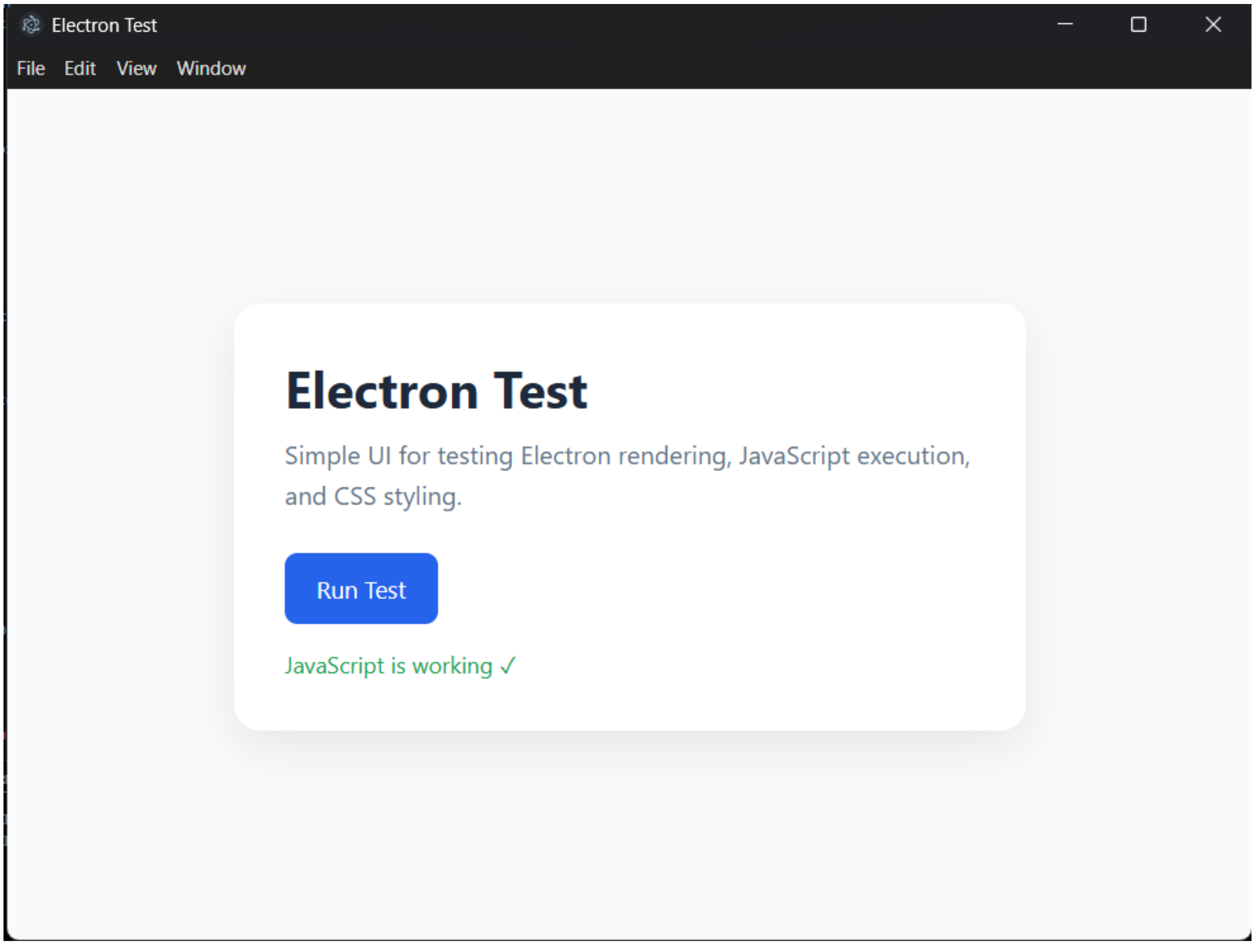
matches your actual HTML filename.

Output:





Description:

Description of Modules Used

1. Node.js

Node.js is a JavaScript runtime environment used to execute JavaScript code outside the browser. It provides the backend environment required for Electron applications. In this project, Node.js is used to initialize the project, install dependencies, and manage packages using npm.

2. npm (Node Package Manager)

npm is the default package manager for Node.js. It is used to install Electron and other required libraries into the project. Commands such as npm init -y, npm install electron --save-dev, and npm start are executed using npm.

3. Electron.js

Electron.js is an open-source framework used for building desktop applications using HTML, CSS, and JavaScript. It combines Chromium and Node.js, allowing web applications to run as desktop software on Windows, Linux, and macOS.

4. main.js

The main.js file is the main entry point of the Electron application. It creates the application window using the BrowserWindow module and controls the lifecycle of the desktop application.

Main functions of main.js:

- **Creates the application window**
- **Loads the HTML file**
- **Handles application startup and closing events**

5. BrowserWindow Module

BrowserWindow is an Electron module used to create and manage desktop windows. It allows the application to display web pages inside a native desktop window.

Features:

- **Set window size**
- **Load HTML pages**
- **Control window behavior**

6. app Module

The app module controls the overall lifecycle of the Electron application. It handles events such as:

- **Application startup**
- **Closing windows**
- **Activating the application**

7. package.json

The package.json file contains project information and configuration details. It stores:

•Project name and version

• Installed dependencies

• Main entry file

• Scripts used to run the application

8. index.html

The index.html file contains the user interface of the Electron application. It is loaded inside the Electron window and displays the frontend content using HTML and CSS.

9. preload.js

The preload.js file acts as a bridge between the renderer process and Node.js environment. It is executed before the web page loads and is commonly used for secure communication between frontend and backend components.

Introduction to Electron.js

Electron.js is an open-source framework used to build cross-platform desktop applications using web technologies such as HTML, CSS, and JavaScript. It was developed and maintained by Electron.js.

Electron combines two major technologies:

• Chromium browser engine for the user interface

• Node.js runtime for backend and system-level operations

This allows developers to create desktop applications using the same technologies used for web development.

Applications built with Electron can run on:

• Windows

• macOS

• Linux

Electron is widely used for developing modern desktop applications because it provides:

- **Easy development process**
- **Cross-platform compatibility**
- **Access to native system features**
- **Support for modern web technologies**

Popular applications built using Electron include:

- **Visual Studio Code**
- **Discord**
- **Slack**
- **WhatsApp Desktop**
- **Spotify Desktop**

In this project, Electron is used to create a desktop application by setting up a Node.js project, installing Electron through npm, and running the application using the Electron runtime.